



UNIVERSITÀ
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Department of Economics and Management

Master Programme in Finance

**Machine Learning in Portfolio Optimization under
the SAFE AI Framework: An Empirical Study of
Expected Return Estimation**

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Thesis

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Academic Year 2025-2026

Abstract

This thesis examines whether machine-learning-based expected return estimates can improve out-of-sample portfolio performance compared with the traditional Historical Mean benchmark. It also studies whether the model with the best financial performance is also the most compliant from a SAFE AI perspective.

The empirical analysis uses a stock universe derived from the S&P 500, based on Bloomberg data, from January 2010 to December 2025. The out-of-sample period covers 2023 to 2025. Four return estimation methods are compared: Historical Mean, Ridge Regression, Neural Network and XGBoost. All models are evaluated within the same constrained portfolio optimization process, using the same covariance estimator, constraints, rebalancing rule and transaction cost assumptions.

The SAFE AI assessment considers robustness, accuracy, fairness and explainability. Since the predictions concern securities rather than individuals, fairness is measured across GICS sectors. The SAFE AI dimensions are then combined into an integrated compliance score and linked to portfolio performance through the SAFE-Performance Frontier.

The results indicate that XGBoost delivers the strongest out-of-sample risk-adjusted performance and the highest SAFE AI compliance score, although its advantage over the Historical Mean benchmark is reduced after transaction costs due to higher turnover.

Questa tesi analizza se le stime dei rendimenti attesi ottenute tramite modelli di machine learning possano migliorare la performance out-of-sample di portafoglio rispetto al benchmark tradizionale della media storica. Inoltre, lo studio verifica se il modello con la migliore performance finanziaria sia anche quello più conforme dal punto di vista SAFE AI.

L'analisi empirica utilizza un universo di titoli derivato dall'S&P 500, costruito con dati Bloomberg, da gennaio 2010 a dicembre 2025. Il periodo out-of-sample considerato va dal 2023 al 2025. Vengono confrontati quattro metodi di stima dei rendimenti: media storica, Ridge Regression, Neural Network e XGBoost. Tutti i modelli sono valutati nello stesso processo di ottimizzazione vincolata di portafoglio, usando lo stesso stimatore della covarianza, gli stessi vincoli, la stessa regola di ribilanciamento e la stessa struttura dei costi di transazione.

La valutazione SAFE AI considera robustezza, accuratezza, fairness e spiegabilità. Poiché le previsioni riguardano titoli finanziari e non individui, la fairness viene misurata tra i settori GICS. Le dimensioni SAFE AI sono poi sintetizzate in un compliance score integrato e collegate alla performance attraverso la SAFE-Performance Frontier.

I risultati indicano che XGBoost ottiene la migliore performance out-of-sample corretta per il rischio e il compliance score SAFE AI più elevato, sebbene il suo vantaggio rispetto alla media storica si riduca dopo i costi di transazione a causa del maggiore turnover.

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1 Introduction

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1.1 Background and motivation

The starting point of this thesis is the mean-variance framework of Markowitz (1952), because it offers a direct link between two elements that are central to portfolio decisions: expected return and risk. In practical applications, however, the framework depends on inputs that are not observed directly. The investor must estimate both the vector of expected returns and the covariance matrix. Among these inputs, the expected-return vector is especially relevant for this study, because even small estimation errors can lead to large changes in the optimized portfolio weights.

Previous studies show that small errors in expected returns can produce large changes in the optimal portfolio weights (Michaud, 1989; Best & Grauer, 1991; Chopra & Ziemba, 1993). For this reason, the way in which expected returns are estimated can strongly affect the final portfolio.

The Historical Mean is one of the simplest methods used for this task. It estimates each stock's future return using its average return over a past time window. This approach is useful because it is simple, transparent and easy to reproduce. For this reason, it is used in this thesis as the main benchmark. However, it also has some important limits. Stock returns are volatile, while the expected component is usually small. For this reason, past averages can be noisy estimates of future performance (Merton, 1980).

Another limit is related to the information used by the Historical Mean. This method uses only past returns and does not include other stock characteristics. The empirical asset pricing literature shows that several characteristics can be related to future returns, such as size, momentum and illiquidity (Fama & French, 1992; Jegadeesh & Titman, 1993; Amihud, 2002). Machine learning models can be useful in this context because they can use different characteristics at the same time and can capture relations that are not necessarily linear. This is consistent with Gu, Kelly and Xiu (2020), who show that machine learning models can be useful in empirical asset pricing when many firm-level characteristics are used.

The main idea of this thesis is therefore to compare the Historical Mean benchmark with machine-learning-based estimates of expected returns. The aim is to understand whether this replacement can improve out-of-sample portfolio performance. This question matters because expected returns are not evaluated only as predictions. They enter the portfolio optimizer and are transformed into portfolio weights. For this reason, the thesis evaluates the return estimation methods also at portfolio level.

Three machine learning models are considered. XGBoost is the main model, while Ridge Regression and the Neural Network are included as robustness checks. This structure makes it possible to check whether the results obtained with XGBoost are specific to this model or whether similar patterns are also obtained with other machine learning approaches.

However, when machine learning models are used to support portfolio decisions, portfolio performance is not the only aspect to consider. A model can produce good investment results but still be unstable, difficult to interpret, or different in its behavior across groups of stocks. For this reason, the thesis also evaluates the machine learning models from a SAFE AI perspective. This adds a second level of analysis, focused on the quality of the models that generate the expected return estimates.

1.2 Moving from return prediction to portfolio construction

Return prediction and portfolio construction are connected, but they are not the same thing. A model can produce useful predictions, but this does not automatically mean that it will generate a good portfolio. The final result also depends on risk estimation, portfolio constraints, the rebalancing rule and transaction costs. For this reason, the thesis does not stop at the prediction level. The expected return estimates produced by the different approaches are used to construct portfolios within the same portfolio construction framework so that the comparison focuses mainly on the expected return estimator.

This also follows the logic of out-of-sample portfolio evaluation. A method can look good before the optimization step, but this does not guarantee that it will generate better realized portfolio results. For this reason, the comparison between models is made after the expected return estimates are used in the optimizer. This is consistent with DeMiguel, Garlappi and Uppal (2009), who show that optimized portfolios must be evaluated carefully out-of-sample.

This creates two connected but distinct levels of evaluation. The portfolio level considers the investment results generated by the different expected return estimates. The model level evaluates the machine learning models that produce those estimates, using the SAFE AI framework.

The empirical analysis is therefore organized around three connected parts. The first part evaluates the portfolios produced by the different return estimators. The second part evaluates the machine learning models through SAFE AI. The third part connects these two analyses through the SAFE-Performance Frontier, which is used in this thesis as an empirical tool to compare SAFE AI compliance and portfolio performance.

These three parts are connected to the two main research questions of the thesis: the first question concerns the comparison between XGBoost and the Historical Mean benchmark at portfolio level, whereas the second question concerns the relation between SAFE AI compliance and portfolio performance.

1.3 Motivation for Responsible AI and SAFE AI

When machine learning models are used to support portfolio decisions, their evaluation should not be limited to the final investment outcome. A model can produce good portfolio results while still being unstable, difficult to interpret, or uneven in its behavior across groups of stocks. Therefore, it is useful to evaluate not only the final portfolio, but also the model that produces the expected return estimates. This idea is consistent with the responsible AI literature, which underlines the importance of robustness, fairness, transparency and explainability when machine learning models are used in relevant decision-making contexts (Rudin, 2019; Arrieta et al., 2020; Barocas, Hardt, & Narayanan, 2023; High-Level Expert Group on AI, 2019).

For this reason, the thesis builds on the SAFE AI framework proposed by Giudici and Raffinetti (2023). SAFE AI is useful because it evaluates models through different dimensions, such as robustness, accuracy, fairness and explainability. In this thesis, these dimensions are adapted to the financial setting, because the models predict future stock returns and not outcomes for individuals.

The SAFE AI evaluation is applied only to the machine learning models. The Historical Mean is not included in this part because it is not a trained machine learning model. However, it remains the benchmark in the portfolio-level comparison. This gives the thesis two levels of evaluation. The first one is the portfolio level, where the final investment results are compared. The second one is the model level, where the machine learning models are evaluated from a SAFE AI perspective.

A specific point concerns the interpretation of fairness. In standard machine learning applications, fairness is often related to demographic groups. In this thesis, the objects of prediction are stocks. For this reason, fairness is interpreted across groups of securities. More specifically, sectors are used as the relevant groups.

The relation between the two levels of evaluation is then studied through the SAFE-Performance Frontier. This makes it possible to understand whether the machine learning model with higher SAFE AI compliance is also the one that produces better portfolio performance.

1.4 Research objectives

Based on the motivation presented in the previous sections, the study has three main objectives. The first objective is to compare the Historical Mean benchmark with machine-learning-based expected return estimates inside the same portfolio optimization framework. This is done to understand whether replacing the Historical Mean with machine learning estimates can improve

out-of-sample portfolio performance. The optimizer, the covariance estimator, the portfolio constraints and the rebalancing rule are kept fixed across models. In this way, the comparison focuses mainly on the method used to estimate the expected return vector.

The second objective is to evaluate Ridge Regression, the Neural Network and XGBoost through a four-dimensional SAFE AI evaluation. The evaluation considers robustness, accuracy, fairness and explainability. In this setting, fairness is interpreted at the sector level because the objects of prediction are stocks rather than individuals.

The third objective is to connect the model-level SAFE AI evaluation with the portfolio-level evaluation. This is done through CS4 and the SAFE-Performance Frontier. CS4 summarizes the SAFE AI profile of each machine learning model, while the SAFE-Performance Frontier is used to study the relation between compliance and risk-adjusted portfolio performance.

These three objectives follow the three parts of the empirical analysis. The first part concerns portfolio performance. The second part concerns SAFE AI evaluation. The third part connects the two through the SAFE-Performance Frontier.

1.5 Research questions

The objectives described in the previous section lead to two main research questions.

RQ1. Does XGBoost generate better out-of-sample portfolio performance than the Historical Mean benchmark when both are used in the same constrained portfolio optimization framework?

RQ2. Is the machine learning model with the highest four-dimensional SAFE AI compliance score also the one that obtains the best risk-adjusted portfolio performance?

RQ1 is addressed in Chapter 4. This chapter compares the out-of-sample performance of the four portfolios: Historical Mean, Ridge Regression, the Neural Network and XGBoost. The

comparison is based on portfolio-level measures such as Sharpe ratio, maximum drawdown, turnover and transaction-cost-adjusted performance.

RQ2 is addressed in Chapters 5 and 6. Chapter 5 evaluates the three machine learning models through the SAFE AI framework and summarizes robustness, accuracy, fairness and explainability through CS4. Chapter 6 then connects this score with portfolio performance through the SAFE-Performance Frontier. This makes it possible to understand whether the model that is more compliant from a SAFE AI point of view is also the one that performs better at portfolio level.

1.6 Overview of methodology

The empirical analysis is based on an investment universe derived from the S&P 500. The data are obtained from the Bloomberg Terminal available at the University of Pavia. The sample covers the period from January 2010 to December 2025 and is divided into a warm-up period, a training period and an out-of-sample test period from 2023 to 2025.

Four expected return estimation approaches are compared: the Historical Mean benchmark, Ridge Regression, the Neural Network and XGBoost. The Historical Mean uses past average returns, while the machine learning models use stock-level characteristics related to reversal, momentum, volatility, illiquidity and size. These variables and the data cleaning procedure are described in detail in Chapter 3.

The expected return estimates are then used to construct portfolios. All models are evaluated within the same constrained portfolio optimization framework. This means that the optimizer, the covariance estimator, the portfolio constraints and the rebalancing rule are kept fixed across models. In this way, the comparison focuses mainly on the method used to estimate the expected return vector.

Portfolio performance is evaluated out-of-sample before and after transaction costs. The analysis considers return, risk, drawdown, turnover and net performance. This makes it possible to understand not only which portfolio performs better, but also whether the strategy is stable and practically implementable.

The three machine learning models are also evaluated through the SAFE AI framework. The four dimensions considered are robustness, accuracy, fairness and explainability. These dimensions are summarized through CS4, a four-dimensional compliance score used in this thesis to compare the SAFE AI profile of the models. Finally, the relation between CS4 and portfolio performance is studied through the SAFE-Performance Frontier. The full methodological details are presented in Chapter 3.

1.7 Contribution of the thesis

This thesis builds on the SAFE AI framework and on the recent literature on rank-based SAFE AI metrics. The contribution of the thesis is mainly methodological and empirical. The thesis applies and combines existing SAFE AI concepts in a stock return prediction and portfolio optimization setting.

The first contribution is the implementation of a four-dimensional SAFE AI evaluation for machine learning models used in expected return estimation. The models are evaluated in terms of robustness, accuracy, fairness and explainability. In this thesis, fairness is adapted to the financial setting. Since the observations are stocks and not individuals, fairness is evaluated across groups of securities. More specifically, the thesis uses a sector-level RGF-based component, denoted as RGF* in the empirical analysis.

The second contribution is the use of a four-dimensional compliance score, denoted as CS4. This score combines the four SAFE AI dimensions into one synthetic measure. The objective

of CS4 is not to replace the individual SAFE AI metrics, but to summarize the overall compliance profile of each machine learning model in a comparable way. The inclusion of the fairness component is useful because it allows the score to consider not only average model quality, but also differences in model behavior across sectors.

The third contribution is the connection between SAFE AI compliance and portfolio performance. This is done through the SAFE-Performance Frontier. The frontier links CS4 to risk-adjusted portfolio performance across different model configurations. In this way, the thesis does not treat SAFE AI evaluation and portfolio evaluation as two separate parts but studies their relation inside the same empirical framework.

Overall, the thesis contributes by combining expected return estimation, constrained portfolio optimization, SAFE AI evaluation and portfolio performance analysis in the same study. The contribution is therefore the empirical use of SAFE AI metrics in a machine-learning-based portfolio construction problem.

1.8 Limits of the analysis

When interpreting the results, some limits must be considered.

The first limit concerns the investment universe. The analysis is based on a cleaned universe derived from the S&P 500. This choice is useful because it makes the dataset consistent through the full empirical analysis. However, the results may not be directly valid for other markets, smaller firms or less liquid stocks. In addition, the use of a cleaned universe may introduce a data-availability bias, because stocks with insufficient data coverage are removed from the sample.

The second limit is related to the test period. The out-of-sample window covers the period from January 2023 to December 2025. This period is useful because it uses recent data, but it remains

relatively short. A different test period or a longer out-of-sample window could lead to different results.

The third limit concerns the portfolio structure. The portfolios are built using a constrained optimization framework with a fixed cardinality constraint. This choice is made to obtain concentrated portfolios and to keep the optimization problem manageable. However, different levels of cardinality or different portfolio constraints could produce different results.

The fourth limit is related to transaction costs. They are modeled as proportional costs applied to turnover. This makes it possible to compare the models in a simple and consistent way. However, the analysis does not include other cost components, such as bid-ask spreads, market impact, commissions or differences in trading costs across stocks.

The fifth limit concerns the set of predictors used by the machine learning models. The analysis uses variables related to reversal, momentum, volatility, illiquidity and size. Other variables, such as accounting data, macroeconomic indicators, analyst forecasts or alternative data, could also be considered and may change the results.

The last limit concerns the adaptation of SAFE AI to the financial setting of this thesis. The SAFE AI framework is general, while this thesis applies it to stock return prediction. For this reason, some dimensions require an empirical interpretation. In particular, fairness is evaluated at the sector level. This is a reasonable choice for this setting, but it is not the only possible one. Fairness could also be studied using groups based on size, liquidity, volatility, country, industry group or ESG characteristics.

For these reasons, the results should be interpreted within the specific setting of this thesis. They provide evidence for the selected universe, period, features, portfolio constraints and SAFE AI implementation, but they should not be considered automatically valid for all markets or all portfolio construction rules.

1.9 Outline of the thesis

The rest of the thesis is organized as follows.

Chapter 2 reviews the literature related to the thesis. It discusses mean-variance optimization, expected return estimation, machine learning in finance, portfolio evaluation, responsible AI and SAFE AI metrics.

Chapter 3 presents the data and methodology. It describes the investment universe, data cleaning procedure, preprocessing pipeline, feature engineering, return estimation models, portfolio optimizer, performance metrics and SAFE AI evaluation procedure. It also explains the construction of CS4 and the SAFE-Performance Frontier.

Chapter 4 presents the out-of-sample portfolio results. It compares the Historical Mean benchmark with Ridge Regression, the Neural Network and XGBoost in terms of return, risk, Sharpe ratio, maximum drawdown, turnover and transaction-cost-adjusted performance.

Chapter 5 reports the SAFE AI evaluation of the three machine learning models. It evaluates robustness, accuracy, fairness and explainability and summarizes these dimensions through CS4.

Chapter 6 introduces the SAFE-Performance Frontier. It studies the relation between CS4 and portfolio performance across different model configurations. The objective is to understand whether higher SAFE AI compliance is associated with better risk-adjusted portfolio performance.

Chapter 7 concludes the thesis. It summarizes the main findings, answers the research questions, discusses the limits of the analysis and suggests possible directions for future research.

2 Literature Review

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2.1 Introduction

This chapter presents the literature used to motivate the thesis.

The chapter starts from the mean-variance framework introduced by Markowitz (1952). This framework is important because it is the basis of the classical portfolio optimization approach. In this framework, the portfolio depends mainly on two inputs: *the expected return vector and the covariance matrix*. The expected return vector is particularly relevant for this thesis, because it is difficult to estimate and because errors in expected returns can strongly affect the final portfolio weights.

For this reason, the first part of the chapter focuses on the role of expected returns in portfolio optimization. Previous studies show that mean-variance portfolios are very sensitive to errors in the estimated expected returns.

The chapter then discusses the Historical Mean approach. This is one of the simplest ways to estimate future returns, but it also has important limitations. This is relevant because the optimizer does not use expected returns only as forecasts, it transforms them into portfolio weights.

The chapter then introduces machine learning as an alternative way to estimate expected returns and introduces the three machine learning models considered in the thesis: XGBoost, Ridge Regression and the Neural Network.

The final part of this section discusses the need to evaluate machine learning models not only from a portfolio-performance point of view, but also from a model-level point of view.

The chapter therefore prepares the two main research questions of the thesis. The first one is whether replacing Historical Mean expected returns with machine-learning-based expected returns improves out-of-sample portfolio performance. The second one is whether the machine

learning model that performs better under the SAFE AI framework is also the one that produces the best portfolio results.

2.2 Markowitz portfolio theory and the role of expected returns

The starting point of this thesis is the mean-variance framework introduced by Markowitz (1952). This framework represents the basis of modern portfolio theory. The main idea is that an investor should not evaluate each asset only in isolation but should consider how all assets behave together inside the portfolio.

In the Markowitz framework, the investor chooses the portfolio weights by considering both expected return and risk. Risk is measured through the variance of portfolio returns. A portfolio can therefore be built by minimizing risk for a given level of expected return, or by maximizing expected return for a given level of risk. This is important because portfolio risk does not depend only on the volatility of each stock, but also on the correlations and covariances between stocks.

The two main inputs of the mean-variance problem are the expected return vector, μ , and the covariance matrix, Σ . The expected return vector contains the expected return of each asset. The covariance matrix describes how the returns of the assets move together. If these two inputs were known with certainty, the optimization problem would be straightforward. In practice, however, they are not known and must be estimated from historical data.

This creates an important problem. The portfolio obtained from mean-variance optimization depends on estimated inputs. If these inputs contain errors, the optimized portfolio can also be affected by these errors. This issue is especially important for expected returns. Michaud (1989) shows that mean-variance optimization can overweight assets with overestimated expected

returns and underweight assets with underestimated expected returns. In this way, the optimizer can amplify estimation errors instead of reducing them.

Best & Grauer (1991) also study this problem. They show that optimal portfolios are very sensitive to changes in expected returns. Even small changes in the estimated mean returns can produce large changes in the optimal weights. At the same time, the portfolio is usually less sensitive to changes in the covariance matrix. This means that errors in expected returns are particularly relevant for portfolio construction.

This point is also discussed by Chopra and Ziemba (1993). They show that errors in mean returns can have a larger impact on optimal portfolio choice than errors in variances and covariances. This result is important for this thesis because it explains why the estimation of expected returns deserves specific attention. If the expected return vector is badly estimated, the final portfolio can be unstable or inefficient, even if the optimization model is correctly specified.

Another reason why expected returns are problematic is that they are difficult to estimate accurately. Merton (1980) explains that estimating expected returns requires long historical samples, because the average return is small compared with return volatility. This means that the historical average return can contain a high level of noise. For this reason, using past average returns as estimates of future returns can be simple, but it can also be unreliable.

The covariance matrix is also important in portfolio optimization. If the number of assets is large, the sample covariance matrix can become unstable. For this reason, the literature proposes shrinkage estimators. Ledoit and Wolf (2004) propose a covariance estimator that combines the sample covariance matrix with a structured target. This helps to obtain a more stable covariance matrix, especially when the investment universe is large.

Overall, the literature shows that the quality of the portfolio depends strongly on the quality of the estimated inputs. Among these inputs, expected returns are especially important because they are difficult to estimate and because the optimizer is very sensitive to them. For this reason, the estimation of μ is one of the central points of this thesis. This motivates the comparison between the traditional Historical Mean approach and machine-learning-based alternatives.

2.3 The historical mean approach and its limitations

The Historical Mean is one of the simplest ways to estimate the expected return vector μ . With this approach, the expected return of each stock is estimated using its average return over a past time window. This method is simple, transparent and easy to apply. For this reason, it is often used as a first benchmark in mean-variance portfolio optimization.

The main advantage of the Historical Mean is its simplicity. It does not need a prediction model, because it only looks at the returns observed in the past. For this reason, it is easy to understand, easy to reproduce and useful as a benchmark.

However, this simplicity also has a cost. Past average returns can be very noisy. Stock returns move a lot over time, while the part that can be interpreted as expected return is usually much smaller. For this reason, the average return calculated over a past period may not be a reliable estimate of future returns. Merton (1980) explains that estimating expected returns is difficult for this reason, because very long samples would be needed to estimate the mean return precisely. Michaud (1989) also shows that this problem becomes very relevant in portfolio optimization, because noisy expected returns can lead to unstable portfolios.

The second limitation is related to the type of information used. The Historical Mean uses only past returns. It does not include other information about the stock. This is an important limitation because the empirical asset pricing literature shows that different stock characteristics

can be related to future returns. Fama & French (1992) show that size and book-to-market help explain differences in average stock returns. Jegadeesh and Titman (1993) document the momentum effect, showing that stocks that performed well in the recent past tend to continue performing well in the short to medium term. Amihud (2002) shows that illiquidity is also related to expected returns. Andersen, Bollerslev, Diebold and Labys (2003) also discuss realized volatility as an important measure of return variability in financial time series.

This point is important for this thesis. The Historical Mean assumes that the past average return of a stock is enough to estimate its future return. However, stocks can also differ in momentum, volatility, liquidity and size. These differences may contain useful information when the objective is to compare stocks inside the same investment universe.

Another important point is that the Historical Mean is not used only as a forecast. In a portfolio optimization problem, the expected returns are transformed into portfolio weights. This means that if the historical averages are noisy, the optimizer can give more weight to stocks whose estimated return is high only because of noise. This makes the limitation of the Historical Mean more relevant at portfolio level.

For these reasons, the Historical Mean is useful as a benchmark, but it may be limited as a method to estimate future returns. This motivates the use of alternative methods that can include more information than past average returns alone. In this thesis, machine learning models are considered for this reason. The objective is to test whether expected returns obtained from machine learning models can improve out-of-sample portfolio performance compared with the Historical Mean benchmark.

2.4 Machine learning as an alternative for expected return estimation

Machine learning provides an alternative way to estimate expected returns. The main difference from the Historical Mean is that machine learning models do not rely only on past average returns. They can use several stock-level characteristics at the same time and combine them into one prediction. This is useful because the relation between stock characteristics and future returns may not be simple. For example, the effect of momentum may depend on the level of volatility, or the effect of liquidity may depend on the size of the firm. A linear model can capture only a linear relation between the variables and future returns. More flexible models can instead capture non-linear relations and interactions between variables.

Gu, Kelly and Xiu (2020) study the use of machine learning models in empirical asset pricing. They compare different methods for return prediction and show that tree-based models and neural networks can generate economic gains compared with more traditional linear approaches. Their results also show that the advantage of these models comes from their ability to capture non-linear interactions between predictors. This is relevant for this thesis because the stock-level features used in the analysis may interact with each other.

At the same time, return prediction remains difficult. Financial returns contain a lot of noise, and the predictable part is usually small. For this reason, machine learning should not be interpreted as a perfect solution to the problem of expected return estimation. It is better to consider it as a more flexible method that can use more information than the Historical Mean.

For this reason, the prediction problem in this thesis is treated mainly as a *ranking problem*. The objective is not only to predict the exact future return of each stock, which is particularly difficult in financial markets, but also to identify which stocks are expected to perform better or worse relative to the others. This ranking logic is consistent with the literature on cross-sectional return signals. For example, the momentum effect documented by Jegadeesh and

Titman (1993) is also based on the idea that the relative position of stocks contains useful information.

However, the portfolio optimizer cannot work only with an ordered list of stocks. It needs a numerical expected return vector μ . For this reason, the scores produced by the machine learning models must be transformed into expected return estimates before they can be used in the portfolio optimization step. This makes it possible to compare machine learning models and the Historical Mean using the same portfolio construction framework.

The next section presents the main families of machine learning models that can be used for this type of return prediction problem, distinguishing between linear regularized models, neural networks and tree-based boosting models.

2.5 Machine learning model families for return prediction

After discussing machine learning as an alternative to the Historical Mean, it is useful to explain the main types of models that can be used for return prediction. Machine learning models do not all work in the same way. Some models are simple and linear, while others are more flexible. The first type is represented by linear models. These models are useful because they are easier to understand. Ridge Regression belongs to this group. It is similar to a standard linear regression, but it adds a penalty to the coefficients. This penalty helps to make the model more stable when the predictors are correlated, as explained by Hoerl and Kennard (1970).

The second type is represented by neural networks. The Neural Network used in this thesis belongs to this family. It can capture relations that are not linear because it uses hidden layers and activation functions. This makes the model more flexible than Ridge Regression. However, this also means that the model can become more difficult to interpret. In addition, when the

target variable is noisy, as in return prediction, a more flexible model can also learn part of the noise.

The third type is represented by tree-based boosting models. XGBoost belongs to this group. It is based on gradient-boosted decision trees and was introduced by Chen and Guestrin (2016). The model builds several trees one after the other; each new tree tries to improve the errors made by the previous trees. This makes XGBoost useful when the relation between the variables and the target is not only linear.

This distinction is useful because return prediction can be studied with models of different complexity. Ridge Regression gives a simple linear reference. The Neural Network gives a non-linear neural-network model. XGBoost gives a non-linear tree-based model. Gu, Kelly and Xiu (2020) show that tree-based models and neural networks can be useful in empirical asset pricing because they can capture interactions between predictors. At the same time, simpler regularized models remain important because they give a clearer benchmark.

2.5.1 Ridge regression as a linear regularized model

Ridge Regression is a linear model with regularization. It was introduced by Hoerl and Kennard (1970). The model adds an L_2 penalty to the regression coefficients. This penalty reduces the size of the coefficients and helps to avoid very large values; this is useful when the predictors are correlated. In that case, a standard linear regression can become unstable. Ridge Regression reduces this problem because it shrinks the coefficients toward zero. In this way, the model remains linear, but it becomes more stable. This is also useful in financial prediction. Future returns are noisy and difficult to predict. For this reason, a model can easily fit noise instead of useful information. Ridge Regression helps to limit this problem because the regularization term controls the size of the coefficients.

The main advantage of Ridge Regression is that it is simple. It assumes a linear relation between the predictors and the target. This makes it easier to interpret than more complex machine learning models. The main limitation is that it cannot capture non-linear relations between variables. For example, if the effect of momentum changes when volatility is high, a linear model may not fully capture this relation.

For this reason, Ridge Regression is useful as a linear benchmark. It shows what can be obtained with a simple regularized model before considering more flexible models such as neural networks or tree-based boosting models.

2.5.2 Multilayer perceptron as a nonlinear neural model

The Multilayer Perceptron, referred to as Neural Network in the running text of this thesis and as MLP in tables, is a Neural Network model. It is usually organized as a feed-forward network, where the information moves from the input layer to one or more hidden layers and then to the output layer. This structure allows the model to learn relations that are not only linear (Goodfellow, 2016; Hastie, 2009).

This is the main difference with Ridge Regression: while Ridge Regression keeps a linear relation between the predictors and the target, the Neural Network is more flexible because it uses hidden layers and activation functions. For this reason, it can be useful when the relation between stock characteristics and future returns is more complex.

However, this flexibility also creates some limits. A Neural Network is less easy to interpret than a linear model. It is also more exposed to overfitting if the model learns noise instead of useful information. This is relevant in financial applications, because return prediction is difficult and the useful signal is usually weak. For this reason, neural networks are often compared with simpler models and with other machine learning methods.

In empirical asset pricing, Gu, Kelly and Xiu (2020) show that neural networks can perform well in return prediction. Their results are important because they show that more flexible models can be useful when many stock-level characteristics are used. At the same time, their work also confirms that return prediction remains difficult, so the use of neural networks must be interpreted with caution.

2.5.3 XGBoost as a tree-based boosting model

XGBoost is a tree-based machine learning model. It was introduced by Chen and Guestrin (2016) and is based on gradient boosting. The model builds several decision trees one after the other. Each new tree tries to improve the errors made by the previous trees. This makes XGBoost different from the other two models discussed before. Ridge Regression is linear. The Neural Network is based on a neural-network structure. XGBoost is instead based on decision trees. A tree model can divide the data into different groups and apply different rules to different parts of the data. This can be useful for stock return prediction because the same characteristic may not have the same effect for all stocks. For example, a variable can be more informative for one part of the market and less informative for another one. A tree-based model can capture this type of difference more naturally than a simple linear model.

The relevance of tree-based models is also supported by the empirical asset pricing literature. Gu, Kelly and Xiu (2020) show that tree-based models are among the strongest machine learning methods for predicting stock returns. They explain that these models are useful because they can capture interactions between predictors that linear models may miss. At the same time, XGBoost must be controlled carefully. Since the target variable is noisy and the model is flexible, there is a risk of overfitting. For this reason, regularization and validation procedures are important, as discussed in the machine learning literature (T. Chen & Guestrin, 2016; Hastie,

2009). For these reasons, XGBoost is useful as a tree-based model for expected return estimation.

2.6 From return forecasts to portfolio optimization

Return forecasting is only one part of the problem. In this thesis, the final objective is not only to obtain a prediction of future returns, but to build a portfolio from those predictions. For this reason, a model cannot be evaluated only by looking at how well it predicts or ranks future returns. It is also necessary to understand what happens when those predictions are used inside the portfolio optimizer.

This point is important because a model can look useful before the optimization step but still produce a weak portfolio. For example, the predicted returns can lead to unstable weights, high turnover or poor out-of-sample performance. This means that good forecasts do not automatically imply good portfolio results.

This problem is also discussed by DeMiguel, Garlappi and Uppal (2009). They show that optimized portfolios can perform worse than a simple equally weighted portfolio when they are tested out-of-sample. This result is relevant because it shows that the optimization step can make the final result very different from what is expected from the inputs alone. For this reason, the expected return estimates must be evaluated after they enter the portfolio construction process. The predicted scores become useful only when they are transformed into an expected return vector μ and used by the optimizer. This type of comparison is clearer when the portfolio construction framework is kept consistent across models. In that case, the main element that changes is the way in which μ is estimated. This is consistent with Gu, Kelly and Xiu (2020), who show that the value of a return prediction model should not be measured only with statistical prediction metrics. In finance, it is also important to verify whether the predictions

generate economic value when they are used in an investment strategy. For this reason, return forecasts must be evaluated also at portfolio level.

2.7 Portfolio-level performance metrics: risk, return, turnover and costs

A portfolio cannot be evaluated only by looking at its return. Two portfolios can have similar returns, but very different levels of risk. One portfolio can be more volatile, or it can have a larger loss during a bad period. For this reason, different performance metrics are needed.

The first metric is the Sharpe ratio. This is one of the most common measures used in portfolio evaluation (Sharpe, 1966). It compares the return of the portfolio with its volatility. A higher Sharpe ratio means that the portfolio obtains more return for each unit of risk. This is useful because it gives a first indication of the relation between return and risk. However, the Sharpe ratio is not enough by itself. It is based on volatility, but volatility does not always show the largest loss that a portfolio can experience. For this reason, maximum drawdown is also considered because it measures the largest fall of the portfolio from a previous peak. This is important because a portfolio can have a good average return but still experience a large loss during the period (Magdon-Ismail & Atiya, 2004). From an investor point of view, this information is relevant because large losses can make a strategy difficult to hold.

Turnover is also important in this thesis. It measures how much the portfolio changes from one rebalancing date to the next one. This is useful because it gives information about the stability of the portfolio over time. A portfolio that changes too much can be more difficult to implement in practice.

Turnover is also directly connected to transaction costs. Each time the portfolio is rebalanced, some trades are needed. If the strategy trades too much, transaction costs can reduce the final performance. This point is particularly relevant for machine learning models, because their

predicted rankings can change from one month to another. When this happens, the portfolio may require more trading.

For this reason, portfolio performance should be evaluated both before and after transaction costs. A model can look good before costs, but part of its performance can disappear after costs are included. This point is also discussed by Novy-Marx and Velikov (2016), who show that many return signals become much weaker once trading costs are considered. This is why net performance is important in portfolio evaluation.

This type of evaluation is also consistent with the out-of-sample portfolio literature. DeMiguel, Garlappi and Uppal (2009) show that optimized portfolios must be evaluated carefully out-of-sample, because a theoretically more sophisticated strategy does not automatically produce better realized performance. For this reason, the portfolio comparison should include not only return, but also risk, drawdown, turnover and transaction costs.

Annualized return and annualized volatility are also reported. These two measures help to understand where the final performance comes from. A portfolio can have a better Sharpe ratio because it has higher returns, lower volatility, or both. For this reason, these measures support the interpretation of the main results.

2.8 Responsible AI and the need for model-level evaluation

The previous sections focused on expected return estimation and portfolio performance. This is important because one objective of the thesis is to understand whether machine-learning-based expected returns can improve the final portfolio. However, portfolio performance is not the only aspect that matters when a machine learning model is used in finance.

A model can produce good portfolio results but still have some problems. It can be difficult to understand, too sensitive to small changes in the input variables, or unstable across time. It can

also work well on average but behave differently across groups of stocks. For this reason, it is useful to look not only at the portfolio, but also at the model itself.

This point is related to the responsible AI literature. Rudin (2019) argues that black-box models should be used carefully in high-stakes decisions. The reason is that, when a model is difficult to understand, it is also difficult to see where errors, bias or unstable relations come from. This is relevant also in finance, because the output of a model can affect investment decisions.

Explainability is one of the main aspects of this discussion. Arrieta et al. (2020) explain that explainability includes different elements, such as transparency, understanding the output of the model and being able to justify the decision. This is important because a model should not only produce a prediction, but should also be understandable enough to be evaluated.

Fairness is another important aspect. Barocas, Hardt and Narayanan (2023) show that a model can perform well on average, but still behave differently across groups. In the usual machine learning literature, these groups are often demographic groups. In the financial setting of this study, the same idea can be adapted to groups of assets. For example, a model can work better in some sectors and worse in others. This type of difference may not be visible if only the average performance is considered.

For this reason, portfolio-level evaluation and model-level evaluation answer two different questions. Portfolio metrics show whether the final strategy performs well. Responsible AI concepts help to understand whether the model is robust, accurate, fair and explainable. These two evaluations are connected, but they are not the same. A model can be strong from a portfolio point of view, but weaker from a responsible AI point of view. The need for this type of evaluation is also consistent with the broader discussion on trustworthy AI. The High-Level Expert Group on AI (2019) identifies robustness, transparency and fairness as important elements of trustworthy AI. In finance, Giudici and Raffinetti (2023) develop this idea through

the SAFE AI framework, which gives a way to evaluate financial AI models beyond predictive performance alone.

2.9 The SAFE AI framework in financial applications

The SAFE AI framework was proposed by Giudici and Raffinetti (2023) for the evaluation of artificial intelligence models in finance. The framework considers four aspects of a model: Sustainability or Robustness, Accuracy, Fairness and Explainability. The main idea is that a model should not be judged only by its predictive performance.

The first aspect is Sustainability/Robustness. In financial applications, this is important because financial variables can contain a lot of noise. A model should not change its predictions too much when the input data change only slightly. If this happens, the model may be too sensitive to noise and less reliable in practice. Giudici and Raffinetti (2023) interpret this dimension as part of the trustworthiness of financial AI models.

The second aspect is Accuracy. In return prediction, accuracy is not only about predicting the exact value of future returns. As discussed in the previous sections, it is often more useful to understand whether a model can rank stocks correctly. For this reason, accuracy can also be interpreted in a ranking sense. This is consistent with the return-prediction setting studied by Gu, Kelly and Xiu (2020), where the economic value of a model depends on how useful its predictions are for investment decisions.

The third aspect is Fairness. In standard machine learning applications, fairness is usually related to demographic groups, as discussed by Barocas et al. (2023). In financial applications, the idea can be adapted. Since the objects of the analysis are stocks, fairness can be interpreted as the ability of the model to behave consistently across groups of securities. Sectors are a natural example of these groups, because they represent different parts of the investment

universe. This point is important because a model can have a good average result, but still behave very differently inside the investment universe. For example, it can rank stocks well in some sectors and poorly in others. If only the aggregate result is observed, this difference can remain hidden.

The fourth aspect is Explainability. This aspect is related to the way the model uses the input information. A model that depends only on a very small number of variables can be easier to describe, but it can also become fragile if those variables stop being useful. For this reason, explainability is useful not only to understand the model, but also to check whether the model depends too much on a limited part of the information set. This is consistent with the broader explainable AI literature, where interpretability and transparency are considered important for the evaluation of machine learning systems (Arrieta et al., 2020; Rudin, 2019).

Overall, SAFE AI is useful for this thesis because it introduces a second level of evaluation. The portfolio evaluation shows what happens after the predictions are used in portfolio construction. SAFE AI instead focuses on the model itself. This distinction is important for the second research question, because the model that looks better under a responsible AI perspective is not necessarily the same model that produces the best portfolio performance.

Recent work by Babaei and Giudici (2025) also develops statistical metrics and an open-source package for SAFE AI evaluation. This is useful because it shows that SAFE AI is not only a general principle, but can also be translated into quantitative metrics.

2.10 Link between SAFE AI compliance and portfolio performance

The previous section presented SAFE AI as a way to evaluate machine learning models from a responsible AI point of view. This creates another important question: a model that looks good under SAFE AI is not necessarily the same model that produces the best portfolio.

Therefore, the best portfolio model and the best SAFE AI model do not necessarily have to coincide. A model can produce strong investment results while being less stable or less explainable; conversely, a model can be more compliant with SAFE AI but weaker in portfolio performance. This is not a contradiction, because the two evaluations measure different dimensions of model quality.

This point is important in finance because prediction quality is not always enough to explain portfolio results. As discussed by Gu, Kelly and Xiu (2020), a return prediction model should also be evaluated in terms of economic value. At the same time, the responsible AI literature shows that model quality also depends on aspects such as robustness, fairness and explainability (Rudin, 2019; Arrieta et al., 2020; Barocas, Hardt, & Narayanan, 2023).

Giudici and Raffinetti (2023) propose SAFE AI as a way to evaluate these aspects in financial applications. More recent work by Giudici and Kolesnikov (2025) develops this idea further by proposing an integrated approach to SAFE AI metrics. This is useful because it gives a way to summarize different responsible AI dimensions and compare models not only on one single aspect, but on a more general compliance profile.

For this reason, the relation between SAFE AI compliance and portfolio performance is not obvious. It must be studied directly. This is the logic behind the second research question of this thesis: whether the model that performs better under the SAFE AI framework is also the model that produces the best portfolio results.

2.11 Research gap and thesis contribution

The literature reviewed in this chapter shows that the main topics of this thesis have already been studied, but mostly separately.

The first topic is mean-variance optimization. The literature shows that this approach is very sensitive to estimation error, especially in the expected return vector μ (Michaud, 1989; Best & Grauer, 1991; Chopra & Ziemba, 1993). This is important because expected returns are one of the main inputs of the optimizer.

The second topic is expected return estimation. The Historical Mean is simple and easy to use, but it has important limits. It uses only past average returns and does not include other stock characteristics. At the same time, empirical asset pricing studies show that characteristics such as size, momentum, liquidity and volatility can contain useful information (Fama & French, 1992; Jegadeesh & Titman, 1993; Amihud, 2002; Gu, Kelly, & Xiu, 2020).

The third topic is portfolio evaluation. A model should not be evaluated only before the optimization step. It is also necessary to see what happens after the predictions are used to build a portfolio. For this reason, out-of-sample performance, risk, turnover and transaction costs are important (DeMiguel, Garlappi, & Uppal, 2009; Novy-Marx & Velikov, 2016).

The fourth topic is responsible AI. The literature shows that machine learning models should not be evaluated only with accuracy. Robustness, fairness and explainability are also important, especially when the model is used in financial decisions (Rudin, 2019; Arrieta et al., 2020; Barocas, Hardt, & Narayanan, 2023; Giudici & Raffinetti, 2023).

The gap is in the connection between these topics. There are many studies on portfolio optimization, many studies on machine learning return prediction, and many studies on responsible AI and SAFE AI metrics. However, fewer studies put these elements together in the same analysis.

This thesis is positioned in this gap. The first contribution is to compare the Historical Mean benchmark with machine-learning-based expected returns inside the same portfolio

optimization setting. This is useful because the comparison focuses on the expected return estimator, while the rest of the portfolio construction is kept consistent.

The second contribution is to evaluate the machine learning models also from a SAFE AI point of view. This makes it possible to study not only which model produces the best portfolio, but also which model is more robust, accurate, fair and explainable.

The final contribution is to connect these two levels. The thesis asks whether the model that performs better under SAFE AI is also the model that produces the best portfolio. This connection is the main element that remains less developed in the existing literature and motivates the empirical design of this thesis.

2.12 Summary of chapter

This chapter reviewed the literature that supports the two research questions of the thesis. The main point is that expected return estimation, portfolio performance and responsible AI should not be treated as separate issues. They are connected in this thesis because the same machine learning models are evaluated from two points of view: the portfolio they produce and their SAFE AI profile.

The chapter first explained why estimating μ is important in mean-variance optimization. It then discussed why the Historical Mean can be limited and why machine learning can be used as an alternative. After this, the chapter moved from return prediction to portfolio evaluation, and then to responsible AI and SAFE AI.

The next chapter uses this literature to build the empirical design of the thesis. It explains the data, the return prediction models, the portfolio optimizer, the performance metrics and the SAFE AI evaluation procedure.

3 *Data and Methodology*

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3.1 Introduction

This chapter introduces the full methodological pipeline used in the thesis, from the construction of the investment universe to the estimation of expected returns, portfolio optimization and model-level SAFE AI evaluation.

This follows a two-level evaluation. The first is the portfolio level, where portfolios generated from different expected return vectors are compared in terms of out-of-sample performance. The second is the model level, where the machine learning models are evaluated through the SAFE AI framework.

Four different return estimation approaches are compared: the Historical Mean Baseline, Ridge Regression, a Neural Network and XGBoost. The Historical Mean Baseline represents the classical Markowitz approach, while the three machine learning models are used in order to verify whether replacing the historical mean with ML-based expected returns improves portfolio performance. XGBoost is the main machine learning model of the thesis. Ridge Regression and the Neural Network are included as robustness checks.

This methodological design is directly connected to the two research questions of the thesis: whether XGBoost improves out-of-sample portfolio performance relative to the Historical

Mean Baseline, and whether the model with the strongest SAFE AI profile is also the one with the best portfolio-level performance.

The SAFE AI framework is applied only to Ridge Regression, the Neural Network and XGBoost. The Historical Mean Baseline is not included in the SAFE AI evaluation because it is not a trained machine learning model, but it remains the benchmark in the portfolio-level comparison.

The remainder of the chapter is organized as follows. Section 3.2 describes the dataset and the investment universe. Section 3.3 presents the preprocessing pipeline. Section 3.4 describes the return estimation models. Section 3.5 presents the portfolio optimizer. Section 3.6 describes the portfolio evaluation framework, including performance metrics, transaction costs and the lambda sensitivity analysis. Section 3.7 introduces the SAFE AI evaluation of the machine learning models. Section 3.8 presents the Integrated Compliance Score. Section 3.9 describes the SAFE-Performance Frontier. Section 3.10 concludes the chapter by summarizing the methodological design.

3.2 Dataset and investment universe

3.2.1 Dataset and its source

The investment universe used in this thesis is derived from the S&P 500 Index.

The data were obtained from the Bloomberg Terminal (Bloomberg Terminal, 2026) available at the University of Pavia.

The S&P 500 was selected for several reasons:

- 1. its composition, since the index is composed of large-cap U.S. stocks, which are usually characterized by high liquidity;*

2. *it represents a standard benchmark in the portfolio literature (Robertson, 2023) and it is widely used in empirical studies for asset pricing and portfolio optimization (Wu et al., 2017);*

3. *it is one of the indices for which long historical data are widely available;*

4. *its size is sufficient to build portfolios with the cardinality constraint used in this thesis.*

3.2.2 Investment universe

The dataset is initially composed of 426 stocks, reduced to 407 stocks after modeling filters. The filtering steps leading to this final universe are described in the next subsection. For each ticker the metadata include sector, following the GICS (Global Industry Classification Standard) (MSCI & S&P Dow Jones Indices, 2023), industry group, country and ESG scores. One ticker did not report its GICS sector, so the missing sector classification was manually added based on the corresponding company information. In addition, four different time-series datasets were imported, all with the same structure. Each dataset contains daily data for the 426 stocks over the period that goes from 01/01/2010 to 31/12/2025, and they are:

1. *Prices;*
2. *Total Return Index, including gross dividends;*
3. *Market Cap;*
4. *Volume.*

These datasets are used to compute returns, liquidity, size and the other stock-level features used in the analysis.

The full sample covers the period from January 2010 to December 2025 and contains 4,174 trading days. For the empirical analysis, this period is divided into two main windows:

1. *Warm-up window: from 01 January 2010 to 31 December 2015 (1,565 trading days) used to compute the 12 months momentum,*
2. *Model window: from 01 January 2016 to 31 December 2025 (2,609 trading days).*

The model window is also divided into a training window, which goes from 2016 to 2022 (1,826 trading days) and an out-of-sample test window, from 2023 to 2025 (783 trading days).

3.2.3 Data cleaning

Several filters were applied to the raw dataset to obtain a consistent and usable investment universe. The filters are ordered according to their function: first, filters addressing structural consistency across files; second, filters dealing with missing classifications; third, filters related to data coverage; and finally, a forward-fill procedure for small gaps in the time series.

Filter A: Cross-file ticker alignment. This filter checks that the same tickers are present across the metadata file and the time-series datasets.

Filter B: Metadata duplicate removal. This filter removes duplicate entries from the metadata file, so that each security is represented only once.

Filter C: Time-series duplicate removal. This filter removes duplicate observations from the time-series files, ensuring that each ticker-date observation appears only once.

Filter D: Missing sector or country. This filter checks whether each security has a GICS sector and country classification. Securities with missing information are not removed automatically, because a missing label alone is not considered sufficient reason to exclude them from the analysis.

Filter E: Insufficient coverage in the model window. Tickers with more than 5% missing trading days in either the Prices or Total Return Index series were removed. This filter may introduce a data-availability bias, because securities with insufficient data coverage are excluded from the final universe. The filter was not intended to remove stocks because they failed or were delisted, but to ensure that the machine learning models and covariance estimation could be applied to complete matrices. As a consequence, the final universe may be biased toward more stable and liquid stocks.

Filter F: Temporal coverage. This filter removes stocks whose first valid price or TRI observation occurs after 1 January 2015 or whose last valid observation occurs before 31 December 2024. The first condition ensures that enough historical data are available to compute the 12-month momentum variable before the beginning of the model window. The second condition requires the stock to have valid data after 31 December 2024, so that securities disappearing before the final year of the out-of-sample period are excluded.

Filter G: Forward-fill. This filter replaces missing values by using the last available value, but only when the gap is short, up to 5 consecutive trading days. It is not applied to volume. This is because volume is used to compute the Amihud illiquidity ratio, and filling missing volume values would create trading volume on days where it is not observed. After this step, tickers with remaining missing values in the selected window are removed.

3.3 Preprocessing pipeline

This section describes the preprocessing steps applied to the cleaned universe obtained after the operations described in Section 3.2. The objective of this phase is to transform the cleaned daily data into the return series, covariance inputs and monthly prediction panel used in the rest of the analysis.

3.3.1 Daily log-returns

Daily log-returns are computed based on the Total Return Index (TRI), rather than daily closing prices. The return for stock i on day t is defined as:

$$r_{i,t} = \ln \left(\frac{\text{TRI}_{i,t}}{\text{TRI}_{i,t-1}} \right) \quad (3.1)$$

where $r_{i,t}$ is the daily log-return of stock i on day t , and $\text{TRI}_{i,t}$ is the Total Return Index of stock i on day t .

The choice of using TRI rather than closing prices reflects the fact that the TRI index includes gross dividends and assumes that they are reinvested shortly after the ex-dividend date. This is important for the selected universe because several stocks distribute dividends during the period considered. The choice of using only closing prices would underestimate the real returns of dividend-paying stocks.

This decision is relevant because the daily log-returns series is applied both as the forecasting target and as an input for covariance estimation. Any distortion in return measure would therefore affect both the forecasting and the portfolio optimization stages.

However, prices are still used during feature engineering when required, particularly for the computation of the Amihud illiquidity ratio, following Amihud (2002).

The final daily log-return matrix includes 4,173 daily observations for 407 stocks. These returns are computed using the full period available in the dataset, from January 2010 to December 2025. In the empirical analysis, however, the actual model window starts in January 2016 and ends in December 2025, for a total of 2,609 trading days. The years from 2010 to 2015 are kept as a warm-up period. They are mainly used to compute lagged variables, such as the 252-day momentum measure, before the start of the model window.

3.3.2 Additional universe filters

Before proceeding to covariance estimation, three additional filters were applied. Differently from the structural filters described in Section 3.2.3, these filters address data-quality issues that become visible only after computing daily returns.

First, stocks with an excessive number of zero returns within the model window were removed. In some cases, Bloomberg TRI data remain unchanged for extended periods, generating sequences of zero returns. These patterns may reflect stale or non-updated observations rather than real market behavior. If these securities were kept, they could artificially reduce estimated variance and distort cross-correlations during both feature construction and covariance estimation. Similar concerns regarding stale prices, nonsynchronous trading, and their effect on return moments are discussed by Lo and MacKinlay (1990).

Therefore, a threshold of 5% was set. If more than 5% of a stock's daily returns within the model window were equal to zero, the stock was removed from the universe. This filter led to the exclusion of 17 stocks.

The second filter concerns only one stock, SATS UW EQUITY. This stock was removed because it showed an abnormal return of +53.21% on 26 August 2025, followed by another return of +14.42% on the next trading day. Since no documented corporate event was found to justify these movements, the observations were treated as likely TRI data-feed errors, and the stock was excluded from the study universe.

Lastly, the S&P 500 Index includes two different share classes of Alphabet Inc., GOOG and GOOGL. During the model window, the pairwise return correlation was 0.9952. To avoid including two almost identical return series in the covariance matrix, GOOGL was removed and only GOOG was retained. This choice is also consistent with the idea that the investment

universe should be carefully screened before applying return-prediction models and portfolio analysis, as discussed by Gu et al. (2020).

After these additional filters, the final investment universe contains 407 stocks.

3.3.3 Covariance matrix estimation

At this stage of the study, the investment universe includes 407 stocks. As a result, the sample covariance matrix contains 83,028 unique elements, including both variances and pairwise covariances. Although the model window contains 2,609 daily observations, estimating a covariance matrix of this size with the sample covariance estimator can still lead to estimation errors. This issue is particularly relevant in portfolio optimization, where estimation error in expected returns and covariances can strongly affect out-of-sample performance, as discussed by DeMiguel et al. (2009).

In order to address this problem, three different covariance estimators are first compared: the standard sample covariance matrix, the Ledoit-Wolf shrinkage estimator, and the Oracle Approximating Shrinkage (OAS) estimator. This comparison is used only as a diagnostic step. The sample covariance matrix and the OAS estimator are used mainly for diagnostic comparison; the Ledoit-Wolf estimator is selected as the main covariance input for the optimization stage.

Following Ledoit & Wolf (2004), the shrinkage estimator is defined as a convex linear combination between the sample covariance matrix and a structured target:

$$\hat{\Sigma}_{LW} = \hat{\delta}F + (1 - \hat{\delta})S \quad (3.2)$$

where S is the sample covariance matrix, F is the shrinkage target, and $\hat{\delta}$ is the estimated shrinkage intensity. In this thesis, the shrinkage target is proportional to the identity matrix:

$$F = \bar{\sigma}^2 I \quad (3.3)$$

where I is the identity matrix and $\bar{\sigma}^2$ is the average sample variance. Therefore, the estimator used in the empirical analysis can also be written as:

$$\hat{\Sigma}_{LW} = \delta \bar{\sigma}^2 I + (1 - \delta) S \quad (3.4)$$

The estimator proposed by Ledoit & Wolf (2004) shrinks the sample covariance matrix toward a structured target, improving conditioning and reducing estimation error in large-dimensional covariance settings.

The Ledoit-Wolf estimator is used during the optimization stage because it can be estimated repeatedly in the rolling out-of-sample framework in a stable way.

The OAS estimator follows a similar shrinkage logic and is also designed to improve covariance estimation in high-dimensional settings, but in this study it is used only for diagnostic comparison (Y. Chen et al., 2010).

In the out-of-sample optimization, the covariance matrix is re-estimated at each rebalancing date using the previous 252 daily return observations. The rebalancing date itself is excluded from the estimation window, so only information available before portfolio formation is used. This avoids look-ahead bias and keeps the portfolio construction fully out-of-sample.

3.3.4 Descriptive statistics

This section presents the main descriptive statistics of the final stock universe over the model window 2016–2025. The statistics are computed using the daily log-return series of the 407 stocks included after the filtering procedure.

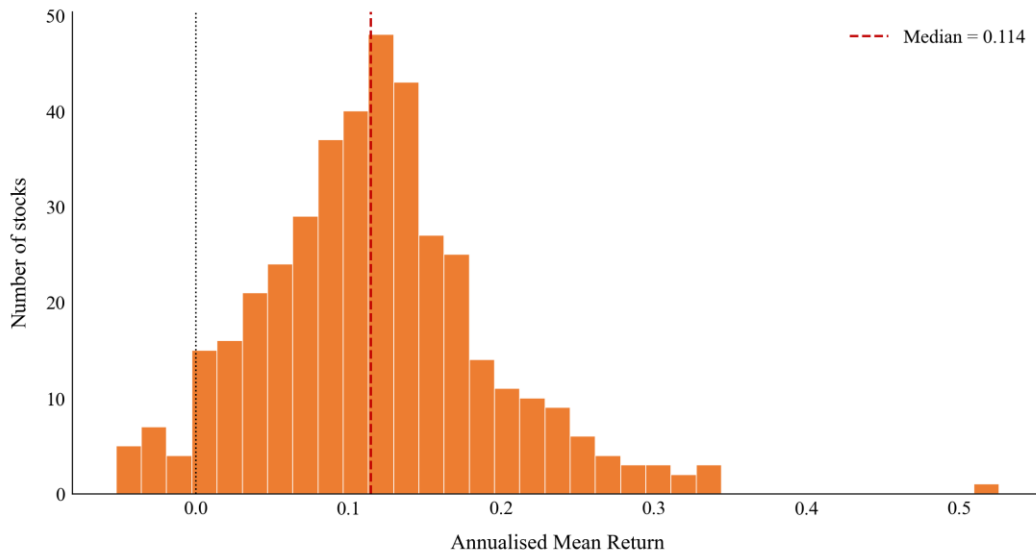


Figure 3.1: Distribution of annualized mean returns across stocks over the 2016–2025 model window.

Figure 3.1 reports the distribution of annualized mean returns for the stocks included in the final universe. The distribution is mostly concentrated on positive values. The median annualized return is 11.44%, while the cross-sectional average is 11.68%. These values show that, during the 2016–2025 model window, most stocks in the final universe recorded positive average performance. Annualized volatility is also computed as part of the descriptive analysis. It has a median value of 29.07% and an average value of 30.72%, showing that the final universe includes stocks with different risk levels.

The distributional statistics also show that daily returns are not normally distributed. The average skewness is -0.668 , meaning that daily returns are, on average, skewed to the left. The average excess kurtosis is 17.877 , suggesting the presence of fat tails and extreme observations. These results are consistent with the stylized facts of financial returns discussed by Cont (2001).

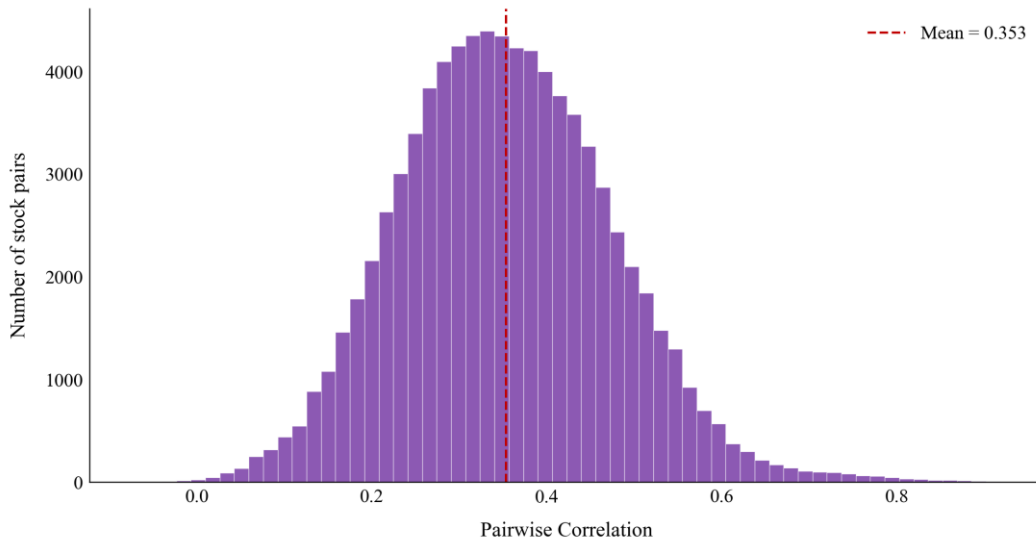


Figure 3.2: Distribution of pairwise stock correlations, estimated from daily log-returns over the model window 2016–2025.

Figure 3.2 shows the distribution of sample pairwise correlations across the stocks in the final universe. The average off-diagonal correlation is 0.3530, with a standard deviation of 0.1218. Negative correlations are rare, while 11.3% of pairs have a correlation above 0.50. This is relevant for portfolio construction because diversification depends not only on the number of stocks included, but also on how strongly their returns move together.

3.3.5 Feature engineering

Feature engineering transforms the daily time series into a structured monthly panel, which is used as the common input for the three machine learning models. The dataset is organized in a long format, in which each row represents one stock at one specific rebalancing date. For each stock-month observation, the predictors are based only on information available before the rebalancing date. The target is the return realized in the following holding period.

The monthly panel runs from January 2016 to December 2025. The rebalancing date is the first trading day of each month. With 407 stocks and 120 monthly rebalancing dates, the final panel contains 48,840 stock-month observations.

In addition, ten stock characteristics are constructed and grouped into five categories:

- Short term reversal: the 1-week cumulative log-returns are included to capture short-term return reversal effect.
- Momentum: the 1-month, 3-month, 6-month and 12-month cumulative log-returns are included to capture medium-term return continuation effects, following the momentum effect documented by Jegadeesh & Titman (1993).
- Annualized realized volatility: realized volatility is calculated over one-month and three-month horizons. In addition, a volatility ratio is included to capture whether recent risk is increasing or decreasing compared with the previous period. Realized volatility is commonly used as a measure of return variability in financial time series (Andersen et al., 2003).
- Illiquidity: the Amihud illiquidity ratio is computed over the previous 21 trading sessions. It is calculated as the average absolute return divided by traded value, measured as price multiplied by volume. This measure is used as a proxy for stock-level illiquidity, since it links price movements to trading volume (Amihud, 2002).
- Size: the natural logarithm of market capitalization is added as a control variable for firm size. This is included because size is one of the standard characteristics used in empirical asset pricing and has been related to differences in average stock returns (Fama & French, 1992).

Each original feature is transformed into a cross-sectional rank within the universe of stocks at each rebalancing date. The ranks are then normalized to the interval $[0,1]$. This step is useful

because the models use variables expressed in different units, such as returns, volatility, illiquidity and market capitalization. Ranking the variables at each date makes the inputs more comparable and also limits the effect of extreme observations. This type of cross-sectional transformation is commonly used in machine learning applications in empirical asset pricing, especially when the objective is to compare stocks within the same period (Gu et al., 2020).

To conclude, the prediction target is the cumulative log-return over the following holding period. It is computed using returns after the rebalancing date t and before the next rebalancing date t_{next} . The return on date t is not included because portfolio weights are set at the close of that day, so they can only be applied from the following day. This also keeps the target consistent with the portfolio-return calculation used later in the analysis.

3.4 Return estimation models

This section compares four models that differ in the way they estimate the expected return vector μ .

The same MIQP portfolio optimizer is applied to the expected return vector estimated by each model. This allows the portfolio construction step to remain fixed, so that the differences between the portfolios depend mainly on the return estimation method.

The four approaches are the Historical Mean Baseline, Ridge Regression, a Neural Network and XGBoost. The Historical Mean Baseline represents the traditional mean-variance approach, where expected returns are estimated from past average returns. The three machine learning models use the stock characteristics described in Section 3.3.5 to produce prediction scores for future stock performance.

For the three machine learning models, one additional step is needed. Ridge Regression, the Neural Network and XGBoost are not used to predict the exact future return of each stock

directly. Instead, they predict a score based on the stock's future position within the cross-sectional ranking. This choice is used because stock returns are very noisy, while the relative ranking of stocks is more useful for portfolio selection. However, the optimizer still needs a numerical expected-return vector. For this reason, the scores produced by the machine learning models are transformed into expected return estimates before they are used in the portfolio optimization step. At each rebalancing date, they are rescaled to have the same cross-sectional mean and standard deviation as the Historical Mean estimates. This makes the comparison fairer, because all models enter the same optimizer with expected return vectors on a comparable scale.

The main comparison of the thesis is between the Historical Mean Baseline and XGBoost. Ridge Regression and the Neural Network are included as robustness checks. Ridge Regression is used as a linear machine learning model, while the Neural Network is used as a non-linear model. This helps to check whether the results obtained with XGBoost are specific to this model or whether similar results are obtained with other machine learning approaches.

3.4.1 Historical Mean Baseline

The Baseline model represents the traditional mean-variance approach introduced by Markowitz (1952). In this model, the expected return vector μ is estimated using historical average returns. More specifically, for each stock and at each rebalancing date, μ is computed as the annualized mean return over the previous 252 trading days.

$$\hat{\mu}_i = \bar{r}_i^{(252)} \times 252 \quad (3.5)$$

where $\bar{r}_i^{(252)}$ is the average daily log-return of stock i over the 252 trading days immediately before the rebalancing date.

After computing the expected returns, cross-sectional winsorization is applied at the 1% and 99% levels. This step is useful because mean-variance optimization is sensitive to input estimates, especially expected returns. Small changes in the estimated means can lead to large changes in the optimal portfolio weights (Michaud, 1989; Best & Grauer, 1991).

The same 252-day window is also used for the estimation of the covariance matrix Σ . The covariance matrix is estimated through the Ledoit-Wolf shrinkage method, which is used to obtain a more stable covariance estimate than the simple sample covariance matrix (Ledoit & Wolf, 2004). Therefore, both μ and Σ are estimated from the same historical window.

The Baseline does not require model training or hyperparameter selection. For this reason, it is used as a simple and transparent reference model. Since the optimizer, the covariance matrix and the constraints are kept fixed across all approaches, the comparison with XGBoost and with the robustness-check models depends mainly on the method used to estimate expected returns.

3.4.2 Ridge Regression

Ridge regression is the first model used as a robustness check. It produces prediction scores using the ten cross-sectional features described in Section 3.3.5. Compared to the Baseline, it uses not only past average returns, but also the information contained in the selected features.

Ridge Regression is a linear model with regularization. This means that the model penalizes large coefficients, which is useful when working with noisy financial data and helps to reduce overfitting (Hoerl & Kennard, 1970).

At each of the 36 out-of-sample rebalancing dates, the model is trained with an expanding window. Therefore, only the data available before the rebalancing date are used. The last six months of the training window are used as a validation set.

The regularization parameter λ is selected through a grid search over $\{0.01, 0.1, 1.0, 10.0, 100.0\}$ using the validation R^2 . After this, the model is trained again on the full training sample,

including the validation period, and then used to produce prediction scores. The predicted scores are then transformed into expected returns using the rescaling procedure described in Section 3.4. After this step, cross-sectional winsorization at the 1% and 99% levels is applied. In the results, $\lambda = 100.0$ is selected in 31 out of 36 rebalancing dates, equal to 86% of the cases. This shows that the model usually needs strong regularization. In the other five dates, $\lambda = 0.01$ is selected. These cases are all between March and July 2023.

3.4.3 *Neural Network*

The Neural Network is the second model used as a robustness check. It uses the same ten cross-sectional features described in Section 3.3.5 and produces prediction scores that are then converted into expected return estimates through the same rescaling procedure.

The model applied is a Multilayer Perceptron, referred to as Neural Network in the running text and as MLP in tables. It has two hidden layers: the first one with 64 neurons and the second one with 32 neurons. The activation function used is ReLU. The Neural Network is included because it is a non-linear model. This means that, unlike Ridge Regression, it does not assume a linear relation between the features and future returns. This is useful because the relation between stock characteristics and returns could be more complex than a simple linear relation (Gu et al., 2020).

The training procedure follows the same expanding window logic used for Ridge Regression. At each of the 36 out-of-sample rebalancing dates from 2023 to 2025, the model is trained only with the data available before that date. This avoids using future information in the training process.

Early stopping is applied using the internal procedure of scikit-learn. This procedure takes 10% of the training data and uses it as validation data. No hyperparameter search is performed for this model, because the architecture is fixed. This choice was made in order to keep the Neural

Network simple and comparable across all the rebalancing dates. The predicted scores are then converted into expected returns using the rescaling procedure described in Section 3.4. The number of iterations before early stopping ranges from 16 to 33 across the 36 rebalancing dates. This shows that the model usually stops quite early, which helps to reduce overfitting.

3.4.4 XGBoost

XGBoost is the main machine learning model of this study. It uses the same ten cross-sectional features described in Section 3.3.5 and it produces prediction scores that are converted into expected return estimates before entering the portfolio optimizer.

XGBoost is based on gradient-boosted decision trees (T. Chen & Guestrin, 2016). It is selected as the main model because it can capture non-linear relations between stock characteristics and future returns. This is important because the relation between stock characteristics and future returns is not assumed to be linear.

For example, the effect of momentum, volatility, illiquidity and size may depend on the level of the other variables. This type of relation is difficult to capture with a linear model. For this reason, XGBoost is used as the main alternative to the Historical Mean Baseline. Previous studies also show that machine learning models can be useful in return prediction when many stock-level characteristics are used (Gu et al., 2020).

The training procedure follows the same expanding-window logic used for Ridge Regression and the Neural Network. At each of the 36 out-of-sample rebalancing dates from 2023 to 2025, the model is trained only with data available before that date. This avoids using future information in the training process.

The last six months of the training window are used as a validation set. This one is used both to choose the best hyperparameters and to apply early stopping.

The grid search is done over three hyperparameters:

1. number of estimators: {100, 300, 500};
2. maximum tree depth: {3, 4, 6};
3. learning rate: {0.01, 0.05}.

The model also uses subsampling and column subsampling, both set equal to 0.8. Early stopping is applied after 20 boosting rounds without improvement on the validation set. After choosing the best parameters, the model is trained again using both the training and validation data. The final model is then used to produce prediction scores for the rebalancing date.

As for the other machine learning models, XGBoost produces predicted scores for the future return ranking. These scores are then converted into expected returns using the rescaling procedure described in Section 3.4. The trained XGBoost models are also used later for the SAFE AI evaluation described in Section 3.7.

The effective number of trees after early stopping ranges from 1 to 69 across the 36 rebalancing dates. In many cases, the model remains small and strongly regularized. This is consistent with the difficulty of stock return prediction, where the useful signal is usually weak compared to the noise (Gu et al., 2020).

3.5 The portfolio optimizer

All four models described previously use the same portfolio optimizer. The covariance matrix, the optimization problem, the constraints, the solver, and the rebalancing calendar are kept the same for all models. The only element that changes is the expected return vector μ .

This makes the comparison between models clearer. Since all the other parts of the portfolio construction are fixed, the differences in the final portfolios are mainly due to the different way in which μ is estimated.

3.5.1 MIQP model

At each rebalancing date, a new portfolio is created by solving a Mixed Integer Quadratic Program (MIQP).

The model follows the mean-variance logic introduced by Markowitz (1952). The objective is to balance portfolio risk and expected return. In this thesis, the optimizer minimizes portfolio variance and, at the same time, rewards higher expected return.

Since the portfolio must contain a fixed number of stocks, binary variables are needed. For this reason, the problem is solved as a MIQP. This type of formulation is commonly used in portfolio optimization problems with cardinality constraints (Bertsimas & Shioda, 2009).

The objective function is:

$$\min_{w,z} w^T \Sigma w - \lambda \mu^T w \quad (3.6)$$

where w is the portfolio weight vector, Σ is the covariance matrix, μ is the expected return vector, z is the binary selection vector, and λ is the risk-return trade-off parameter. During the study, λ is fixed equal to 1.0 for all models. This choice is made to keep the optimizer setting the same for all portfolios. In this way, the comparison between models does not depend on a different value of λ .

The optimizer is subject to the following constraints:

1. *Full investment constraint:*

$$\sum_{i=1}^N w_i = 1 \quad (3.7)$$

This means that all the capital is invested in the portfolio.

2. *Cardinality constraint:*

$$\sum_{i=1}^N z_i = K = 10 \quad (3.8)$$

where z_i is a binary variable equal to 1 if stock i is selected, and 0 otherwise.

In this thesis, K is fixed equal to 10. This means that each portfolio contains exactly 10 stocks at every rebalancing date. This choice was made to obtain a concentrated portfolio, consistent with an active investment strategy. The same cardinality level is also considered in previous studies on cardinality-constrained mean-variance portfolios, for example in Dehghan Hardoroudi et al. (2017).

3. *Weight bounds:*

$$0.01z_i \leq w_i \leq 0.20z_i \quad (3.9)$$

This means that each selected stock must have a weight between 1% and 20%. If a stock is not selected, its weight is equal to zero.

The lower bound avoids very small positions, while the upper bound avoids excessive concentration in one single stock. These types of lower and upper bounds are common in practical portfolio optimization problems with integer constraints (Mencarelli & d'Ambrosio, 2019).

4. *Sector concentration constraint:*

$$\sum_{i \in S} w_i \leq 0.30 \quad (3.10)$$

This means that the total weight invested in each sector cannot be higher than 30%. This constraint is used to limit sector concentration, but it does not force the portfolio to be sector neutral.

The reason for this constraint is to avoid that the optimizer puts too much weight in one single sector. Similar sector or group constraints are used in constrained portfolio optimization to control the exposure to specific parts of the market (Soleimani et al., 2009).

3.5.2 Estimating the covariance matrix

The covariance matrix Σ is estimated at each rebalancing date using the Ledoit-Wolf shrinkage method (Ledoit & Wolf, 2004). The estimation is done using daily log-returns over the previous 252 trading days.

The reason for using Ledoit-Wolf is that the sample covariance matrix can be unstable, especially when many stocks are included. The shrinkage method helps to obtain a more stable estimate of the covariance matrix.

Only information available before each rebalancing date is used. This is done in order to avoid using future information in the portfolio construction. The same covariance matrix is used for all four models at each rebalancing date. Therefore, the only element that changes across portfolios remains the expected return vector μ .

3.5.3 Solver

The optimization problems are solved using Gurobi 13 (Gurobi Optimization, 2026). Gurobi is used because the problem is a MIQP and therefore requires a solver able to manage integer variables.

In this thesis, the solver is configured with:

1. a time limit of 60 seconds;
2. a MIP optimality gap of 1%.

The same solver and the same settings are used for all models to ensure that the differences between portfolios are not due to the optimization procedure, but mainly to the different

expected return estimates. Across the 36 out-of-sample rebalancing dates from 2023 to 2025, all problems reached OPTIMAL status. The average solving time was 0.3 seconds, and no problem reached the time limit.

3.5.4 Rebalancing and portfolio returns

The portfolios are rebalanced on the first trading day of each calendar month. In the out-of-sample test window, from January 2023 to December 2025, this gives 36 rebalancing dates.

At each rebalancing date, the portfolio weights are set equal to the optimized target weights. Between two rebalancing dates, the portfolio is not traded. The weights change only because of the realized returns of the stocks. This follows a buy-and-hold logic.

The daily portfolio log-return is computed as:

$$r_t^p = \ln \left(\sum_{i=1}^N w_{i,t-1}^{\text{drift}} \cdot e^{r_{i,t}} \right) \quad (3.11)$$

where $w_{i,t-1}^{\text{drift}}$ is the weight of stock i after the drift from the previous rebalancing date, and $r_{i,t}$ is the daily log-return of stock i on day t .

The return on the rebalancing date itself is excluded because the new weights are set at the close of that day. Therefore, the new weights can be invested only from the following trading day.

The same rule is applied to all four portfolios. This makes the comparison consistent across models, following the out-of-sample portfolio evaluation logic used by DeMiguel et al. (2009).

3.5.5 Turnover and transaction costs

Monthly turnover is computed at each rebalancing date. It measures how much the portfolio must be traded in order to move from the previous drifted weights to the new target weights.

Monthly turnover is defined as:

$$TO_t = \frac{1}{2} \sum_{i=1}^N |w_{i,t}^{\text{target}} - w_{i,t}^{\text{pre}}| \quad (3.12)$$

where $w_{i,t}^{\text{target}}$ is the new optimized weight at rebalancing date t , and $w_{i,t}^{\text{pre-trade}}$ is the drifted weight immediately before trading. The first rebalancing date is excluded from the turnover calculation because there is no previous portfolio.

Transaction costs are implemented as proportional costs on traded turnover. They are deducted from portfolio returns on the first trading day after each rebalancing date, starting from the second rebalance. Three cost levels are considered: 10, 20 and 30 basis points. This implementation rule is kept identical for all four portfolios, so that differences in net performance depend on the return estimation method and on the trading activity generated by each strategy.

3.6 Portfolio evaluation framework

This section explains how the portfolios are evaluated after the optimization step. The objective of this part is to answer the *first research question* of the thesis: whether replacing the Historical Mean expected returns of the classical Markowitz framework with machine-learning-based expected returns improves out-of-sample portfolio performance.

The evaluation is done on the out-of-sample test window, from January 2023 to December 2025. Since the portfolios are rebalanced on the first trading day of each calendar month, this gives 36 rebalancing dates.

The four portfolios compared are the Historical Mean Baseline, Ridge Regression, the Neural Network and XGBoost. As explained in the previous sections, all these portfolios are constructed using the same MIQP optimizer. The covariance matrix, the constraints, the solver settings and the rebalancing calendar are also kept the same for all models. For this reason, the

main element that changes across portfolios is the expected return vector μ . This is important because it allows the comparison to focus on the different return estimation methods.

The main comparison is done with $\lambda = 1.0$. This is the same value used in the optimizer described in Section 3.5. In addition, the analysis also includes lambda sensitivity analysis. This is done in order to check whether the portfolio results depend strongly on the chosen value of λ , or whether the relative behaviour of the models remains similar when the risk-return trade-off parameter changes.

3.6.1 *Performance metrics*

The portfolio performance is evaluated using three main metrics: Sharpe ratio, maximum drawdown and average turnover.

The **Sharpe ratio** is used because it gives a first measure of the relation between return and risk. It is one of the most common measures in portfolio evaluation, starting from Sharpe (1966). It is computed as:

$$\text{Sharpe} = \frac{R_p - r_f}{\sigma_p} \quad (3.13)$$

where R_p is the annualized portfolio return, r_f is the risk-free rate, and σ_p is the annualized volatility of the portfolio.

In this thesis, the risk-free rate is set equal to zero. This is done because the aim is not to measure returns relative to a specific risk-free asset, but to compare the four portfolio strategies under the same assumption. Since the same rule is applied to all models, the comparison between them remains consistent. This approach is also in line with the logic used in out-of-sample portfolio comparison studies, such as DeMiguel et al. (2009).

The second metric is **maximum drawdown**. This measure is used to understand the largest loss that the portfolio experiences from a previous peak. It is calculated using the portfolio wealth index:

$$\text{MaxDD} = \max_t \left(\frac{\text{Peak}_t - W_t}{\text{Peak}_t} \right) \quad (3.14)$$

where W_t is the value of the portfolio at time t , and Peak_t is the highest value reached by the portfolio up to that date. A lower maximum drawdown means that the portfolio had a smaller worst loss during the test period. This measure is useful because volatility alone does not show how large the worst decline of the portfolio was.

The third metric is **average turnover**. Turnover is computed at each rebalancing date, as described in Section 3.5.5. It measures how much the portfolio has to be traded in order to move from the drifted weights before rebalancing to the new optimized weights. The first rebalancing date is excluded, because there is no previous portfolio; for this reason, average turnover is computed over 35 rebalancing dates.

Turnover is important because it gives information on how stable the portfolio is over time. A portfolio can have good returns before costs, but if it changes too much every month, part of this performance can disappear once transaction costs are included. This point is particularly important for the machine learning portfolios, because their predicted rankings can change from one rebalancing date to another.

In addition to these three main metrics, other measures are also reported, such as the Sortino ratio and Calmar ratio (Sortino & Price, 1994; Young, 1991).

Finally, annualized return and annualized volatility are shown separately, in order to understand whether the final performance comes mainly from higher returns, lower risk, or both.

3.6.2 Transaction costs and net performance

Transaction costs are added after computing gross portfolio returns. This is done in order to check whether the performance of each strategy remains good also after considering the trading needed to rebalance the portfolio.

The cost calculation is based on the turnover measure defined in Section 3.5.5. The same three levels are used: 10 basis points, 20 basis points and 30 basis points per unit of traded turnover. After deducting these costs, the performance metrics are computed again using net returns. In this way, it is possible to compare the four portfolios both before and after transaction costs. This step is important because a portfolio can perform well before costs, but part of this performance can disappear when costs are included. This is especially relevant for the machine learning portfolios, because their predicted rankings can change from one month to another and this can increase turnover. For this reason, a model with good gross performance is not necessarily the best one after transaction costs.

The inclusion of transaction costs is consistent with portfolio studies that evaluate out-of-sample performance after trading costs, such as DeMiguel et al. (2009) and Novy-Marx & Velikov (2016).

3.6.3 Portfolio comparison criteria

The portfolio comparison is not reduced to a single overall rank score. In this thesis, these measures are interpreted separately.

The Sharpe ratio is used as the main metric for comparing portfolio performance. A higher Sharpe ratio means that the portfolio obtains a better return per unit of total risk. For this reason, the model with the highest Sharpe ratio is considered the strongest one in terms of risk-adjusted performance.

Maximum drawdown is then used to check the downside behavior of the portfolios. This is important because a portfolio can have a good Sharpe ratio but still experience a large loss during the test period. A lower maximum drawdown is therefore preferred.

Average turnover is used to interpret how much trading is required by each strategy. A lower turnover means that the portfolio is more stable across rebalancing dates. This is not a performance measure in the same sense as the Sharpe ratio, but it is useful because it affects the practical implementation of the strategy and the impact of transaction costs.

The other measures, such as Sortino ratio, Calmar ratio, annualized return and annualized volatility, are used to support the interpretation. They help to understand whether the results are driven by higher returns, lower volatility, lower downside risk, or lower drawdowns.

Later in the thesis, the portfolio results are compared with the SAFE AI evaluation. This comparison is useful because it helps to understand whether the model that performs better at portfolio level is also the model that behaves better according to the SAFE AI framework.

3.6.4 Lambda sensitivity analysis

The sensitivity analysis is done for all four portfolios: the Historical Mean Baseline, Ridge Regression, the Neural Network and XGBoost. Ridge Regression and the Neural Network are included as robustness checks. The main comparison remains between the Historical Mean Baseline and XGBoost. In the main analysis, the portfolios are built using $\lambda = 1.0$. This value is used for all four models, so the comparison is done under the same optimizer setting. However, λ is still a parameter chosen in the model. For this reason, it is useful to check what happens when this value changes. λ controls the balance between risk and expected return in the MIQP objective function. When λ is low, the optimizer gives more importance to the variance term. In this case, the portfolio is more focused on reducing risk. When λ is high, the

optimizer gives more importance to the expected return term, and the portfolio depends more on the expected returns estimated by the model.

For this reason, the same optimization procedure is repeated using the following values:

$$\lambda \in \{0.1, 0.2, 0.5, 1.0, 2.0, 5.0, 10.0\}$$

The sensitivity analysis is done for all four portfolios: the Historical Mean Baseline, Ridge Regression, the Neural Network and XGBoost. For each value of λ , the optimizer is solved again. The universe, covariance matrix, constraints, solver settings and rebalancing dates are kept the same as in the main analysis. The only element that changes is the value of λ .

This step is useful because it shows whether the results depend too much on the value chosen for λ . If the behavior of the models is similar across the different values, then the main result with $\lambda = 1.0$ is more stable. If instead the results change a lot, then the portfolio comparison must be interpreted with more caution.

The comparison is made using the same main metrics introduced in Section 3.6: Sharpe ratio, maximum drawdown and average turnover. The Sharpe ratio is used to see how risk-adjusted performance changes. Maximum drawdown is used to see whether the worst loss becomes larger or smaller. Average turnover is used to understand whether the portfolio becomes more unstable when more weight is given to expected returns.

This analysis is particularly important because the expected return vector μ is the only element that changes across the four models. If higher values of λ make one model perform much better or much worse, this means that the optimizer is reacting strongly to that model's expected return estimates. This is relevant for the thesis because the main objective is to understand whether replacing the Historical Mean expected returns with machine-learning expected returns improves portfolio performance.

The value $\lambda = 1.0$ remains the reference case. The other values are used only to check the sensitivity of the results. After defining the portfolio evaluation framework, the analysis moves to the second level of evaluation: the model-level assessment of the machine learning methods through SAFE AI.

3.7 SAFE AI evaluation of return prediction models

This section moves from the portfolio-level evaluation to the model-level evaluation of the machine learning models. While the previous sections explain how portfolios are built and evaluated, this section focuses on the quality of the models that generate the expected return estimates. The evaluation of machine learning models should not be based only on predictive accuracy. This is important especially in financial applications, where a model can have good predictive performance but still be unstable, difficult to interpret or different in its behavior across groups of stocks. For this reason, this thesis applies the SAFE AI framework.

The SAFE AI framework, proposed by Giudici and Raffinetti (2023), evaluates AI models through four dimensions: Sustainability/Robustness, Accuracy, Fairness and Explainability. In this thesis, the first dimension is interpreted as robustness of the predicted ranking, following the RGR metric used in the *safeaipackage*.

In this thesis, SAFE AI is adapted to a cross-sectional stock return prediction problem. The models do not predict outcomes for individuals, but future returns for stocks. For this reason, some parts of the framework need to be adapted. In particular, Fairness is not interpreted in terms of protected social groups. It is interpreted as similarity of predictive accuracy across groups of stocks.

SAFE AI is applied only to the machine learning models described in Section 3.4: Ridge Regression, Neural Network and XGBoost. The Historical Mean Baseline is not included

because it is not a trained machine learning model and does not use the same input features. It is instead used only as a benchmark in the portfolio performance analysis.

The SAFE AI metrics used in this thesis are rank based. This is useful because the exact value of future stock returns is difficult to estimate, while the ranking of stocks is central in the portfolio construction process. A model can be useful if it ranks future winners above future losers, even if the predicted return values are not perfectly accurate.

The following subsections explain how the four SAFE AI dimensions are adapted and computed in this thesis.

3.7.1 Adaptation of the SAFE AI framework

In this study, the SAFE AI framework is applied to models that produce cross-sectional predictions of future stock returns. This is different from standard applications where the predictions concern individuals. For this reason, some adaptations are needed.

In this thesis, the S dimension of SAFE AI is evaluated in terms of robustness of the predicted ranking. This choice is consistent with the RGR metric implemented in the *safeai* package, which measures the robustness of model predictions to perturbations in the input variables (Babaei & Giudici, 2025). Therefore, Sustainability is interpreted in this application as robustness to input perturbations, and not as ESG sustainability.

The Accuracy dimension is measured through Rank Graduation Accuracy, or RGA. In this thesis, RGA is used to evaluate whether the predicted stock ranking is close to the realized future return ranking. This choice is consistent with the rank-based nature of RGA, which compares the ranks of the actual values with the ranks of the predicted values (Giudici & Raffinetti, 2025).

For the Fairness dimension, predictive accuracy is evaluated across GICS sectors, which are used as financial groups. In a stock return prediction problem, there are no protected social

groups. Therefore, sector membership is used as the closest financial grouping variable. This choice makes it possible to check whether the models have a similar predictive accuracy across different parts of the market.

The Explainability dimension is measured through Rank Graduation Explainability, or RGE. In the package, RGE is based on the comparison between the original predictions and the predictions obtained after neutralizing one variable or one group of variables (Babaei et al., 2025). In this thesis, RGE is used to understand which groups of input variables contribute more to the predicted stock ranking.

The ten input features are divided into five economic groups: Momentum, Short-term Reversal, Volatility, Liquidity and Size. These groups are used for both RGR and RGE. This choice is made because the objective is not only to evaluate single variables, but also broader types of financial information.

These adaptations are specific to this thesis. In this thesis, SAFE AI is not used as part of the portfolio optimization process. It is used only to evaluate the machine learning models that produce the expected return vectors. The portfolio-level evaluation is carried out separately in Section 3.6.

3.7.2 Monthly prediction panel used for SAFE AI evaluation

The SAFE AI evaluation is based on a monthly prediction panel.

Each row of the panel represents one stock at one specific rebalancing date.

For each observation, the panel contains:

1. the ticker.
2. the rebalancing date.
3. the GICS sector.
4. the realized future return.

5. the expected return from Ridge Regression.
6. the expected return from the Neural Network.
7. the expected return from XGBoost.
8. the ten features described in Section 3.3.5.

The realized future return is observed after each rebalancing date and is used as the target during the training and evaluation of the machine learning models. The expected return estimates from Ridge Regression, the Neural Network and XGBoost are instead produced before portfolio formation. These estimates are already rescaled according to the procedure described previously, so that they are on a comparable scale and can be used within the same portfolio optimization framework.

The GICS sector is included only for the Fairness analysis. It is not used as an input feature of the prediction models. The ten input features are instead used for the Sustainability/Robustness and Explainability analyses, where the effect of perturbing or neutralizing groups of variables is evaluated.

All SAFE AI metrics are computed only on out-of-sample predictions, using the same test period considered in the portfolio analysis.

3.7.3 Sustainability/Robustness: rank graduation robustness

The Sustainability/Robustness dimension is measured using Rank Graduation Robustness, or RGR. It is one of the Rank Graduation metrics implemented in the *safeaipackage* (Babaei & Giudici, 2025). In this thesis, Sustainability does not refer to ESG sustainability, instead it refers to the robustness of the ranking produced by the models. The idea is that a model is more robust if its stock ranking does not change too much when the input features are perturbed. This is important because financial variables can contain noise. For this reason, a model whose predictions change too much after perturbations may be less reliable. RGR compares the

original ranking of predicted returns with the ranking obtained after perturbing a group of input variables. Each input feature is perturbed separately, so that it is possible to evaluate how sensitive the predicted ranking is to changes in each variable. The perturbation follows the procedure implemented in the *safeai* package.

As described in Section 3.3.5, the ten input features are:

Feature group	Variables
Momentum	1-month, 3-month, 6-month and 12-month cumulative returns
Short-term Reversal	1-week cumulative return
Volatility	1-month volatility, 3-month volatility and volatility ratio
Liquidity	Amihud illiquidity
Size	Log market capitalization

Table 3.1: Economic classification of the input features used in the SAFE AI analysis.

In addition, RGR is also computed after perturbing all input features at the same time. This helps to understand whether the model ranking remains stable when the whole set of information used by the model is affected by noise.

In this thesis, RGR is computed for each model, each valid out-of-sample rebalancing date and each input feature. It is also computed after perturbing all features simultaneously. The value obtained by perturbing all features simultaneously is used as the final RGR value in the scalar SAFE AI comparison.

3.7.4 Accuracy: rank graduation accuracy

The Accuracy dimension is measured using Rank Graduation Accuracy, or RGA. This metric is used to evaluate how well each model reproduces the ranking of realized future returns (Babaei & Giudici, 2025).

In this thesis, RGA compares the ranking of the realized future returns with the ranking of the predicted returns. A higher RGA means that the model is better at ordering stocks in a way that is close to the realized return ranking. This point is important because, in this thesis, the models are not used only to predict the exact value of future returns. They are also used to rank stocks before the portfolio optimization.

RGA is computed cross-sectionally at each of the 36 out-of-sample rebalancing dates. For each date, the realized future returns are compared with the predicted returns produced by each model.

Finally, the Accuracy value for each model is obtained by averaging RGA across the valid out-of-sample rebalancing dates.

3.7.5 Fairness: Sectoral RGA parity gap

The Fairness dimension is adapted to the financial context of this thesis.

In many AI applications, Fairness is evaluated across protected groups, such as demographic groups. Here, the prediction problem is different because the models predict future returns of stocks and not outcomes for individuals. Therefore, there are no protected social groups. In this thesis, GICS sectors are used as financial groups. This choice is made because sectors represent different parts of the market and are available for all stocks in the final universe. The objective is to check whether the models have similar predictive accuracy across sectors.

The metric is based on Rank Graduation Accuracy. Following the RGA-based parity logic used in the *safeaipackage*, RGA is computed separately within each group and the differences across groups are used to evaluate Fairness (Babaei & Giudici, 2025).

In this analysis, the groups are represented by the GICS sectors.

For each rebalancing date and each model, RGA is computed separately within each GICS sector. The sectoral fairness gap is defined as:

$$FG_{m,t}^{\text{sector}} = \max_{s \in S} RGA_{m,t,s} - \min_{s \in S} RGA_{m,t,s} \quad (3.15)$$

where S is the set of GICS sectors and $RGA_{m,t,s}$ is the RGA of model m at rebalancing date t , computed only on stocks belonging to sector s . A lower value means that predictive accuracy is more similar across sectors.

The GICS sector is not used as an input feature of the prediction models. It is used only to split the stocks into groups for the Fairness analysis.

For each model, the final Fairness value is obtained by averaging the sectoral RGA gap across the out-of-sample rebalancing dates. Since this measure is a gap, lower values indicate higher Fairness.

3.7.6 Explainability: rank graduation explainability

The Explainability dimension is measured using Rank Graduation Explainability, or RGE, and it is implemented in the *safeaipackage* (Babaei & Giudici, 2025).

RGE evaluates how much the predicted ranking changes when the information contained in an input feature is neutralized. In the scalar SAFE AI evaluation, this step is computed through the external SAFE AI implementation adopted in this thesis. For this reason, RGE is interpreted as a rank-based measure of the sensitivity of the model ranking to each feature.

The original predicted ranking is then compared with the ranking obtained after feature neutralization. If the ranking changes a lot, the feature has a stronger role in the model prediction. For this reason, higher RGE values indicate that the variable contributes more to the predicted stock ranking.

RGE is computed for each machine learning model, for each valid out-of-sample rebalancing date, and for each input feature separately. This makes it possible to understand which variables are more important for the ranking produced by Ridge Regression, the Neural Network and XGBoost. In addition, RGE is also computed after neutralizing all input features at the same time. This gives a general indication of how much the predicted ranking depends on the full set of information used by the model.

The final Explainability value for each model is obtained by averaging the RGE values across input features and valid rebalancing dates.

As with the other SAFE AI metrics, Explainability is interpreted in a rank-based sense. It does not measure causal importance. It measures how much the predicted stock ranking changes when the information contained in one or more input features is neutralized. The four SAFE AI dimensions provide separate information about model quality. The next section explains how these dimensions are transformed into compliance vectors and combined into an Integrated Compliance Score.

3.8 Integrated Compliance Score

3.8.1 From single SAFE AI values to compliance vectors

The SAFE AI evaluation presented in Section 3.7 gives one final value for each model and for each dimension. These final values are obtained by averaging the corresponding metric across

the 36 out-of-sample rebalancing dates. This first step is useful because it gives a simple comparison between Ridge Regression, the Neural Network and XGBoost.

This section explains how the SAFE AI values are combined into one compliance score. The construction follows the Compliance Score approach of Giudici & Kolesnikov (2025), which is used as reference. In their approach, the score is based on three dimensions: Accuracy, Robustness and Explainability. In this thesis, the same logic is used, but a fourth dimension is added in order to include Fairness.

For this reason, two scores are considered. The first one is CS3, which includes Accuracy, Robustness and Explainability. The second one is CS4, which also includes Fairness through the RGF* vector.

Their approach is based on rank-based metrics derived from the Lorenz and concordance curve framework. This is consistent with the objective of this thesis, because the machine learning models are mainly used to rank stocks rather than to predict the exact numerical value of future returns.

The scalar SAFE AI metrics are transformed into compliance vectors. Each vector contains eleven values. These values describe the behavior of one SAFE AI dimension across a sequence of steps. The first value refers to the initial situation, where no additional stress is applied. The following values are obtained by gradually increasing the level of removal, perturbation or feature neutralization. All the vectors are computed using only the out-of-sample predictions from the 36 monthly rebalancing dates in the test period. Therefore, the calculation is based only on information available before or at the rebalancing date, without using future information.

3.8.2 *Compliance vectors for accuracy, robustness and explainability*

Each compliance vector has eleven elements. For each model and for each rebalancing date, a date-level vector is first computed. Then, each element of the vector is averaged across the 36 out-of-sample rebalancing dates. In this way, each model obtains one final vector for Accuracy, one for Robustness and one for Explainability.

The RGA compliance vector is used to see how the ranking accuracy changes when part of the stock universe is removed. At each rebalancing date, the stocks are ordered from the highest to the lowest predicted return. After this, the ranked cross-section is divided into ten approximately equal segments.

When $k = 0$, no segment is removed and RGA is computed using the full stock universe available at that date. At each following step, one more segment is removed from the top of the predicted ranking, and RGA is computed again using only the remaining stocks. When $k = 10$, all segments have been removed and RGA is set equal to zero by definition.

This vector shows how much ranking accuracy remains after progressively removing the stocks that the model ranked highest. This is important for this thesis because the highest-ranked stocks are the most relevant for portfolio construction, since they are the stocks that the model identifies as the most attractive before the optimization step.

The RGR compliance vector is used to evaluate how stable the predicted ranking is when the input variables are increasingly perturbed. At each rebalancing date, group RGR is computed for eleven perturbation levels:

$$p \in \{0\%, 5\%, 10\%, 15\%, 20\%, 25\%, 30\%, 35\%, 40\%, 45\%, 50\%\}.$$

When $p = 0$, no perturbation is applied and RGR is equal to 1 by definition. For positive values of p , all ten input features are perturbed at the same time. The perturbation follows the percentile-swapping procedure implemented in the *safeai* package (Babaei et al., 2025). This

means that values from the lower and upper tails of the feature distribution are exchanged according to the selected perturbation level. The resulting vector shows how the model ranking changes when the input data becomes more disturbed. If RGR decreases slowly, the model is considered more robust. This means that small changes in the explanatory variables do not strongly change the ranking predicted by the model.

The RGE* compliance vector is used to check how much of the original predicted ranking remains when the input features are progressively neutralized. In this thesis, RGE and RGE* are used for two different purposes. RGE measures the contribution of each feature to the predicted ranking. RGE*, instead, measures how stable the original ranking remains after the features are neutralized step by step.

At each rebalancing date, the ten features are ordered from the least explainable to the most explainable according to their RGE value. When $k = 0$, no feature is neutralized and RGE* is equal to 1 by definition. At each following step, the k least explainable features are replaced with their training-time mean values. The model is then applied again to the modified test data, producing a new set of predictions.

RGE* is computed by comparing the ranking of the original predictions with the ranking of the predictions obtained after feature neutralization:

$$\text{RGE}^* = \text{RGA}(\hat{y}_{\text{original}}, \hat{y}_{\text{modified}}) \quad (3.16)$$

The scalar RGE described in Section 3.7.6 measures how much each feature contributes to the predicted ranking. The RGE* vector is used for a different purpose. It shows how much of the original ranking is preserved when features are removed step by step. Since the role of each feature can change over time, the order of neutralization is computed separately for each rebalancing date.

All three vectors are positively oriented, so higher values indicate better behavior of the model. RGR and RGE* start from 1 because, in the first step, there is no perturbation and no feature neutralization. RGA starts with the full-cross-section accuracy value and then changes as parts of the predicted ranking are removed.

3.8.3 Sector-level RGF* compliance vector

This section introduces the fourth compliance vector used in this thesis, called RGF*. This vector is added in order to include the Fairness dimension in the Integrated Compliance Score. Since this thesis does not deal with individuals, Fairness is not interpreted in terms of demographic groups. Instead, it is adapted to the financial setting and is interpreted as sector-level group parity, as already explained in Section 3.7.5.

The idea is to check whether a model has a similar predictive accuracy across GICS sectors. A model can have a good overall RGA, but this does not necessarily mean that it works in the same way for all sectors. For example, the model could rank stocks well in one sector and much worse in another one. For this reason, the sector dimension is included separately. The construction of RGF* starts with the sectoral RGA values. At each rebalancing date and for each model, RGA is computed separately within each GICS sector. After this, the sectors are ordered from the highest to the lowest RGA. The sector with the highest RGA is used as the reference sector, because it is the sector where the model obtains the best ranking accuracy at that date. The RGF* value for the sector in position j is then computed as:

$$\text{RGF}^*(j) = 1 - (\text{RGA}_{\text{best}} - \text{RGA}_j) \quad (3.17)$$

where RGA_{best} is the RGA of the best sector, and RGA_j is the RGA of the sector in position j . For the first sector, the model is compared with itself. Therefore, the gap is equal to 0 and RGF* is equal to 1. As sectors with lower RGA are considered, the gap from the best sector becomes

larger and RGF* decreases. If the model has a similar accuracy across sectors, the curve decreases only slightly. If, instead, the model performs much better in some sectors than in others, the curve decreases more clearly.

In this thesis, the investment universe covers the 11 GICS sectors. For this reason, the raw RGF* curve already has 11 points, like the other compliance vectors. The first point is the best sector compared with itself, while the other ten points measure the difference between the best sector and the remaining sectors.

The idea of interpolation is kept only to make the procedure general. If in another application the number of valid sectors were different from 11, the raw RGF* curve could be transformed into an eleven-point vector using linear interpolation. This would make it comparable with RGA, RGR and RGE*. In the empirical application of this thesis, this step was not needed, because the sector-level curve already had the same length as the other compliance vectors.

The final RGF* vector for each model is then obtained by averaging each element across the 36 out-of-sample rebalancing dates. RGF* is positively oriented. Higher values indicate stronger sector-level parity, because they mean that the sectoral RGA values are closer to each other. This makes the vector consistent with the other three compliance vectors, where higher values always indicate better model behavior.

3.8.4 Four-dimensional compliance score

After constructing the four compliance vectors, they are combined into one final score. The four vectors are the eleven-element vectors for RGA, RGR, RGE* and RGF*. Each vector represents one dimension of the SAFE AI evaluation.

Since each vector contains eleven values, the score is computed by considering all possible combinations of the vector elements. Therefore, the indices are:

$$i, j, k, l \in \{0, \dots, 10\}$$

For each combination (i, j, k, l) , the four values RGA_i , RGR_j , RGE_k^* and RGF_l^* are taken together and aggregated. The final four-dimensional Compliance Score, denoted as CS4, is the average of this aggregation over all possible combinations. In this sense, the score can be interpreted as a normalized volume over the four-dimensional compliance tensor. Higher value means that the model keeps better values across the four SAFE AI dimensions and across the different points of the compliance vectors.

Three aggregation methods are used. The first one is the arithmetic mean:

$$M_A(i, j, k, l) = \frac{RGA_i + RGR_j + RGE_k^* + RGF_l^*}{4} \quad (3.18)$$

The second one is the geometric mean:

$$M_G(i, j, k, l) = (RGA_i \cdot RGR_j \cdot RGE_k^* \cdot RGF_l^*)^{1/4} \quad (3.19)$$

The third one is the root mean square:

$$M_{RMS}(i, j, k, l) = \sqrt{\frac{RGA_i^2 + RGR_j^2 + (RGE_k^*)^2 + (RGF_l^*)^2}{4}} \quad (3.20)$$

The Compliance Score is then computed as:

$$CS4 = \frac{1}{11^4} \sum_{i=0}^{10} \sum_{j=0}^{10} \sum_{k=0}^{10} \sum_{l=0}^{10} M(i, j, k, l) \quad (3.21)$$

where M can be the arithmetic mean, the geometric mean or the RMS.

The division by 11^4 normalizes the score, because there are 11^4 possible combinations in the four-dimensional tensor. In this way, the final value is the average compliance level across the full grid and not the sum of all tensor values.

The three versions are reported because each one combines the four dimensions in a slightly different way. The arithmetic mean gives a simple average of the four compliance vectors. The

geometric mean is more sensitive to low values, because a weak value in one dimension reduces the final score more strongly. The RMS is added as an additional comparison, since it gives more weight to higher values.

To understand the effect of adding RGF*, a three-dimensional score is also computed. This score is called CS3 and includes only RGA, RGR and RGE*, which are the three vectors used in the original integrated Compliance Score logic proposed by Giudici and Kolesnikov (2025). In this case, the same tensor logic is applied, but only on three dimensions. The difference between the four-dimensional and the three-dimensional score is defined as:

$$\Delta = \text{CS4} - \text{CS3} \quad (3.22)$$

This difference is used to see how much the sector-parity dimension changes the final score. If Δ is positive, the inclusion of RGF* increases the Compliance Score. If it is negative, the sector-parity dimension reduces it.

Finally, the models are ranked according to their CS4 value. For each aggregation method, the model with the highest CS4 is considered the best model according to the Integrated Compliance Score used in this thesis. Once the Integrated Compliance Score is obtained, it can be compared with portfolio performance. This comparison is developed through the SAFE-Performance Frontier described in the next section.

3.9 SAFE-Performance Frontier

3.9.1 Reasoning and design

The SAFE AI evaluation described in the previous section gives one Compliance Score for each of the three machine learning models, following the SAFE AI framework proposed by Giudici

& Raffinetti (2023) and the integrated Compliance Score logic developed by Giudici & Kolesnikov (2025).

At this point, the analysis gives three points in the compliance-performance space: one for Ridge Regression, one for the Neural Network and one for XGBoost. These three points are useful for the comparison between the main models. However, they are not enough to understand if there is a general relation between compliance and portfolio performance. With only three points, it is difficult to say if a higher Compliance Score usually leads to a better portfolio. It is also difficult to see if a model with a higher Compliance score loses something in terms of financial performance. For this reason, this thesis also builds a SAFE-Performance Frontier.

The idea is to repeat the same analysis for many different configurations of the same model family. Therefore, instead of using only one Ridge Regression, one Neural Network and one XGBoost model, several versions of each model are tested.

For each configuration, the model is trained again. Then, the model produces prediction scores, which are transformed into an expected return vector using the rescaling procedure described in Section 3.4. This vector is used in the MIQP optimizer, and the resulting portfolio performance is computed using the realized returns in the following period. This keeps the portfolio construction consistent with the mean-variance framework introduced by Markowitz (1952), while allowing the expected return vector to change across model configurations. After this, both the Compliance Score and the portfolio performance measures are computed. In this way, each configuration becomes one point in the compliance-performance space.

This analysis helps to understand if compliance and performance move in the same direction when the model specification changes. If the configurations with higher Compliance Score also have better portfolio performance, this means that compliance does not reduce financial

performance. If, instead, they have lower performance, this suggests that there is a trade-off between the two.

For all configurations, the portfolio optimization parameter λ is fixed equal to 1.0, as in the main analysis. The covariance estimation method, the MIQP optimizer, the portfolio constraints and the monthly rebalancing calendar are also kept the same. This is consistent with out-of-sample portfolio evaluation studies, where the comparison between strategies should depend on the forecasting input and not on changes in the optimization setting (DeMiguel et al., 2009). The only element that changes is the model configuration. This is important because the aim is to isolate the effect of the hyperparameters, without changing the portfolio construction rule.

3.9.2 Grids of configurations

The SAFE-Performance Frontier is based on 150 configurations in total. More specifically, 50 configurations are created for Ridge Regression, 50 for XGBoost and 50 for the Neural Network.

For **Ridge Regression**, the parameter that changes is the regularization parameter α , following the approach introduced by Hoerl & Kennard (1970). This parameter is tested over 50 values between 10^{-4} and 10^4 . The range is quite wide because the goal is to see how the model behaves with different levels of regularization. Small values of α make the model close to a standard linear model, while larger values force the coefficients to be more strongly penalized. For **XGBoost**, 50 configurations are created using random search. This model is based on gradient-boosted decision trees and is included because it can capture non-linear relations between the input variables and future return ranks (T. Chen & Guestrin, 2016). The random seed is fixed equal to 42, so that the same configurations can be reproduced. The search includes different values for tree depth, learning rate, number of estimators, subsampling, column

subsampling and regularization parameters. Duplicate configurations are removed, so that each XGBoost configuration is used only once.

For the **Neural Network**, 50 configurations are also created using random search with the same seed. The model is a Neural Network implemented in scikit-learn (Pedregosa et al., 2011). In this case, the configurations differ in the number and size of hidden layers, the regularization parameter, the initial learning rate and the batch size. Also in this case, duplicate configurations are removed before running the analysis. For every configuration, the same pipeline used in the main analysis is repeated. First, the model is trained with the expanding-window procedure. Then, the model produces out-of-sample prediction scores. These scores are transformed into expected returns using the same rescaling procedure described in Section 3.4. After this, the MIQP optimizer is solved with $\lambda = 1.0$.

Finally, the portfolio performance is computed using the realized returns over the out-of-sample period, and the SAFE AI metrics are computed for the same configuration. At the end, each configuration has its own SAFE AI component values, its own Compliance Score, and its own portfolio performance measures.

3.9.3 Frontier construction and visualization

For each of the 150 configurations, three versions of the four-dimensional Compliance Score are computed: arithmetic, geometric and RMS. These three aggregation methods follow the same integrated Compliance Score structure described in Section 3.8, where the four SAFE AI dimensions are combined into one final score. The four SAFE AI dimensions are also reported separately. These are Accuracy, Robustness, Fairness and Explainability, based on RGA, RGR, RGF* and RGE* (Giudici & Kolesnikov, 2025).

The portfolio performance is evaluated mainly with the annualized Sharpe ratio, Maximum drawdown and Turnover. The Sharpe ratio is used as the main risk-adjusted performance

measure, following its standard use in portfolio evaluation (Sharpe, 1966). Maximum drawdown measures the largest loss from a previous peak. Turnover measures how much the portfolio changes at each rebalancing date. This is important because high turnover can reduce net performance once transaction costs are considered, as discussed in empirical portfolio evaluation studies (DeMiguel et al., 2009; Novy-Marx & Velikov, 2016).

The SAFE-Performance Frontier is represented with scatter plots. In each graph, the Compliance Score is placed on the x-axis, while one portfolio measure is placed on the y-axis. Separate graphs are created for the Sharpe ratio, maximum drawdown and average turnover. The figures are also repeated for the three aggregation methods of the Compliance Score: arithmetic, geometric and RMS.

The interpretation of the graphs depends on the metric used. For the Sharpe ratio, higher values are better. For Maximum drawdown and turnover, lower values are better. Therefore, the best trade-off point is selected differently depending on the graph. This analysis is useful because it shows whether the most compliant configurations are also good from a portfolio point of view. If the points with high Compliance Score are close to the best performance points, then compliance and performance seem to move together. If instead they are far from each other, then there is a trade-off between the two objectives. For this reason, the SAFE-Performance Frontier gives more information than the comparison of the three main models alone.

3.10 Chapter summary

This chapter presented the full methodological pipeline used in the thesis, starting from the raw data and arriving at the construction and evaluation of the portfolios.

First, the dataset and the investment universe were described. The analysis is based on the S&P 500 stocks obtained from Bloomberg. After applying the data cleaning filters and the additional

universe filters, the final universe is used to compute the daily log-returns, the covariance matrix and the stock-level features used in the rest of the study.

Then, the chapter described the return estimation models. Four approaches are compared: the Historical Mean Baseline, Ridge Regression, the Neural Network and XGBoost. The Historical Mean Baseline represents the classical Markowitz approach, while the three machine learning models are used to check whether replacing the historical mean with ML-based expected returns improves out-of-sample portfolio performance.

After this, the same MIQP portfolio optimizer was presented. The optimizer is kept fixed for all models, together with the covariance matrix estimation method, the constraints, the solver settings and the monthly rebalancing calendar. In this way, the differences between the portfolios depend mainly on the expected return vector estimated by each model.

The chapter also presented the two levels of evaluation used in the thesis. At portfolio level, the four portfolios are compared using Sharpe ratio, maximum drawdown, average turnover, transaction costs and lambda sensitivity analysis. At model level, Ridge Regression, the Neural Network and XGBoost are evaluated through the SAFE AI framework, adapted to a stock return prediction problem.

Finally, the Integrated Compliance Score and the SAFE-Performance Frontier were introduced. These two parts are used to connect the SAFE AI evaluation with the portfolio results, in order to understand whether the model that behaves better in terms of compliance is also the one that performs better at portfolio level.

The next chapter presents the empirical results obtained from this methodology over the out-of-sample test window from January 2023 to December 2025.

4 Portfolio Performance

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4.1 Introduction

This chapter presents the results of the portfolio performance analysis. The analysis is done on the out-of-sample test window, from January 2023 to December 2025. As described in Chapter 3, the portfolios are rebalanced on the first trading day of each calendar month. This gives 36 rebalancing dates.

The four portfolios analyzed are the Historical Mean Baseline, Ridge Regression, the Neural Network and XGBoost. These are the same four approaches presented in the methodology chapter. The Historical Mean Baseline represents the traditional mean-variance approach introduced by Markowitz (1952). XGBoost is the main machine learning model of the thesis and is based on gradient-boosted decision trees, following T. Chen & Guestrin (2016). Ridge Regression and the Neural Network are included as robustness checks.

All the portfolios are constructed using the same portfolio optimizer. The covariance matrix is estimated in the same way for all models. The constraints, the solver settings and the rebalancing calendar are also the same. For this reason, the comparison is done by changing only one element, which is the expected return vector μ .

Optimized portfolios can be very sensitive to the expected return estimates (Best & Grauer, 1991; Michaud, 1989). This is important because the objective is to understand if the way in which μ is estimated can change the final portfolio performance. In this thesis, one of the main questions is whether the expected returns produced by XGBoost can generate a better out-of-sample portfolio than the expected returns based on the Historical Mean Baseline. Ridge Regression and the Neural Network are used to check whether the results are only related to XGBoost or whether other machine learning models show a similar behavior.

The first part of the chapter presents the portfolio results for the main case, where λ is equal to 1.0. After this, the analysis considers transaction costs, in order to see whether the performance remains good after trading costs are included. Then turnover is analyzed separately, because it gives information on how much the portfolio changes from one rebalancing date to another. The last part of the chapter presents the lambda sensitivity analysis. This is done to check whether the results depend on the value chosen for λ .

4.2 Portfolio performance at $\lambda = 1.0$

This section presents the main portfolio results obtained with λ equal to 1.0. As described in Chapter 3, this is the value used in the main analysis for all four models. The other values of λ are used later only to check the sensitivity of the results.

Table 4.1 reports the performance measures of the four portfolios over the out-of-sample test window, from January 2023 to December 2025. The four portfolios are the Historical Mean

Baseline, Ridge Regression, the Neural Network and XGBoost. The main comparison remains between the Historical Mean Baseline and XGBoost. Ridge Regression and the Neural Network are included as robustness checks.

	Baseline	XGBoost	Ridge	MLP
Ann. Return	20.93%	21.60%	13.39%	5.45%
Ann. Volatility	25.39%	23.98%	21.76%	18.04%
Sharpe ratio	0.8243	0.9007	0.6151	0.3021
Sortino Ratio	1.1532	1.3578	0.8759	0.4178
Calmar Ratio	0.8156	0.9225	0.5927	0.2131
Max Drawdown	25.66%	23.41%	22.58%	25.57%
Avg. Turnover	46.46%	78.15%	79.86%	93.15%

Table 4.1: Out-of-sample portfolio performance at $\lambda=1.0$ (January 2023 – December 2025).

From Table 4.1, XGBoost is the portfolio with the best overall performance in the main $\lambda = 1.0$ setting. It has the highest Sharpe ratio, which is the main measure used in this thesis to compare return and risk (Sharpe, 1966). The improvement over the Historical Mean Baseline is not very large in terms of annualized return, but it is relevant because XGBoost also has lower volatility. This means that the better Sharpe ratio comes from both sides: a slightly higher return and a lower level of risk.

The other performance measures confirm the same result. XGBoost also has the highest Sortino ratio and the highest Calmar ratio. These two measures are useful because they give additional information about downside risk and drawdown (Sortino & Price, 1994; Young, 1991). The

maximum drawdown gives a similar indication. XGBoost has a lower drawdown than the Historical Mean Baseline and the Neural Network. Ridge Regression has the lowest maximum drawdown, but this comes together with a much lower return. For this reason, Ridge is not the strongest portfolio in terms of overall performance, even if it has the smallest worst loss.

The main weak point of XGBoost is turnover. The XGBoost portfolio changes more than the Historical Mean Baseline from one rebalancing date to another. This is important because higher turnover means more trading, and more trading can reduce final performance when transaction costs are included. This point is analyzed in Section 4.3 and then discussed more directly in Section 4.4, also following the importance of trading costs in portfolio evaluation discussed by DeMiguel et al. (2009) and Novy-Marx & Velikov (2016).

The Neural Network is the weakest portfolio in this first comparison. It has the lowest return and the lowest Sharpe ratio, while also having the highest turnover. This shows that using a non-linear machine learning model is not enough by itself to improve portfolio performance. Ridge Regression performs better than the Neural Network, but it remains below XGBoost and the Historical Mean Baseline in terms of risk-adjusted performance.

Figure 4.1 shows the cumulative wealth index of the four portfolios during the test period.

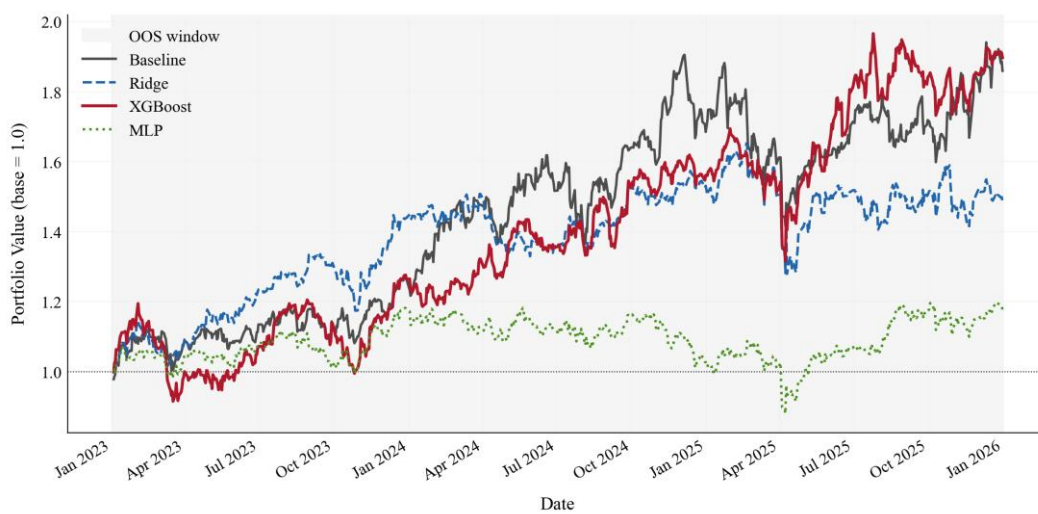


Figure 4.1: Cumulative portfolio wealth, January 2023–December 2025.

The wealth index confirms the results of Table 4.1. XGBoost and the Historical Mean Baseline have a similar path for most of the test period. In the final part, XGBoost ends above the Baseline. Ridge Regression performs better at the beginning of the period, but after that it remains below the two main strategies. The Neural Network remains close to the initial level for most of the period and does not show a clear improvement.

Figure 4.2 shows the drawdown series of the four portfolios.

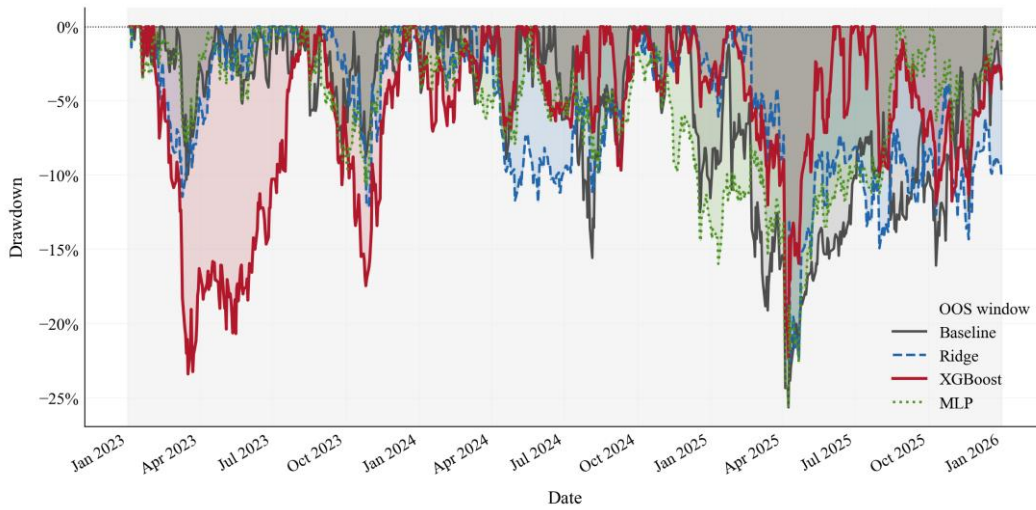


Figure 4.2: Portfolio drawdowns, January 2023–December 2025.

The drawdown figure is useful because it shows how much the portfolios lose from their previous peak during the test period. Also from this figure, XGBoost has a better result than the Historical Mean Baseline and the Neural Network. Ridge Regression has the lowest maximum drawdown, but this result must be read together with the return. In fact, Ridge has a lower drawdown, but also a much lower annualized return. For this reason, it is not better than XGBoost in the full comparison.

To conclude, when λ is equal to 1.0, XGBoost is the portfolio with the best result among the four strategies. It has better risk-adjusted performance than the Historical Mean Baseline and also better downside measures. The result is not only due to higher return, because XGBoost

also has lower volatility. However, XGBoost has higher turnover. For this reason, the next section checks whether this advantage remains after transaction costs.

4.3 Transaction costs

The results presented in Section 4.2 are gross results, so they do not include transaction costs. This point is important because XGBoost has a higher turnover than the Historical Mean Baseline. As shown in Table 4.1, XGBoost obtains a better Sharpe ratio, but it also requires more trading at each rebalancing date. Since transaction costs can reduce the performance of strategies with higher turnover, this step is important in out-of-sample portfolio evaluation (DeMiguel et al., 2009; Novy-Marx & Velikov, 2016). For this reason, it is useful to check if the advantage of XGBoost remains also after transaction costs. Transaction costs are applied using the same procedure described in Chapter 3. Three cost levels are considered: 10, 20 and 30 basis points. The costs are applied to the turnover of each portfolio, and the Sharpe ratio is then computed again using net returns. This is done for all four portfolios, so the comparison remains the same across models.

	Baseline	XGBoost	Ridge	MLP
Gross (0 bps)	0.8243	0.9007	0.6151	0.3021
10 bps	0.8025	0.8621	0.5717	0.2411
20 bps	0.7807	0.8234	0.5281	0.1800
30 bps	0.7588	0.7846	0.4846	0.1190

Table 4.2: Net Sharpe ratios after transaction costs at $\lambda=1.0$. Three cost levels are applied:

10, 20 and 30 basis points per unit of traded turnover.

The decrease is stronger for the portfolios with higher turnover, especially for XGBoost, Ridge Regression and the Neural Network. However, XGBoost remains the portfolio with the highest Sharpe ratio at all cost levels. The gap between XGBoost and the Historical Mean Baseline becomes smaller when costs increase, because XGBoost has higher turnover. However, even with 30 basis points, XGBoost is still above the Baseline. This means that the higher gross performance of XGBoost is strong enough to compensate the additional trading costs considered in this analysis.

Ridge Regression and the Neural Network remain below the Historical Mean Baseline. This confirms the result already shown in Section 4.2. Ridge Regression performs better than the Neural Network, but it does not improve the Baseline. The Neural Network remains the weakest one, because it has low performance and the highest turnover.

To conclude, transaction costs reduce the performance of all strategies, but they do not change the main result in the $\lambda = 1.0$ setting. XGBoost remains above the Historical Mean Baseline also after costs. At the same time, the smaller gap with the Baseline shows that turnover is important for the interpretation of the results. For this reason, turnover is analyzed separately in the next section.

4.4 Turnover analysis

The previous sections showed that the machine learning portfolios have higher turnover than the Historical Mean Baseline. This point was also important in Section 4.3, because transaction costs are applied on turnover. For this reason, this section looks more directly at the trading activity of the four portfolios.

Figure 4.3 shows the monthly turnover during the test period. The first rebalancing date is not included, because there is no previous portfolio before that date. For this reason, turnover is computed over 35 rebalancing dates.

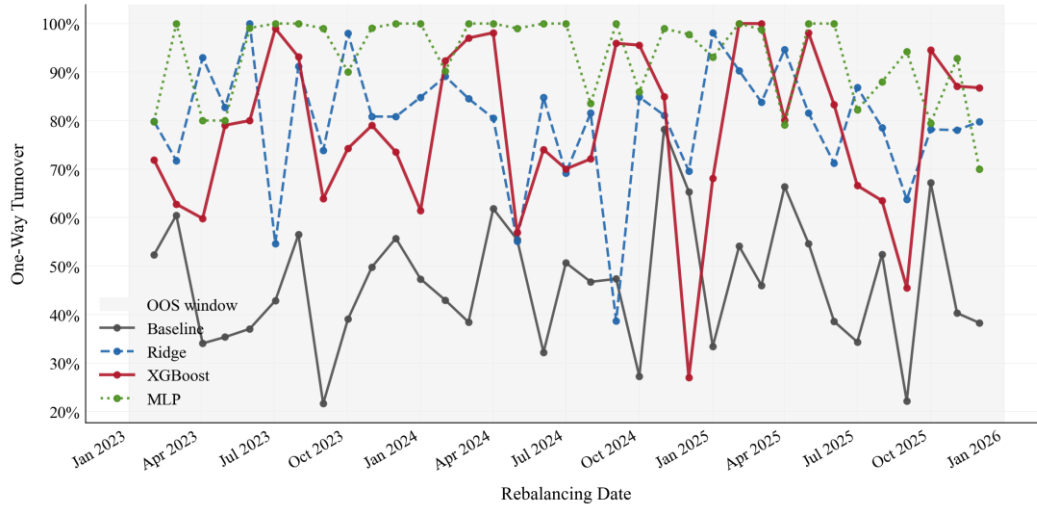


Figure 4.3: Monthly portfolio turnover, January 2023–December 2025.

The figure shows that the Historical Mean Baseline is the portfolio with the lowest turnover. The three machine learning portfolios are instead at a higher level. XGBoost and Ridge Regression have similar average turnover, while the Neural Network has the highest one. This means that the machine learning portfolios change more from one month to another.

This result is linked to the way in which the expected return vector is estimated. The Historical Mean Baseline uses the average return over the previous 252 trading days. This measure changes slowly when new observations are added. For this reason, the ranking of stocks is more stable from one rebalancing date to another. When the ranking changes less, the optimizer also needs to change the portfolio less.

The machine learning models are different. They use the stock characteristics described in Chapter 3, such as momentum, short-term reversal, volatility, liquidity and size. Some of these variables can change more quickly over time. For this reason, the expected return vector

produced by the machine learning models can change more from one month to another. This explains why the optimizer selects more different portfolios and why turnover is higher. This point is important because high turnover is not a negative thing by itself. It becomes a problem if the additional trading is not followed by better performance. This is also the reason why transaction costs are usually considered in out-of-sample portfolio evaluation, especially for strategies with higher trading activity (DeMiguel et al., 2009; Novy-Marx & Velikov, 2016).

Figure 4.4 shows the relation between average monthly turnover and Sharpe ratio.

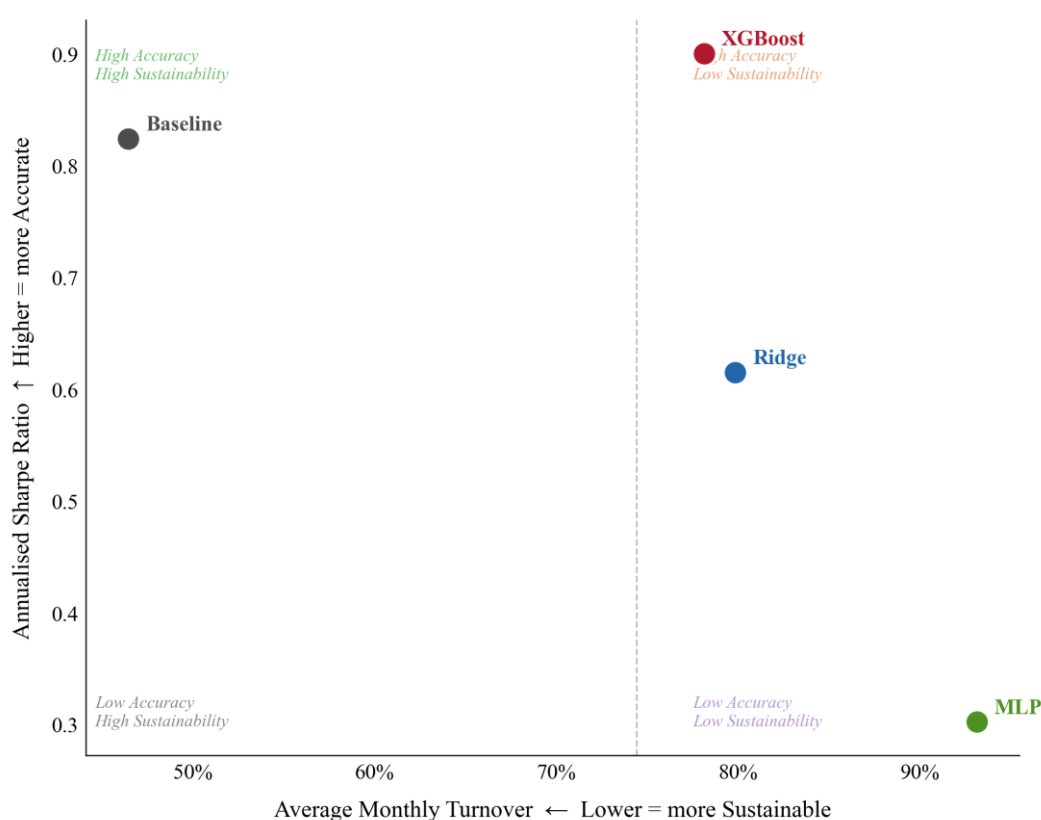


Figure 4.4: Sharpe ratio and average monthly turnover.

The scatter plot helps to read turnover together with performance. The Historical Mean Baseline has lower turnover and a good Sharpe ratio. XGBoost has higher turnover, but also the highest Sharpe ratio. Ridge Regression and the Neural Network also have high turnover, but they do not obtain the same performance.

This means that the higher turnover of XGBoost is different from the one of the other machine learning models. In the case of XGBoost, the additional trading is associated with better portfolio performance. For Ridge Regression and the Neural Network, instead, the higher turnover is not compensated by a better Sharpe ratio.

To conclude, turnover helps to explain the cost of using machine-learning-based expected returns. XGBoost gives a better portfolio in the main setting, but it also requires more trading than the Historical Mean Baseline. Ridge Regression and the Neural Network also require more trading, but without improving the Baseline. For this reason, turnover is one of the main elements needed to interpret the portfolio results.

4.5 Lambda sensitivity analysis

This section presents the lambda sensitivity analysis. The previous sections used λ equal to 1.0. This is the value used in the main analysis. However, λ is a parameter chosen in the optimization problem. For this reason, it is useful to see what happens when this value changes. This is done to check whether the portfolios behave in a similar way also with other values of λ . In the MIQP model, λ decides the relation between risk and expected return. This follows the mean-variance logic introduced by Markowitz (1952). When λ is low, the optimizer gives more importance to the variance part, so the portfolio is more focused on reducing risk. When λ is high, the optimizer gives more importance to the expected return vector μ . Therefore, the portfolio depends more on the expected returns estimated by the model.

The sensitivity analysis is done using these values:

$$\lambda \in \{0.1, 0.2, 0.5, 1.0, 2.0, 5.0, 10.0\}$$

Values lower than 1.0 are used to see what happens when the optimizer gives less importance to expected returns. Values higher than 1.0 are used to see what happens when expected returns

have more weight in the objective function. A similar idea is also used by Mäkinen & Toivanen (2024), who optimize for different values of the risk-return parameter.

Also in this study, the same optimization is repeated for the four portfolios: Historical Mean Baseline, Ridge Regression, Neural Network and XGBoost. The universe, the covariance matrix, the constraints and also the rebalancing dates are the same. The only thing that changes is λ . The main comparison is still between the Historical Mean Baseline and XGBoost, while Ridge Regression and the Neural Network are kept as robustness checks.

Figure 4.5 shows the Sharpe ratio, maximum drawdown and average turnover for the four portfolios across the different values of λ .

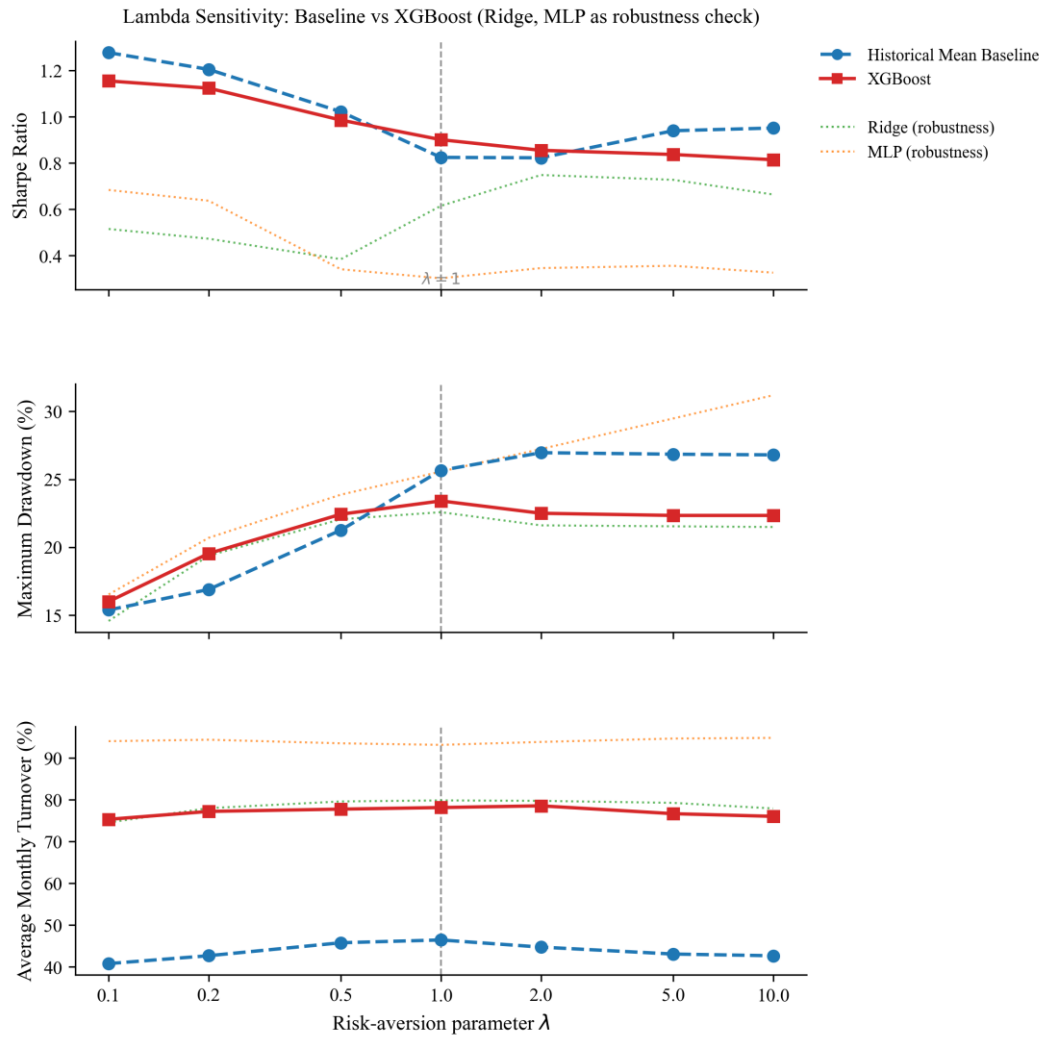


Figure 4.5: Lambda sensitivity analysis:

Sharpe ratio, maximum drawdown and average turnover.

The full tables with Sharpe ratio, maximum drawdown and average turnover are reported in Appendix A. In the main text, Table 4.3 reports only the ranking summary.

λ	Best Sharpe ratio	Lowest max drawdown	Lowest avg turnover
0.1	Historical Mean Baseline	Ridge Regression	Historical Mean Baseline
0.2	Historical Mean Baseline	Historical Mean Baseline	Historical Mean Baseline
0.5	Historical Mean Baseline	Historical Mean Baseline	Historical Mean Baseline
1.0	XGBoost	Ridge Regression	Historical Mean Baseline
2.0	XGBoost	Ridge Regression	Historical Mean Baseline
5.0	Historical Mean Baseline	Ridge Regression	Historical Mean Baseline
10.0	Historical Mean Baseline	Ridge Regression	Historical Mean Baseline

Table 4.3: Best-performing model per metric across the lambda grid $\lambda \in \{0.1, 0.2, 0.5, 1.0, 2.0, 5.0, 10.0\}$. XGBoost cells are highlighted. Ridge and MLP are robustness checks.

Table 4.3 shows that the result changes when λ changes. XGBoost has the best **Sharpe ratio** when λ is equal to 1.0 and 2.0. The Historical Mean Baseline performs better when λ is low and also when λ is high. For example, when λ is equal to 0.1, the Sharpe ratio of the Historical Mean Baseline is 1.2767, while the Sharpe ratio of XGBoost is 1.1544. When λ is equal to 1.0, the result changes. XGBoost has a Sharpe ratio of 0.9007, while Baseline has a Sharpe ratio of 0.8243. This means that the advantage of XGBoost is present in the main setting of the thesis, but it is not stable across the full lambda grid.

This point is important because XGBoost and the Historical Mean Baseline do not create the same ranking of stocks. For this reason, an additional check was done by comparing the expected return vector of XGBoost with the expected return vector of the Baseline. The average Spearman correlation between the two vectors is equal to

-0.28 during the test period. This means that the two models often rank stocks in a different way. Stocks with a high value for the Historical Mean Baseline are often ranked lower by XGBoost.

This difference can also be linked to the variables used by XGBoost. The 1-month return variable has an importance value of 11.4%, which suggests that recent returns have a relevant role in the model. In this study, XGBoost seems to capture part of a short-term reversal signal. Stocks that performed well in the previous month can receive a lower score from XGBoost. This helps to explain why XGBoost does not behave like the Historical Mean Baseline across the lambda grid.

This result is also linked to the role of μ in the optimizer. When λ is low, the expected return vector has less importance in the optimizer. For this reason, the difference between the models is less visible in the final portfolio. When λ is equal to 1.0 and 2.0, μ becomes more important, but it does not completely drive the portfolio. In this case, the XGBoost signal is useful and the Sharpe ratio is higher.

When λ is high, the optimizer gives much more importance to μ . This can be a problem because mean-variance portfolios are sensitive to expected return estimates, especially when these estimates change from one rebalancing date to another (Best & Grauer, 1991; Michaud, 1989). In this scenario, the Historical Mean Baseline becomes more competitive again. This does not mean that XGBoost is not useful. It means that the result depends on the value chosen for λ and on the way in which μ changes.

For **maximum drawdown**, Ridge Regression often has the lowest value. However, this result must be read together with the Sharpe ratio. Ridge Regression has good drawdown values, but it does not obtain the same risk-adjusted performance as XGBoost or the Historical Mean

Baseline. For turnover, the Historical Mean Baseline is always the lowest one. This confirms that the Baseline portfolio is more stable across rebalancing dates.

Turnover does not change much across the lambda grid. This means that turnover depends more on the expected return vector than on λ . The Historical Mean Baseline uses a trailing historical mean, which changes slowly. The machine learning models instead produce expected return vectors that can change more from one month to another. This confirms the result already discussed in Section 4.4.

Ridge Regression and the Neural Network do not change the main interpretation. Ridge Regression sometimes has good drawdown values, but it does not improve the main comparison, while the Neural Network remains the weakest strategy. This confirms that the better result is specific to XGBoost and is not obtained by all machine learning models.

To conclude, XGBoost performs better when λ is equal to 1.0 and also when λ is equal to 2.0. However, it is not better than the Historical Mean Baseline for all values of λ . For this reason, the result is positive but conditional. XGBoost improves the portfolio in the main setting of the thesis, but the result cannot be generalized to every possible value of λ .

4.6 Chapter summary

This chapter presented the portfolio results of the four strategies over the out-of-sample test window, from January 2023 to December 2025. The objective was to answer the *first research question* of the thesis, which is whether replacing the Historical Mean expected returns with machine-learning expected returns improves portfolio performance.

The first results were presented with λ equal to 1.0. This is the reference case used in the thesis. In this case, *XGBoost is the portfolio that performs better*. It has a higher Sharpe ratio than the Historical Mean Baseline. It also has better downside measures, because the maximum

drawdown is lower. This means that the result is not only due to higher returns, but also to a better risk profile. However, XGBoost also has higher turnover. This is an important point because the portfolio changes more from one rebalancing date to another. For this reason, transaction costs were added after the gross performance analysis. The results show that transaction costs reduce the Sharpe ratio of all portfolios. However, XGBoost remains above the Historical Mean Baseline also when the highest cost level is used. This means that the higher turnover reduces part of the advantage, but it does not remove it completely. Ridge Regression and the Neural Network were included as robustness checks. Their results are different from XGBoost: Ridge Regression performs better than the Neural Network, but it does not improve the Historical Mean Baseline, while the Neural Network is the weakest strategy, because it has low performance and high turnover. This shows that using a machine learning model is not enough by itself. The important point is the quality of the expected return vector produced by the model.

The lambda sensitivity analysis gives a more careful interpretation of the result. *XGBoost is the best portfolio when λ is equal to 1.0 and when it is equal to 2.0, but it is not the best one for every value of λ .* When λ is low, the optimizer gives less importance to the expected return vector. In this case, the Historical Mean Baseline performs better. When λ is high, the optimizer places more weight on μ . In this case, the Historical Mean Baseline becomes more competitive again. This result is also clearer when looking at the expected return vectors. XGBoost and the Baseline do not rank stocks in the same way. During the test period, the average Spearman correlation between their expected return vectors is negative. This suggests that XGBoost is not simply reproducing the signal of the Historical Mean, but is producing a different ranking of stocks. This is also linked to the 1-month return variable, which suggests that XGBoost uses part of a short-term reversal signal.

For this reason, the *first research question* has a positive but conditional answer. With λ equal to 1.0, XGBoost performs better than the Historical Mean Baseline, also after transaction costs. However, the same result is not found for all values of λ . In addition, XGBoost has higher turnover than the Baseline. This means that the result is not valid in every possible setting. This chapter shows that XGBoost improves the portfolio in the benchmark case used in this thesis. It does not show that all machine learning models improve portfolio performance. Ridge Regression and the Neural Network do not show the same improvement. The next chapter moves from the portfolio-level analysis to the model-level analysis. It evaluates Ridge Regression, the Neural Network and XGBoost through the SAFE AI framework, in order to answer the second research question of the thesis: *whether the machine learning model with the best SAFE AI score is also the one with the best portfolio performance*.

5 *SAFE AI Model Assessment*

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5.1 Introduction

This chapter presents the results of the SAFE AI evaluation. The analysis is done for the three machine learning models used in this thesis: Ridge Regression, the Neural Network and XGBoost. The Historical Mean Baseline is not included in this part because it is not a trained machine learning model. It is used only in the portfolio performance analysis.

As explained in Chapter 3, the SAFE AI framework evaluates the models through four dimensions: Sustainability/Robustness, Accuracy, Fairness and Explainability (Giudici & Raffinetti, 2023). In this study, these dimensions are adapted to the stock return prediction problem. The models do not predict outcomes for individuals, but future returns for stocks. For this reason, some parts of the framework need to be interpreted in a financial context.

The objective of this chapter is to compare the three machine learning models at prediction level. This means that the chapter looks at the predictions before they are used in the portfolio optimizer. This is useful because a model can have good portfolio performance, but this does not necessarily mean that it is also the best model in terms of SAFE AI evaluation.

This chapter is also connected to the second research question of the thesis. The question is *whether the machine learning model with the best SAFE AI score is also the one with the best portfolio performance*. In this chapter, the first part of this comparison is done. The relation between SAFE AI and portfolio performance is then analyzed in Chapter 6 through the SAFE-Performance Frontier.

The chapter is organized as follows. Section 5.2 presents the scalar results for the four SAFE AI dimensions. Section 5.3 presents the compliance vectors and the Integrated Compliance Score. Section 5.4 compares the SAFE AI ranking with the portfolio performance results presented in Chapter 4. Section 5.5 concludes the chapter with a summary of the main findings.

5.2 Scalar SAFE AI results

This section presents the scalar results of the SAFE AI evaluation. The results are reported for the three machine learning models: Ridge Regression, the Neural Network and XGBoost.

As explained in Chapter 3, each SAFE AI dimension is measured with a specific rank-based metric from the *safeai* package (Babaei & Giudici, 2025). Accuracy is measured with RGA, Sustainability/Robustness is measured with RGR, Fairness is measured with the sectoral RGA gap, and Explainability is measured with RGE.

These metrics are rank-based because, in this thesis, the models are used to rank stocks before the portfolio optimization. The objective is not to predict perfectly the future return of each stock, but to understand whether the model gives a useful ranking.

Each value is computed as the average across the 36 out-of-sample rebalancing dates, from January 2023 to December 2025. The scalar results are shown in Figure 5.1.

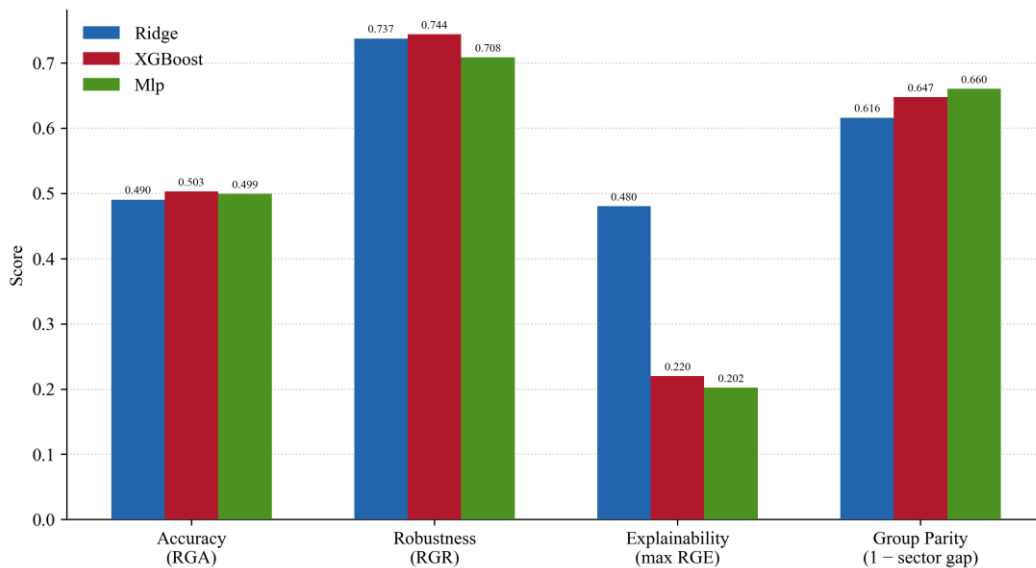


Figure 5.1: Scalar SAFE AI results across the four dimensions.

Higher values indicate better results for all dimensions. Fairness is reported as 1 minus the sectoral RGA gap. This means that a higher value means a lower difference across sectors. Explainability is reported as the maximum single-feature RGE value.

Figure 5.1 shows that the models behave differently across the four SAFE AI dimensions. XGBoost has the highest value for Accuracy and Robustness. The Neural Network has the highest value for Fairness. Ridge Regression has the highest value for Explainability.

This result is useful because it shows that no model is the best in all dimensions. For this reason, the four dimensions are discussed separately in the next subsections.

5.2.1 Accuracy

The Accuracy dimension is measured using Rank Graduation Accuracy, or RGA. As explained in Section 3.7.4, RGA compares the ranking of realized future returns with the ranking of predicted returns. This metric is useful in this thesis because the models are used to rank stocks before portfolio optimization.

Table 5.1 reports the Accuracy results for the three models.

Model	RGA
Ridge Regression	0.4904
XGBoost	0.5026
Neural Network	0.4991

Table 5.1: Mean RGA across 36 rebalancing dates (January 2023 – December 2025).

The random benchmark for RGA is equal to 0.5000. This means that a model that ranks stocks randomly would obtain, on average, a value close to 0.5000. If the value is above 0.5000, the

model ranks stocks slightly better than a random ranking. If the value is below 0.5000, the ranking is worse than the random benchmark (Giudici & Raffinetti, 2025).

The three models obtain very similar values. XGBoost has the highest RGA, equal to 0.5026. The Neural Network follows with 0.4991, while Ridge Regression has the lowest value, equal to 0.4904.

The differences between the three models are very small. This confirms that predicting cross-sectional stock returns is difficult. This is consistent with Gu et al. (2020), who show that stock return prediction with firm-level characteristics is difficult because the signal is usually weak compared to the noise.

Even if the values are close to 0.5000, this result is still useful for the thesis. The models are not used to predict perfectly the future returns. They are used to give a ranking of stocks. For this reason, also a small improvement in the ranking can be relevant, because it can affect the stocks selected by the optimizer.

Among the three models, XGBoost obtains the highest value in terms of Accuracy. However, the difference with the other models is limited. For this reason, this result should not be interpreted as strong evidence of the predictive superiority of XGBoost.

5.2.2 Sustainability/Robustness

The Sustainability/Robustness dimension is measured using Rank Graduation Robustness, or RGR. As explained in Section 3.7.3, in this thesis Sustainability does not refer to ESG sustainability. It is interpreted as robustness of the predicted ranking. The idea is to check whether the ranking produced by the model changes too much when the input features are perturbed. This is important because financial variables can contain noise. For this reason, a model whose ranking changes too much after perturbation can be considered less stable.

Table 5.2 reports the Sustainability/Robustness results for the three models.

Model	RGR (group)
Ridge Regression	0.7374
XGBoost	0.7439
Neural Network	0.7082

Table 5.2: Group-level RGR computed by perturbing all features simultaneously at 5% perturbation.

In this analysis, RGR is computed after applying a simultaneous 5% perturbation to all input features. A higher RGR value means that the ranking remains more stable after the perturbation. Therefore, higher values indicate higher Robustness.

XGBoost has the highest RGR value, equal to 0.7439. Ridge Regression follows with a value equal to 0.7374. The Neural Network has the lowest value, equal to 0.7082.

This means that XGBoost is the most robust model when all features are perturbed at the same time. Ridge Regression is very close to XGBoost, while the Neural Network is less robust. In other words, the ranking produced by the Neural Network changes more after the perturbation of the input data.

Looking at the single-feature RGR values, Ridge Regression is more stable for almost all variables. For example, for `ret_1w`, Ridge Regression has an RGR equal to 0.9995, while XGBoost has 0.9909 and the Neural Network has 0.9447. This means that Ridge Regression is very stable when only one feature is perturbed.

However, when all features are perturbed at the same time, XGBoost obtains the highest value. This means that Ridge Regression is more stable at single-feature level, while XGBoost is stronger when the whole information set is disturbed.

5.2.3 Fairness

The Fairness dimension is measured using the sectoral RGA parity gap. As explained in Section 3.7.5, in this thesis Fairness is not interpreted as fairness between protected groups. This is because the models do not predict outcomes for individuals. They predict future returns for stocks. For this reason, the groups used in this analysis are the GICS sectors.

The objective is to check whether the model has a similar predictive accuracy across the different sectors. For each rebalancing date, RGA is computed separately inside each GICS sector. After this, the difference between the sector with the highest RGA and the sector with the lowest RGA is computed. This gives the sectoral RGA gap.

Table 5.3 reports the Fairness results for the three models.

Model	Fairness (1 – sector gap)
Ridge Regression	0.6160
XGBoost	0.6473
Neural Network	0.6601

Table 5.3: Fairness reported as 1 – mean sector RGA gap across the 11 GICS sectors and 36 rebalancing dates.

In Table 5.3, Fairness is reported as 1 minus this gap. This means that higher values indicate better Fairness. If the value is higher, the difference between sectors is lower. If the value is lower, the model has more different levels of accuracy across sectors. This follows the rank-based logic used in the SAFE AI framework (Babaei et al., 2025; Giudici & Raffinetti, 2023). The Neural Network has the highest Fairness value, equal to 0.6601. XGBoost follows with a value equal to 0.6473. Ridge Regression has the lowest value, equal to 0.6160.

This means that the Neural Network is the model with the most similar ranking accuracy across the 11 GICS sectors. XGBoost is also close to the Neural Network. Ridge Regression instead has the lowest value, so its ranking accuracy changes more from one sector to another. This result is useful because a model can have a good average Accuracy, but this does not mean that the model works in the same way in all sectors. A model can rank stocks better in some sectors and worse in other sectors. For this reason, the Fairness dimension adds information that is not visible from the average RGA alone.

The lower Fairness value of Ridge Regression can be connected to the Explainability results presented in the next section. Ridge Regression depends mainly on one input feature, which is `log_mktcap`. For this reason, the model may work better in sectors where market capitalization is more useful for ranking stocks, and worse in sectors where other variables are more important.

To conclude, the Fairness result shows that the Neural Network obtains the highest value in this dimension. XGBoost is second, while Ridge Regression has the lowest value in terms of sectoral parity.

5.2.4 Explainability

The Explainability dimension is measured using Rank Graduation Explainability, or RGE. As explained in Section 3.7.6, RGE measures how much the predicted ranking changes when one input feature is neutralized. If the ranking changes a lot, this means that the feature has an important role in the model prediction. If the ranking changes only a little, this means that the feature has a smaller role. This follows the Rank Graduation Explainability approach used in the SAFE AI metrics (Babaei et al., 2025).

In this thesis, Explainability is reported as the maximum single-feature RGE value across the ten input features (Raffinetti, 2025). This means that, for each model, the table reports the feature with the strongest effect on the predicted ranking.

Table 5.4 reports the Explainability results for the three models.

Model	Max RGE	Main feature
Ridge Regression	0.4803	log_mktcap
XGBoost	0.2196	log_mktcap
Neural Network	0.2021	vol_1m

Table 5.4: Maximum single-feature RGE value across the 10 input features. The main feature column reports the feature with the highest mean RGE for each model.

Ridge Regression has the highest Explainability value, equal to 0.4803. This value comes mainly from log_mktcap. The RGE of log_mktcap is close to 0.480, while the other variables have much lower values. This means that Ridge Regression uses mainly market capitalization to produce the stock ranking.

XGBoost has a lower maximum RGE value, equal to 0.2196. Also for XGBoost, the most important feature is log_mktcap. However, XGBoost also uses other information. For example, ret_1m has an RGE equal to 0.111. This means that XGBoost still depends a lot on size, but it also uses some momentum information.

The Neural Network has the lowest maximum RGE value, equal to 0.2021. In this case, the most important feature is vol_1m. However, the RGE values are more distributed across the ten variables. The values go from around 0.129 for ret_1m to 0.202 for vol_1m. This means that no

single variable dominates the Neural Network in the same way as `log_mktcap` dominates Ridge Regression.

This result has to be interpreted carefully. A higher maximum RGE does not always mean that the model is better. Ridge Regression has the highest value because one variable explains a large part of its ranking. This makes the model simple to interpret, but it also shows that the model uses the available information in a very concentrated way. The Neural Network has the lowest maximum RGE. This does not necessarily mean that it is worse. It means that the model uses the information in a more distributed way across the input features. In this thesis, this is also relevant because the objective is to rank stocks using different types of information, such as momentum, volatility, liquidity and size.

For this reason, Ridge Regression obtains the highest scalar Explainability value. However, this result does not mean that Ridge Regression uses the information in a richer way. It means that its ranking is easier to explain because it depends mainly on one feature. XGBoost and the Neural Network instead use a broader set of signals, especially the Neural Network.

5.2.5 Summary and dimension-level ranking

This section summarizes the results obtained for the four SAFE AI dimensions. The objective is to understand which model performs better in each dimension before computing the Integrated Compliance Score.

Table 5.5 reports the ranking of the three models for each SAFE AI dimension. For all dimensions, higher values indicate better results. For Fairness, this is because the value is computed as 1 minus the sectoral RGA gap. Therefore, a higher value means that the model has lower differences in predictive accuracy across sectors.

Dimension	Rank 1	Rank 2	Rank 3
Accuracy	XGBoost	Neural Network	Ridge Regression
Robustness	XGBoost	Ridge Regression	Neural Network
Fairness	Neural Network	XGBoost	Ridge Regression
Explainability	Ridge Regression	XGBoost	Neural Network

Table 5.5: Ranking of the three models across the four SAFE AI dimensions. Rank 1 indicates the best-performing model for each dimension.

The table shows that no model obtains the best result in all four dimensions. XGBoost obtains the best result in Accuracy and Robustness. This means that it has the highest average ranking accuracy and the most stable ranking when all input features are perturbed together.

The Neural Network is the best model in Fairness. This means that its ranking accuracy is more similar across the 11 GICS sectors. However, it is the weakest model in Robustness and Explainability.

Ridge Regression is the best model in Explainability. This result is mainly due to the strong role of `log_mktcap` in its predictions. However, Ridge Regression has the lowest value in Accuracy and Fairness. This means that, even if the model is easier to explain, it does not have the best ranking accuracy and it is less balanced across sectors.

This comparison shows that the SAFE AI evaluation cannot be reduced to only one dimension. A model can be strong in one dimension and weak in another one. For this reason, the next section transforms the scalar SAFE AI values into compliance vectors and then combines them into the Integrated Compliance Score, following the approach described in Chapter 3 and based on Giudici and Kolesnikov (2025).

5.3 Compliance vectors and Integrated Compliance Score

The results presented in Section 5.2 show that the three models behave differently across the four SAFE AI dimensions. XGBoost has the best value for Accuracy and Robustness, the Neural Network has the best value for Fairness, while Ridge Regression has the best value for Explainability. For this reason, the scalar results are useful, but they are not enough in order to decide which model is the best overall model according to SAFE AI. A model can be good in one dimension and weaker in another one. Therefore, it is necessary to combine the four dimensions into one final measure.

This is the objective of the Integrated Compliance Score. In this thesis, the Compliance Score is used to obtain a final SAFE AI ranking of the three machine learning models. This ranking is then compared with the portfolio performance results in the next sections of the thesis.

Before computing the final score, the scalar SAFE AI values are transformed into compliance vectors. Each vector contains eleven values and they show how the model behaves when the evaluation changes step by step. The first value refers to the situation where no additional stress is applied, while the following ones are obtained by increasing the level of stress. The last value refers to the maximum stress level.

This step is useful because the final score should not depend only on one scalar value. It should also consider how the model behaves across the full compliance curve. For this reason, each compliance vector is summarized through the Area Under the Curve, or AUC. A higher AUC means that the model keeps better values across the different points of the curve.

The compliance vectors used in this thesis follow the Compliance Score approach of Giudici & Kolesnikov (2025). The RGA, RGR and RGE* compliance vectors are based on this framework. In addition, this thesis also uses the RGF* compliance vector in order to include the Fairness dimension.

The construction of the compliance vectors and of the Integrated Compliance Score was already explained in Chapter 3; for this reason, this section focuses on the results. CS3 refers to the score based on RGA, RGR and RGE*, while CS4 refers to the score that also includes RGF*. CS4 is the main compliance score used in this thesis. The objective is to move from the single SAFE AI dimensions to one final compliance ranking of Ridge Regression, the Neural Network and XGBoost.

5.3.1 RGA compliance vector

The RGA compliance vector is used to see how Accuracy changes when stocks are removed from the top of the predicted ranking.

In the first element of the vector, no stock is removed. For this reason, RGA is computed using all the 407 stocks in the investment universe. After this, the stocks are ordered from the highest to the lowest predicted return. At each following step, one additional decile is removed from the top of the ranking. In the last element, all stocks have been removed, and RGA is set equal to zero by construction.

The idea of this vector is to check whether the model puts the best stocks in the first part of the ranking. This is important because the stocks at the top of the ranking are the most relevant ones for the portfolio optimizer. If a model has most of its useful signal in the top-ranked stocks, the curve should decrease more when these stocks are removed. If instead the signal is more distributed across the full ranking, the curve should decrease more slowly.

This type of vector follows the compliance vector logic described in Chapter 3 and the approach proposed by Giudici & Kolesnikov (2025)

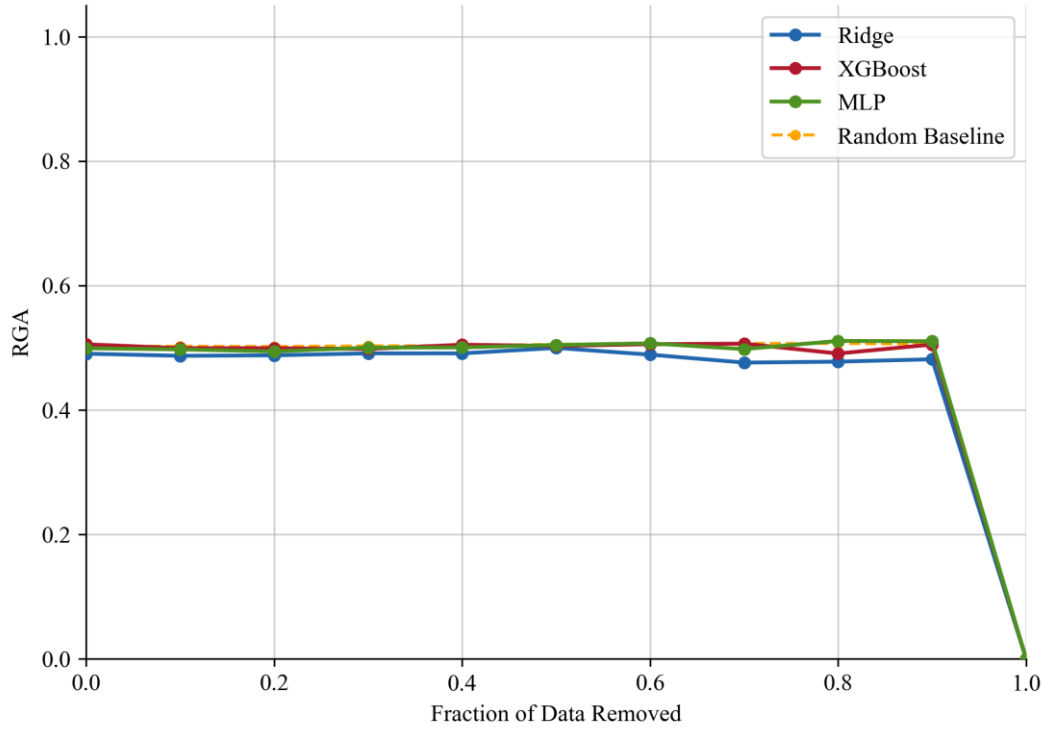


Figure 5.2: RGA compliance vector for Ridge Regression, XGBoost and the Neural Network.

Figure 5.2 shows that the three curves are very similar. The values remain close to 0.5000 for most of the curve. As explained in Section 5.2.1, this is interpreted as the random benchmark for RGA. This means that none of the three models shows a strong predictive signal concentrated in the top part of the ranking. In other words, Ridge Regression, XGBoost and the Neural Network do not seem to predict the top-ranked stocks much better than the other parts of the ranking. This result is consistent with the scalar RGA values presented in Section 5.2.1, also in that section, the three models had values close to 0.5000. For this reason, the RGA compliance vector confirms that predicting cross-sectional stock returns is difficult. The predictive signal is weak and it is not concentrated in one specific part of the predicted ranking. This is also consistent with Gu et al. (2020), who show that stock return prediction is difficult because the signal is usually weak compared to the noise.

5.3.2 RGR compliance vector

The RGR compliance vector is used in order to see how Robustness changes when the input features are increasingly perturbed. As explained in Section 3.7.3, RGR measures how stable the predicted ranking remains after perturbing the input variables.

In the first element of the vector, no perturbation is applied. For this reason, RGR is equal to 1 by construction. After this, the perturbation level increases step by step, from 5% to 50%. A higher RGR value means that the ranking remains more stable after the perturbation. This vector is useful because the scalar RGR value presented in Section 5.2.2 refers only to the 5% perturbation level. The full vector instead shows what happens when the perturbation becomes stronger.

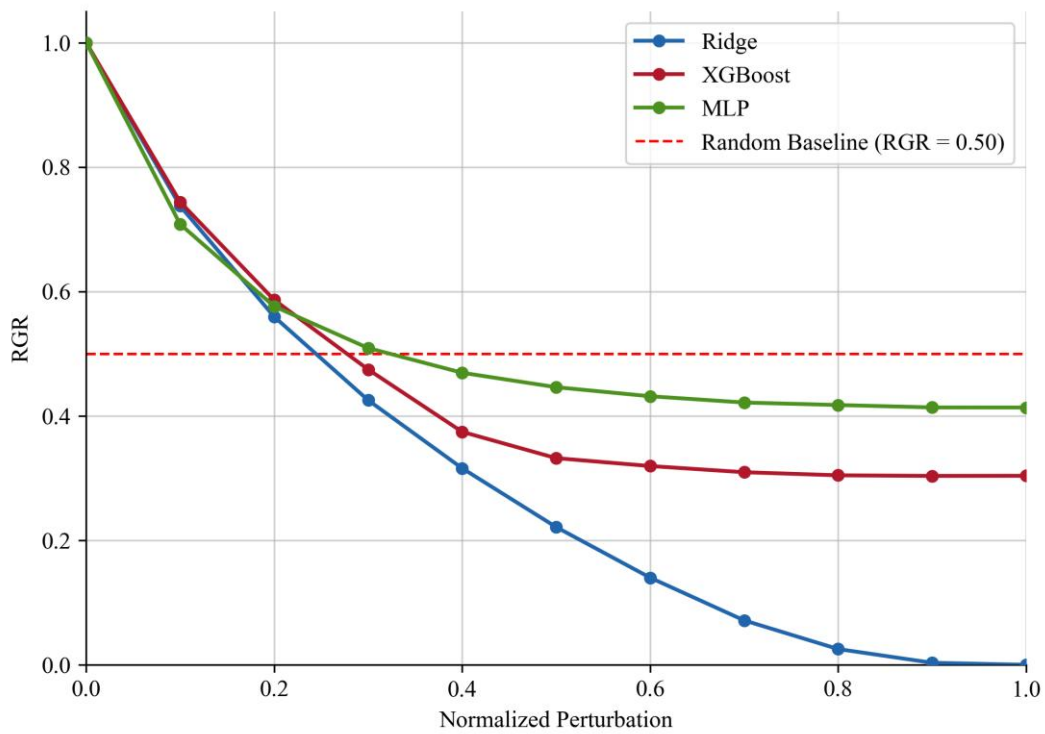


Figure 5.3: RGR compliance vector for Ridge Regression, XGBoost and the Neural Network.

Figure 5.3 shows that the three models behave differently when the perturbation level increases.

At the beginning, Ridge Regression and XGBoost are close to each other; this is consistent with

the scalar RGR values in Section 5.2.2. However, when the perturbation becomes stronger, Ridge Regression decreases more quickly. Around the 30% perturbation level, its RGR goes below 0.5000 and then continues to decrease. This means that Ridge Regression becomes less stable when the input data are strongly perturbed. XGBoost decreases more slowly than Ridge Regression. This means that it keeps a more stable ranking at higher perturbation levels. The Neural Network shows the most stable behavior at high perturbation levels. Even if it had the lowest scalar RGR at 5%, its curve remains higher when the stress becomes stronger.

This result is important because at 5% perturbation, XGBoost seemed to be the most robust model. However, the full compliance vector shows that the Neural Network is more stable under stronger perturbations. Ridge, instead, is the weakest model when the input data are highly disturbed.

5.3.3 *RGE* compliance vector*

The RGE* compliance vector is used in order to see how much of the original predicted ranking remains when the input features are progressively neutralized.

As explained in Chapter 3, RGE* is different from the scalar RGE. RGE is used to understand the role of one single feature. RGE* is instead used to see what happens to the full ranking when the features are removed step by step.

In the first element of the vector, no feature is neutralized. For this reason, RGE* is equal to 1 by construction. After this, one feature at a time is replaced with its training-time average value. The order goes from the least important feature to the most important feature. In the last element, all ten features are neutralized, and the value moves toward 0.5000, which is the random benchmark.

This vector is useful because it shows when the model starts to lose its original ranking. If the model depends mainly on one or two features, the curve remains high until these features are removed. If instead the model uses many features together, the curve decreases more continuously.

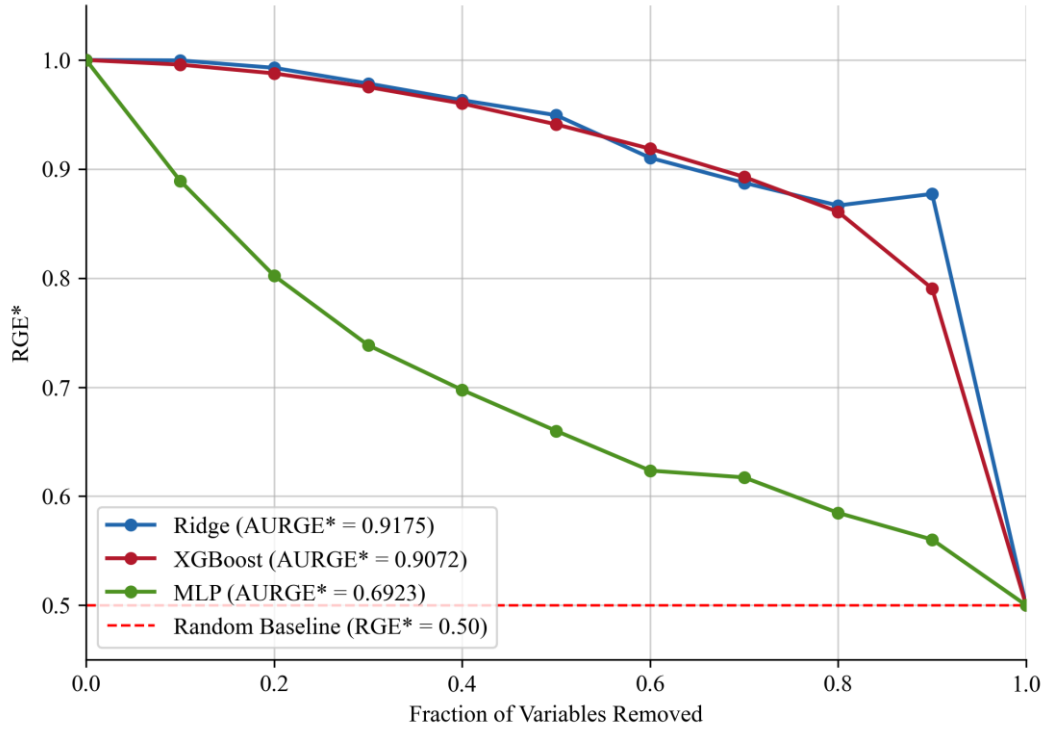


Figure 5.4: RGE compliance vector for Ridge Regression, XGBoost and the Neural Network.*

Figure 5.4 shows that Ridge Regression and XGBoost have very similar curves. The two curves decrease slowly and remain above 0.86 even after 8 out of 10 features have been neutralized. This means that Ridge Regression and XGBoost keep most of their original ranking even when many features are removed. This result is consistent with Section 5.2.4, where both models were mainly influenced by `log_mktcap`. Since this variable is one of the most important ones, it is removed only in the last part of the sequence. The Neural Network has a different behavior. Its curve decreases faster and reaches a value close to 0.70 already in the first part of the

sequence. This means that the Neural Network loses part of its ranking earlier when the features are neutralized. This happens because the Neural Network uses the input features in a more distributed way. There is not one single variable that dominates the model in the same way as `log_mktcap` does for Ridge Regression. For this reason, when the features are removed one by one, the Neural Network loses information more gradually.

The Area Under the RGE* curve gives one final value for this vector. Ridge Regression obtains the highest value, equal to 0.9175. XGBoost follows with 0.9072. The Neural Network obtains the lowest value, equal to 0.6923. This result has to be interpreted carefully. A higher RGE* value does not always mean that the model is better. Ridge Regression and XGBoost have higher values because their ranking depends mainly on few important features. The Neural Network has a lower value because it uses the information in a more distributed way.

5.3.4 RGF* compliance vector

The RGF* compliance vector is used to include the Fairness dimension in the Compliance Score. As explained in Chapter 3, the Compliance Score approach of Giudici and Kolesnikov (2025), used as reference in this thesis, is based on three dimensions: Accuracy, Sustainability/Robustness and Explainability. In this thesis, one additional vector is used in order to include Fairness because the objective is also to check whether the models behave in a similar way across sectors. During the study the models predict future returns for stocks. For this reason, Fairness is not interpreted in terms of demographic groups. It is interpreted as sector-level group parity, using the GICS sectors.

The idea is to check whether a model has similar predictive accuracy across sectors. A model can have a good overall RGA, but this does not mean that it works in the same way for all sectors. For example, it can rank stocks well in one sector and worse in another one. For this reason, the sector dimension is included separately through RGF*. The construction of RGF*

starts from the sectoral RGA values. At each rebalancing date and for each model, RGA is computed separately within each GICS sector. After this, the sectors are ordered from the highest to the lowest RGA. The sector with the highest RGA is used as the reference sector, because it is the sector where the model obtains the best ranking accuracy at that date.

For the first sector, the model is compared with itself. Therefore, the gap is equal to zero and RGF^* is equal to 1. After this, the other sectors are compared with the best sector following the ranking order. As sectors with lower RGA are considered, the gap from the best sector becomes larger and RGF^* decreases. This means that higher RGF^* values indicate stronger sector-level parity. If the model has similar accuracy across sectors, the curve decreases only slightly. If instead the model works much better in some sectors than in others, the curve decreases more clearly.

In this thesis, the investment universe covers the 11 GICS sectors. For this reason, the RGF^* curve has 11 points, like the other compliance vectors. The first point is the best sector compared with itself, while the other ten points measure the difference between the best sector and the remaining sectors. Therefore, in the empirical application of this thesis, interpolation is not needed.

As explained in Chapter 3, interpolation is an idea kept only to make the procedure general.

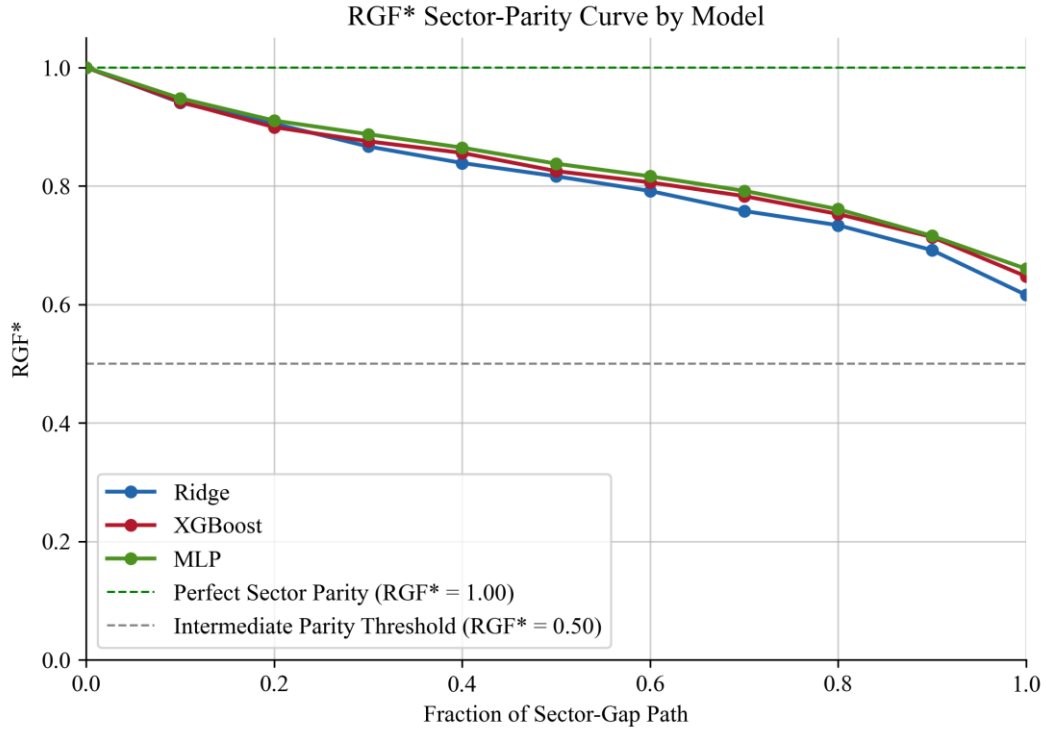


Figure 5.5: RGF compliance vector for Ridge Regression, XGBoost and the Neural Network.*

Figure 5.5 shows that the three curves decrease during the path. However, the order of the models is always the same. The Neural Network is the highest curve. XGBoost is the second one. Ridge Regression is the lowest one. At the final point, the Neural Network reaches 0.6601. XGBoost reaches 0.6473, while Ridge Regression reaches 0.6160. These are the same Fairness values discussed in Section 5.2.3.

This means that the Neural Network has the most similar ranking accuracy across the 11 GICS sectors. XGBoost is close to the Neural Network. Ridge Regression instead has the largest difference between sectors. This result is useful because the average Accuracy values in Section 5.2.1 were very close for the three models. However, the sector analysis shows that the models are not equally balanced across sectors. For this reason, RGF* is included in the Compliance Score, in order to include also the sector-level behavior of the models.

5.4 *Integrated Compliance Score*

5.4.1 *CS4 results and model ranking*

The previous sections presented the four compliance vectors: RGA, RGR, RGE* and RGF*. These vectors are useful because they show how each model behaves in each SAFE AI dimension. However, they do not give one final model ranking.

For this reason, the Integrated Compliance Score is computed. The objective is to combine the four SAFE AI dimensions into one final value. In this way, it is possible to understand which model has the best SAFE AI result overall.

As explained in Section 3.8, the Compliance Score follows the integrated approach proposed by Giudici & Kolesnikov (2025). The score is computed using the four compliance vectors, and each vector has eleven values. Since there are four vectors, the score is based on all possible combinations of RGA, RGR, RGE* and RGF*. This gives $11^4 = 14,641$ combinations. For each combination, the four values are combined with an aggregation method. Three aggregation methods are used: arithmetic mean, geometric mean and root mean square. For this reason, three versions of CS4 are obtained: CS4 arithmetic, CS4 geometric and CS4 RMS.

The three methods are used because the final ranking should not depend only on one aggregation rule. The arithmetic mean gives a simple average. The geometric mean penalizes more when one dimension has a low value. The RMS gives more weight to higher values. For this reason, the three scores are reported together.

Model	CS4 arithmetic	CS4 geometric	CS4 RMS
Ridge Regression	0.619	0.433	0.690
XGBoost	0.659	0.572	0.703
Neural Network	0.629	0.563	0.658

Table 5.6: CS4 scores under three aggregation methods (arithmetic, geometric, root mean square). Bold values indicate the best result for each aggregation method.

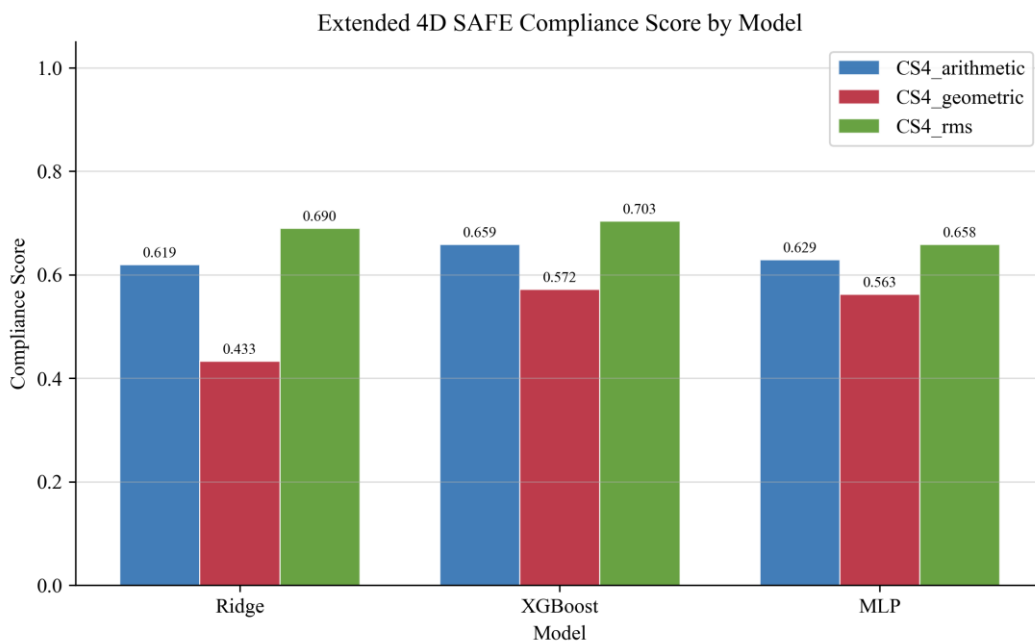


Figure 5.6: Four-dimensional Compliance Score by model.

Table 5.6 and Figure 5.6 show that XGBoost has the highest score in all three cases.

- With the arithmetic mean, XGBoost is first, the Neural Network is second, while Ridge Regression is third.
- With the geometric mean, XGBoost is again first and Ridge Regression last with 0.433. Ridge Regression is penalized more in this case because the geometric mean is more

sensitive to low values. This is connected to the RGR curve of Ridge Regression, which decreases a lot when the perturbation level increases, as shown in Section 5.3.2.

- With the RMS method, XGBoost is still first, with 0.703. Ridge Regression is second, the Neural Network is third.

This means that XGBoost obtains the best result according to the four-dimensional Compliance Score. This result is stable, because XGBoost is first with all three aggregation methods. This is important because in Section 5.2 there was not one model that was best in all dimensions. XGBoost was best in Accuracy and Robustness, the Neural Network was best in Fairness, and Ridge Regression was best in Explainability. The Compliance Score is useful because it combines these different results and gives one final SAFE AI ranking.

5.4.2 Impact of the inclusion of Fairness

In order to see the effect of Fairness, a three-dimensional Compliance Score is also computed. This score is called CS3. It includes only RGA, RGR and RGE*. Therefore, it includes Accuracy, Sustainability/Robustness and Explainability, following the reference structure of Giudici and Kolesnikov (2025).

After this, CS4 is compared with CS3. The difference is computed as:

$$\Delta = \text{CS4} - \text{CS3}$$

This value shows how much the inclusion of RGF* changes the final Compliance Score. If Δ is positive, Fairness increases the score. If Δ is negative, Fairness reduces the score.

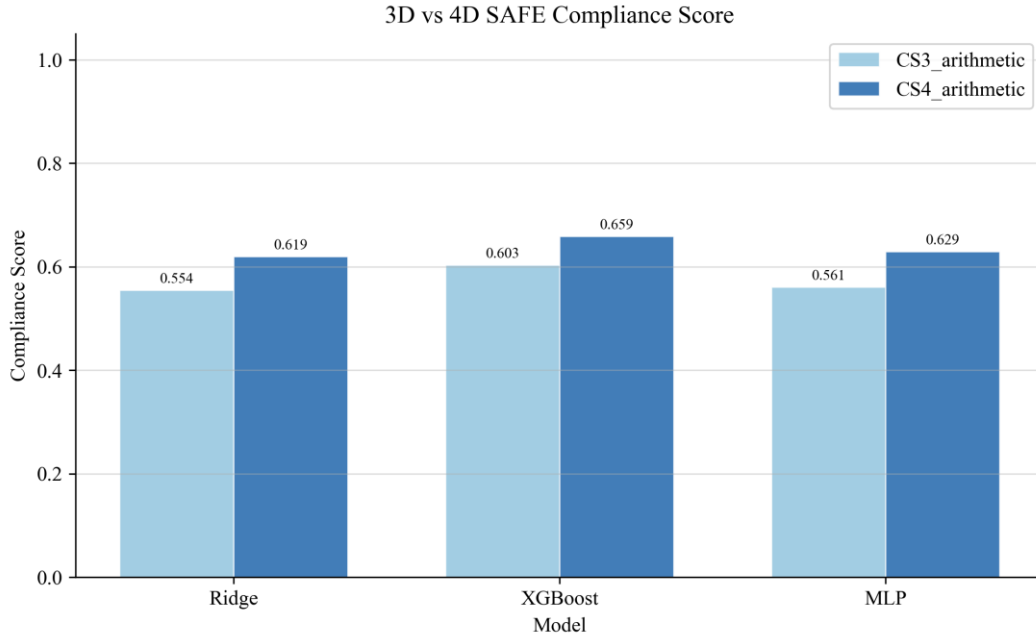


Figure 5.7: Comparison between CS3 and CS4 arithmetic scores.

Figure 5.7 shows that CS4 is higher than CS3 for all three models. This means that the inclusion of Fairness increases the Compliance Score in all cases.

The increase is not the same for all models. The Neural Network has the largest increase, with $\Delta = 0.069$. Ridge Regression follows with $\Delta = 0.065$. XGBoost has the smallest increase, with $\Delta = 0.056$. However, the inclusion of Fairness does not change the final ranking. XGBoost remains the best model also with CS4. For this reason, Fairness improves the scores, especially for the Neural Network, but it does not change the main result. Therefore, according to the Integrated Compliance Score, XGBoost is the best model in the SAFE AI evaluation.

5.4.3 Discussion

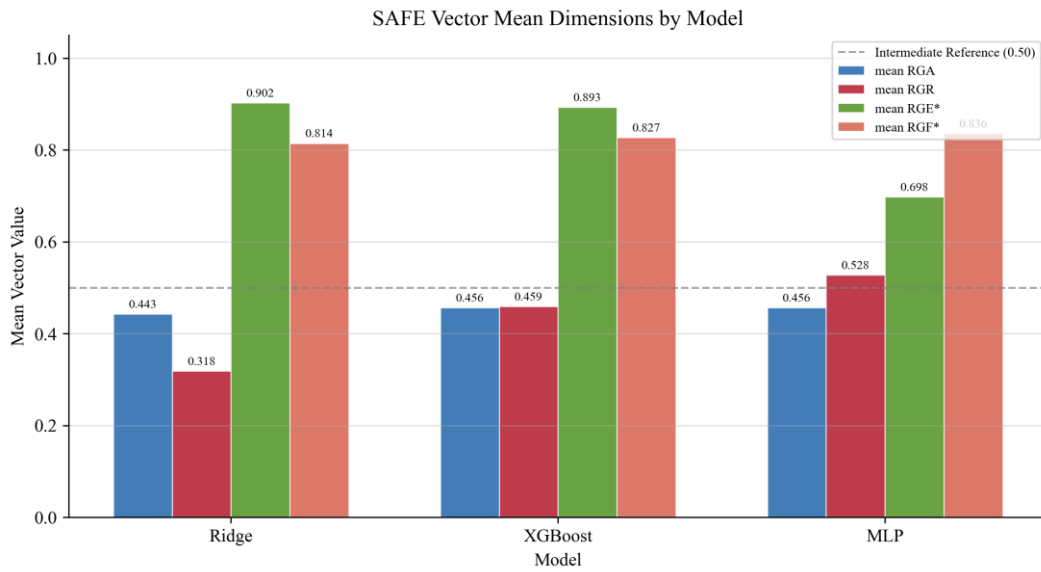


Figure 5.8: Mean compliance values by model and dimension.

Figure 5.8 helps to explain the final CS4 ranking. It shows the average value of each compliance vector for the three models.

The first result is that XGBoost is the most balanced model. It is not the best model in every single dimension, but it does not have one dimension where the result is very weak. This is why it obtains the highest CS4 score.

Ridge Regression has a different behavior. It performs well in Explainability, but it is weak in Robustness. This is linked to the RGR curve, where Ridge Regression decreases a lot when the perturbation level increases. For this reason, Ridge Regression is penalized in the final Compliance Score, especially with the geometric mean. The Neural Network performs better in Fairness and in Robustness when the perturbation becomes stronger. However, it is weaker in RGE*.

For this reason, the Integrated Compliance Score result gives the final SAFE AI ranking of the three machine learning models. XGBoost is the best model according to the Integrated

Compliance Score. Ridge Regression and the Neural Network have some strong dimensions, but also some weaker parts.

The next step is to compare this SAFE AI ranking with the portfolio performance results. This comparison is done in Chapter 6 through the SAFE-Performance Frontier.

5.5 Chapter summary

This chapter presented the SAFE AI evaluation of Ridge Regression, XGBoost and the Neural Network. The analysis was done over the 36 out-of-sample rebalancing dates, from January 2023 to December 2025.

First, the three models were compared using the scalar SAFE AI values. The results showed that there is not one model that is the best in all dimensions. XGBoost was the best in Accuracy and Sustainability/Robustness. The Neural Network was the best in Fairness, while Ridge Regression was the best in Explainability.

After this, the compliance vectors were analyzed. This was useful because the scalar values give only one result, while the vectors show how the models behave across the full curve. The RGA vector showed similar behavior for the three models. The RGR vector showed that Ridge Regression becomes weak when perturbation increases. The RGE* vector showed that Ridge Regression and XGBoost keep more of their ranking when features are neutralized. The RGF* vector showed that the Neural Network is more balanced across sectors.

Finally, the Integrated Compliance Score was computed in order to combine the four dimensions. XGBoost obtained the highest score with the three aggregation methods. This means that XGBoost has the strongest SAFE AI profile among the three machine learning models.

The inclusion of the Fairness dimension increased the Compliance Score for all models, but it did not change the main result. XGBoost remained the best model also after including RGF*. The next chapter connects this SAFE AI result with the portfolio performance results through the SAFE-Performance Frontier.

6 *SAFE-Performance Frontier*

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6.1 *Introduction*

Chapter 4 answered the *first research question* of the thesis. The objective of that chapter was to understand *whether replacing the Historical Mean expected returns with machine-learning-based expected returns improves out-of-sample portfolio performance*.

In this thesis, the SAFE-Performance Frontier is interpreted as an empirical frontier. It is not a theoretical optimum frontier, but a graphical and empirical representation of the relation between SAFE AI compliance and portfolio performance across the model configurations considered in the analysis. Each point on the frontier corresponds to an observed model configuration, not to a theoretical optimum portfolio.

The results showed that XGBoost obtained the best portfolio performance in the main case, where λ was equal to 1. The lambda sensitivity analysis also showed how the results changed when different values of λ were used. This was important because it is the parameter that

controls the balance between portfolio variance and expected return in the MIQP optimizer, as described in Chapter 3 .

Chapter 5 evaluated the three machine learning models through the SAFE AI framework. In that chapter, the objective was not to evaluate the final portfolios, but to compare Ridge Regression, the Neural Network and XGBoost at model level. The results showed that XGBoost also had the strongest SAFE AI profile according to the Integrated Compliance Score.

However, this is not enough to fully answer the second research question of the thesis. The second research question asks *whether the machine learning model with the best SAFE AI evaluation is also the one with the best portfolio-level performance*. Chapter 4 and Chapter 5 show that, in the main configuration, XGBoost is the best model from both points of view. However, this conclusion is based only on the configuration chosen.

For this reason, Chapter 6 extends the analysis to a larger number of configurations: if only the three main machine learning models are compared, the analysis is based only on three points: one point for Ridge Regression, one point for the Neural Network and one point for XGBoost. This comparison is useful, but it does not show whether the relation between SAFE AI evaluation and portfolio performance remains valid when the model configuration changes.

For each model family, different configurations are therefore created by changing the hyperparameters and the Integrated Compliance Score is computed. In addition, the portfolio performance measures are computed using the same portfolio construction procedure used in Chapter 4. In this way, it is possible to compare compliance and performance not only for the three main models, but also for different versions of the same model family.

The chapter is organized as follows. Section 6.2 describes how the SAFE-Performance curves are built. Section 6.3 presents the relation between Integrated Compliance Score and the Sharpe ratio. Section 6.4 presents the relation between Integrated Compliance Score and maximum

drawdown. Section 6.5 presents the relation between Integrated Compliance Score and average turnover. Section 6.6 answers the second research question. Section 6.7 summarizes the chapter.

6.2 Construction of the empirical SAFE-Performance curves

For each of the three model families, 50 configurations are created. This gives 150 configurations in total. These configurations are used to construct the empirical SAFE-Performance curves.

For Ridge Regression, the parameter that changes is the regularization parameter α . This parameter is tested over a log-spaced grid from 0.0001 to 10,000. Ridge Regression is used because it is a linear model with regularization, and the regularization term helps to reduce the risk of overfitting when the data are noisy (Hoerl & Kennard, 1970).

For XGBoost, different combinations of hyperparameters are used; these include tree depth, learning rate, number of estimators, subsampling ratios and regularization parameters. XGBoost is included because it is based on gradient-boosted decision trees and can capture non-linear relations between the input variables and future return ranks (T. Chen & Guestrin, 2016).

For the Neural Network, different configurations are also used. In this case, the parameters that change are the hidden layers, the L2 regularization strength, the learning rate and the batch size. The model is a Neural Network implemented with the same logic described in Chapter 3. Neural networks are included because they can capture non-linear relations between stock characteristics and future returns (Gu et al., 2020).

For every configuration, the model is trained again. The same expanding-window procedure described in Chapter 3 is used. After this, the model produces the expected return vector μ for each rebalancing date. The expected return vector is then used in the same MIQP optimizer

used in the main portfolio analysis. The covariance matrix, the constraints, the solver settings and the rebalancing calendar are kept the same for all configurations. The only element that changes is the configuration of the model used to estimate the expected returns. This is important because the objective is to compare the effect of different model configurations. If the optimizer, the constraints or the rebalancing rule changed, the comparison would not depend only on the model configuration. For this reason, all the portfolio construction steps are kept fixed. This is consistent with the out-of-sample portfolio comparison logic used by DeMiguel et al. (2009), where different strategies are compared using the same evaluation framework.

For each configuration, three portfolio performance measures are computed. These are the annualized Sharpe ratio, maximum drawdown and average monthly turnover. Therefore, it does not represent a theoretical set of optimal portfolios. It represents the relation observed in this thesis between the Integrated Compliance Score and the portfolio metrics obtained by each configuration.

For each configuration, the Integrated Compliance Score is also computed using the three aggregation methods described in Chapter 5: arithmetic mean, geometric mean and root mean square (Giudici & Kolesnikov, 2025).

The relation between Integrated Compliance Score and the portfolio measures is shown using scatter plots. A LOWESS line is added for each model family; this is used to show the general direction of the relation between compliance and performance.

The Spearman rank correlation coefficient is also reported; it is used to understand whether configurations with higher Integrated Compliance Score also tend to have higher or lower portfolio performance values. For the Sharpe ratio, the best value is the highest one. For maximum drawdown and average turnover, the best value is the lowest one. The analysis is presented in two steps. First, all 150 configurations are considered together. The *aggregate*

analysis is useful because the second research question compares the three machine learning models across the full set of configurations. It shows whether higher Integrated Compliance Score values are linked to better portfolio performance.

After that, each model family is analyzed separately. The *within-family analysis* checks whether the same pattern is also present within each model. This step is important because the aggregate result can depend on the fact that the three model families are different from each other. For this reason, the within-family analysis is used to understand whether the Integrated Compliance Score is also useful inside the same model family.

6.3 Integrated Compliance Score and Sharpe ratio

This section presents the relation between the Integrated Compliance Score and the annualized Sharpe ratio. The Sharpe ratio is used because it measures portfolio performance in relation to total risk, and it is one of the standard measures used in portfolio evaluation (Sharpe, 1966). The objective is to understand whether the configurations with a higher compliance score also obtain a higher Sharpe ratio.

Figure 6.1 shows the relation between the Integrated Compliance Score and the annualized Sharpe ratio for all the 150 configurations.

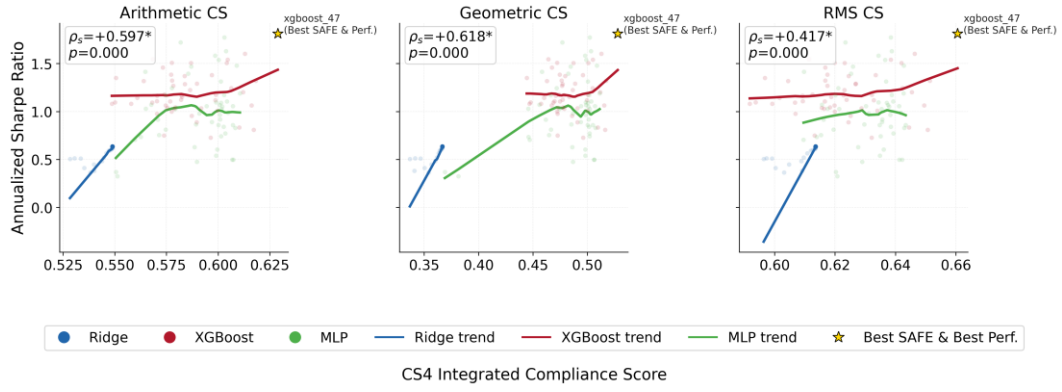


Figure 6.1: Integrated Compliance Score and annualized Sharpe ratio — all 150 configurations.

The three panels report the arithmetic mean, geometric mean and root mean square versions of the Integrated Compliance Score. The LOWESS line is shown separately for each model family.

The relation is positive for the three aggregation methods. The Spearman rank correlation is equal to 0.597 for the arithmetic mean, 0.618 for the geometric mean and 0.417 for the root mean square. In all cases, the p-value is lower than 0.001. Spearman correlation is used because it measures whether two variables move together in terms of ranks, without requiring a linear relation between them (Spearman, 1904). This means that, when all configurations are considered together, the configurations with a higher Integrated Compliance Score also tend to have a higher Sharpe ratio. Therefore, at aggregate level, compliance and Sharpe ratio move in the same direction. This is the first evidence that SAFE AI compliance is connected not only to model quality, but also to portfolio performance. This result is important, but it has to be interpreted carefully. The figure shows that the three model families are not distributed in the same way. XGBoost is mainly positioned in the upper part of the graph, while Ridge Regression is more concentrated in the lower part. The Neural Network has a similar compliance range to XGBoost, but its Sharpe ratios are more different across configurations.

For this reason, the aggregate relation does not describe only what happens inside each model family, but it also includes the differences between Ridge Regression, the Neural Network and XGBoost. This is important because the objective of this chapter is not only to compare the three model families, but also to understand whether the Integrated Compliance Score can help to identify better configurations inside the same family.

In the aggregate graph, the configuration **xgboost_47** is the best point: it has both the highest Integrated Compliance Score and the highest Sharpe ratio among all the 150 configurations. This means that, in the full set of configurations, the most compliant configuration is also the one with the best Sharpe ratio.

The LOWESS line is used in the figures only to show the general direction of the relation between the Integrated Compliance Score and the Sharpe ratio. This method is useful because it gives a smooth representation of the relation without imposing a linear form (Cleveland, 1979).

After this, the same relation is analyzed separately for each model family. This step is useful because it makes it possible to distinguish the aggregate relation from the within-family relation.

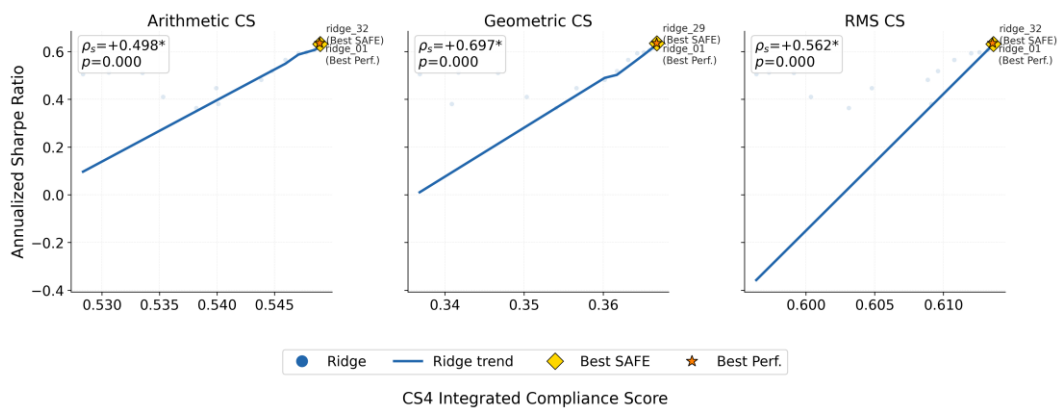


Figure 6.2: Integrated Compliance Score and annualized Sharpe ratio — Ridge Regression.

For Ridge Regression, the relation between the Integrated Compliance Score and the Sharpe ratio is positive and significant. The Spearman correlation is equal to 0.498 for the arithmetic mean, 0.697 for the geometric mean and 0.562 for the root mean square. In all cases, the p-value is lower than 0.001. The LOWESS line increases. This means that, for Ridge Regression, configurations with a higher Integrated Compliance Score also tend to have a higher Sharpe ratio. In this case, the Integrated Compliance Score gives useful information for the selection of the configurations.

Figure 6.2 shows the within-family relation for Ridge Regression. However, the range of the Integrated Compliance Score is narrow. Under the arithmetic mean, it goes approximately from 0.529 to 0.549. For this reason, the result should not be interpreted as a large difference in compliance. It shows only that, inside this small range, higher compliance is associated with higher Sharpe ratio.

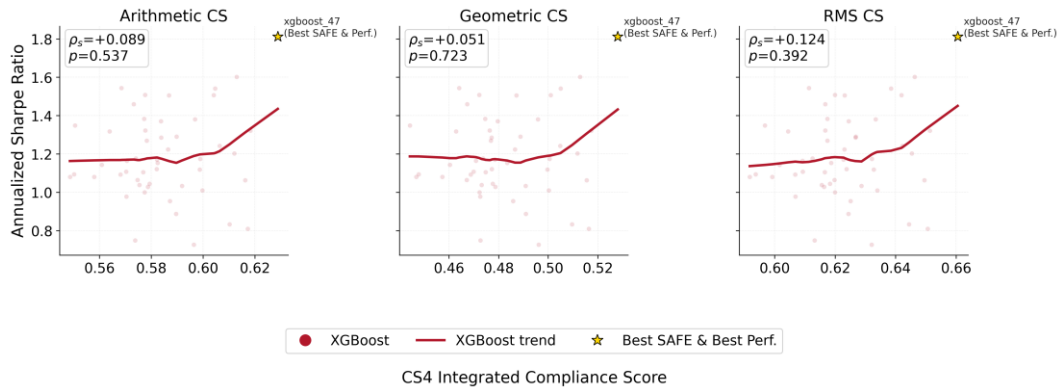


Figure 6.3: Integrated Compliance Score and annualized Sharpe ratio — XGBoost.

For XGBoost, the relation between the Integrated Compliance Score and the Sharpe ratio is not significant. For the arithmetic mean, the Spearman correlation is equal to 0.089, with a p-value equal to 0.537. This means that, for XGBoost, a higher Integrated Compliance Score does not always mean a higher Sharpe ratio.

As shown in Figure 6.3, the points are quite spread. Some configurations have similar Integrated Compliance Score but different Sharpe ratios. This means that the Integrated Compliance Score alone is not sufficient to explain the differences in Sharpe ratio across the XGBoost configurations.

However, **xgboost_47** is the configuration with both the highest Integrated Compliance Score and highest Sharpe ratio in the XGBoost family. This means that, even if the relation is not significant for all XGBoost configurations, the best compliance configuration is also the best Sharpe configuration.

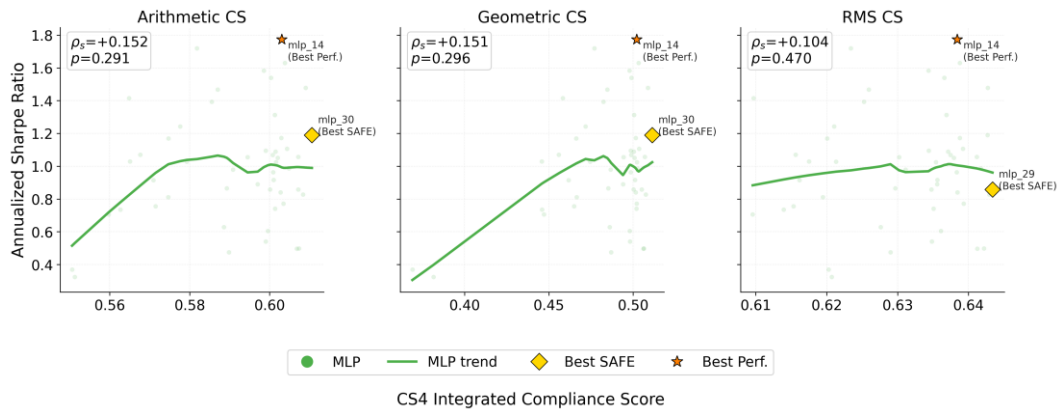


Figure 6.4: Integrated Compliance Score and annualized Sharpe ratio — Neural Network.

For the Neural Network, the relation between the Integrated Compliance Score and the Sharpe ratio is also not significant. For the arithmetic mean, the Spearman correlation is equal to 0.152, with a p-value equal to 0.291. Therefore, also in this case, a higher Integrated Compliance Score does not clearly correspond to a higher Sharpe ratio. As shown in Figure 6.4, the LOWESS line first goes upward and then becomes almost flat, and the configuration with the highest Integrated Compliance Score and the configuration with the highest Sharpe ratio are not the same. This means that, for the Neural Network, the Integrated Compliance Score is not enough to identify the configuration with the best Sharpe ratio.

To conclude, this section shows that the Integrated Compliance Score and the Sharpe ratio are positively related when all configurations are analyzed together. This gives support to the idea that SAFE AI compliance and risk-adjusted performance can move in the same direction. However, this is mainly an aggregate result. Inside the single model families, the relation is clear only for Ridge Regression. For XGBoost and the Neural Network, the relation is not statistically significant. For this reason, the Integrated Compliance Score gives useful information at aggregate level, but it is not always sufficient to select the best-performing configuration inside each model family.

6.4 Integrated Compliance Score and maximum drawdown

This section is used to understand whether, in the empirical frontier, configurations with a higher Integrated Compliance Score also have lower maximum drawdown.

For maximum drawdown, lower values are better. For this reason, a negative Spearman correlation means that configurations with a higher Integrated Compliance Score tend to have lower maximum drawdown.

Figure 6.5 shows the relation between the Integrated Compliance Score and maximum drawdown for all the 150 configurations.

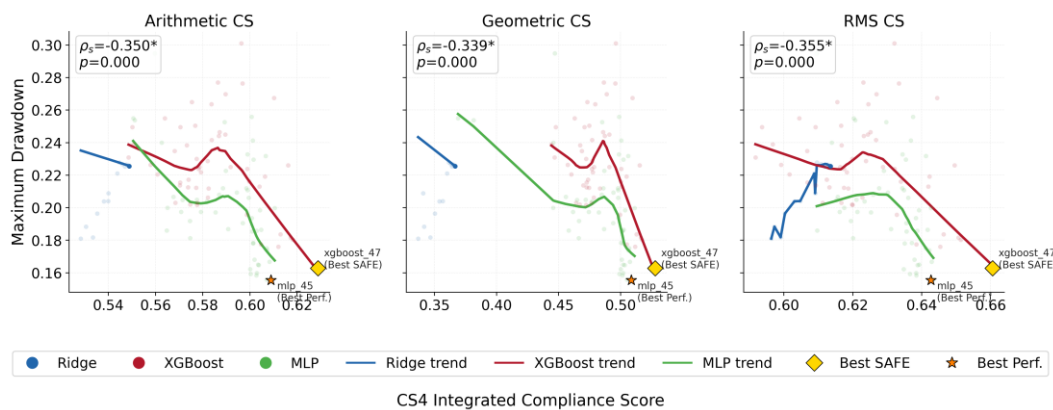


Figure 6.5: Integrated Compliance Score and maximum drawdown — all 150 configurations.

The relation is negative and significant for the three aggregation methods. The Spearman correlation is equal to -0.35 for the arithmetic mean, -0.34 for the geometric mean and -0.36 for the root mean square. In all cases, the p-value is lower than 0.01. This means that, when all configurations are considered together, configurations with a higher Integrated Compliance Score usually have lower maximum drawdown. Therefore, at aggregate level, the relation goes in the expected direction: a higher Integrated Compliance Score corresponds to a lower maximum drawdown value. This reinforces the result found for the Sharpe ratio, because higher compliance is associated not only with better risk-adjusted performance, but also with lower downside risk.

In Figure 6.5, **xgboost_47** is again an important configuration. It has the highest Integrated Compliance Score and a maximum drawdown equal to 16.3%. The configuration with the lowest maximum drawdown is **mlp_45**, with a maximum drawdown equal to 15.5%. However, **mlp_45** has a lower Integrated Compliance Score than **xgboost_47**.

After this, the same relation is analyzed separately for each model family.

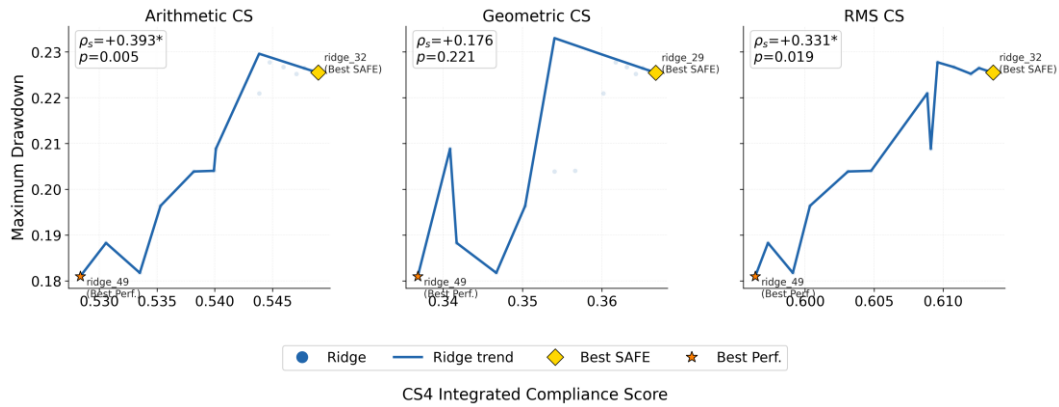


Figure 6.6: Integrated Compliance Score and maximum drawdown — Ridge Regression.

For Ridge Regression, the result goes partly in the opposite direction. The Spearman correlation is positive and significant under the arithmetic mean and root mean square methods. Under the arithmetic mean, the correlation is equal to 0.39, with a p-value equal to 0.05. This means that,

for Ridge Regression, configurations with a higher Integrated Compliance Score tend to have slightly higher maximum drawdown. Therefore, in this case, the relation is not the expected one.

Figure 6.6 shows the relation for Ridge Regression. However, the range of maximum drawdown for Ridge Regression is small. It goes approximately from 20% to 22%. For this reason, the practical difference between the configurations is limited.

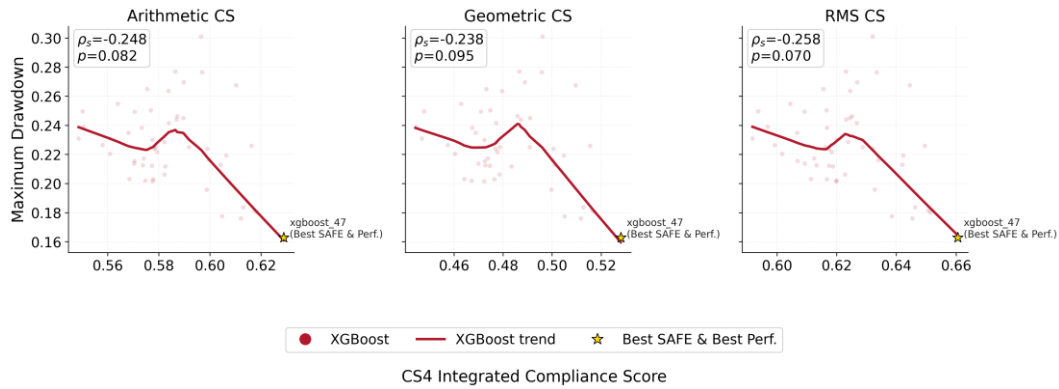


Figure 6.7: Integrated Compliance Score and maximum drawdown — XGBoost.

For XGBoost, the relation is negative, but it is not significant. The Spearman correlation is equal to -0.25, with a p-value equal to 0.08. This means that, for XGBoost, the direction is the expected one, because higher Integrated Compliance Score is associated with lower maximum drawdown. However, the relation is not statistically significant.

Figure 6.7 shows the relation for XGBoost. xgboost_47 is still the main configuration: it has the highest Integrated Compliance Score and also the lowest maximum drawdown inside the XGBoost family, equal to 16.3%. Even if the relation is not significant for all XGBoost configurations, the configuration with the highest compliance is also the configuration with the lowest drawdown inside the XGBoost family.

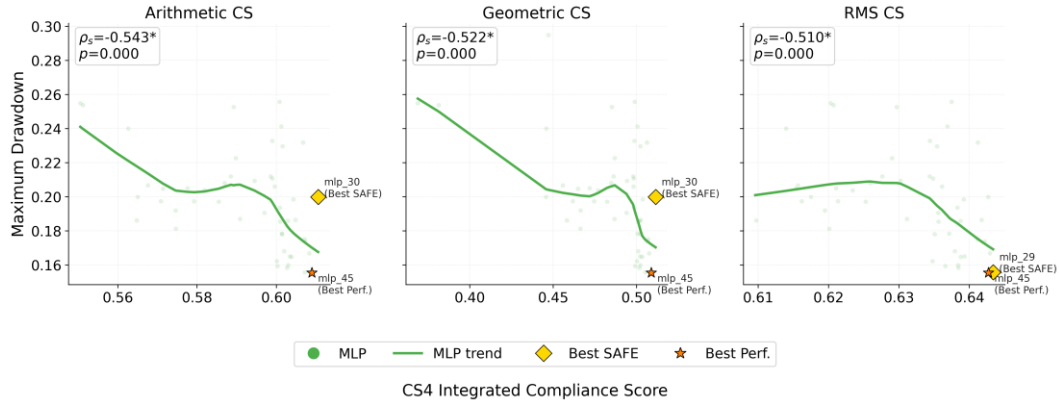


Figure 6.8: Integrated Compliance Score and maximum drawdown — Neural Network.

Figure 6.8 shows the relation for the Neural Network. For the Neural Network, the relation is negative and significant: the Spearman correlation is equal to -0.54, with a p-value lower than 0.01. This is the strongest relation inside the model families in this section. This means that, for the Neural Network, configurations with a higher Integrated Compliance Score tend to have lower maximum drawdown. In this case, the Integrated Compliance Score gives useful information for the selection of the configurations.

To conclude, this section shows again the difference between the *aggregate analysis* and the *within-family analysis*. At aggregate level, the relation between the Integrated Compliance Score and maximum drawdown is negative and significant. This means that higher compliance is associated with lower maximum drawdown when all configurations are considered together. Inside the model families, the result is not the same. The relation is negative and significant for the Neural Network. For XGBoost, the relation is negative but not significant. For Ridge Regression, the result is less clear. In two cases, higher compliance is associated with slightly higher maximum drawdown, which is not the expected direction. However, the effect is small, and it is not found for all aggregation methods. For this reason, the result for maximum drawdown is useful, but it is not the same for all model families.

6.5 Integrated Compliance Score and average turnover

This section helps to understand the empirical relation between the Integrated Compliance Score and turnover in order to assess the practical cost of selecting more compliant configurations.

Figure 6.9 shows the aggregate relation across all the 150 configurations.

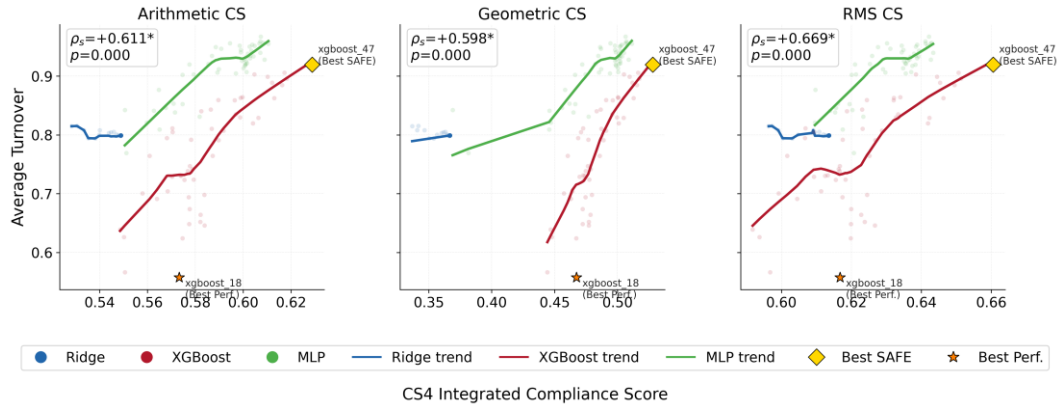


Figure 6.9: Integrated Compliance Score and average turnover — all configurations.

The Spearman correlation is positive and significant under the three aggregation methods. It is equal to 0.611 for the arithmetic mean, 0.598 for the geometric mean and 0.669 for the root mean square. In all cases, the p-value is lower than 0.01.

This means that configurations with higher Integrated Compliance Score tend to trade more. Therefore, at aggregate level, higher compliance is associated with higher average turnover. Figures 6.10, 6.11 and 6.12 show the within-family relations between the Integrated Compliance Score and average turnover.

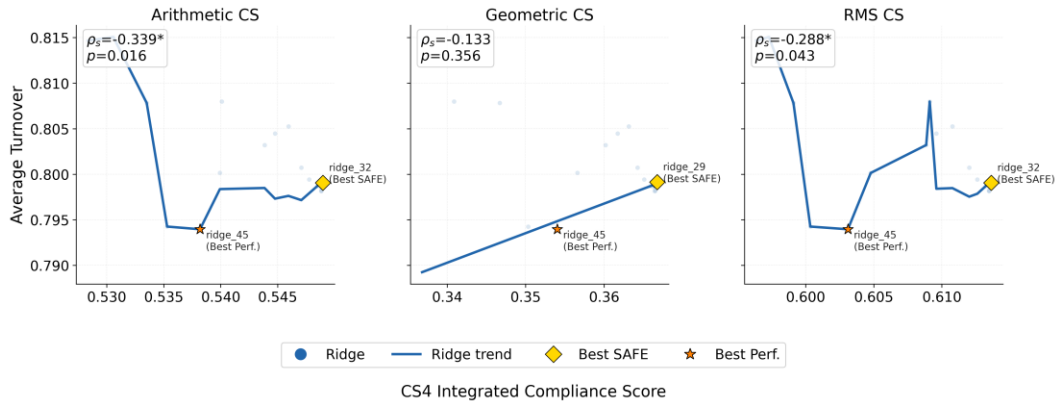


Figure 6.10: Integrated Compliance Score and average turnover — Ridge Regression.

Figure 6.10 shows the within-family relation for Ridge Regression. For Ridge Regression, the result is different. The relation is negative under the arithmetic mean and root mean square methods. However, the range of turnover is very narrow, from about 79% to 82%. For this reason, the result is not very relevant for Ridge Regression. The turnover values are too close to each other.

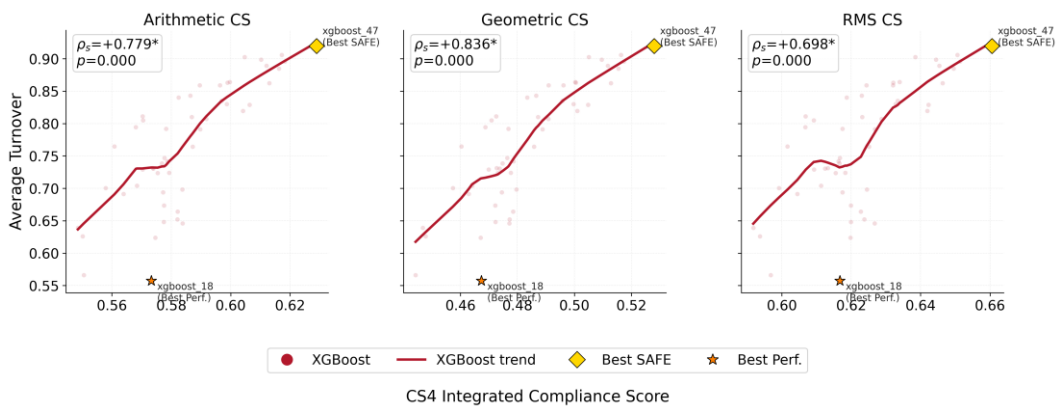


Figure 6.11: Integrated Compliance Score and average turnover — XGBoost.

For XGBoost, the within-family relation is positive and strongly significant. The Spearman correlation is equal to 0.779, with a p-value lower than 0.01.

xgboost_47 has the highest Integrated Compliance Score and an average turnover of 91.9%. The lowest compliance configuration in the XGBoost family is **xgboost_18**, and it has an average turnover of 55.7%.

Figure 6.11 shows the relation for XGBoost. Therefore, for XGBoost, higher compliance is also associated with more trading. This introduces an important trade-off: the configurations with stronger SAFE AI profiles can also be more costly to implement.

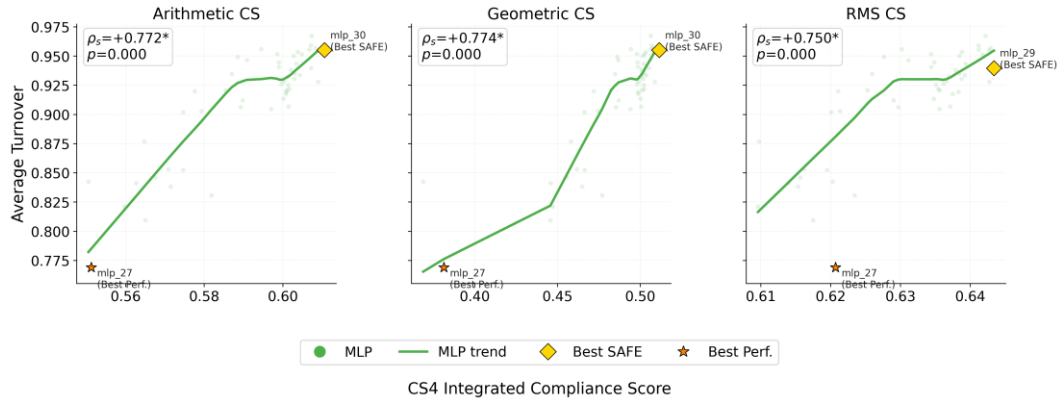


Figure 6.12: Integrated Compliance Score and average turnover — Neural Network.

For the Neural Network, the result is also positive and strongly significant. The Spearman correlation is equal to 0.772, with a p-value lower than 0.01.

The most compliant Neural Network configuration is **mlp_30**, and it has an average turnover of 95.5%. This means that higher compliance is also associated with higher turnover for the Neural Network.

The result of this section is important because it shows that a higher Integrated Compliance Score also has practical costs. In some cases, especially for XGBoost and the Neural Network, more compliant configurations also have higher turnover. This means that they require more trading. As discussed in Chapter 4, higher turnover can reduce portfolio performance when transaction costs are considered. For this reason, the Integrated Compliance Score should not be used alone to select the best configuration. It should be considered together with the Sharpe ratio, maximum drawdown, turnover and transaction costs.

6.6 Answer to the second research question

This section answers the second research question, which asks *whether the machine learning model with the strongest SAFE AI profile is also the one with the best portfolio-level performance*.

The answer is positive in the main case, but it cannot be considered as a general rule for all model families. In Chapter 4, XGBoost obtained the best portfolio performance among the machine learning models, in the main case where λ was equal to 1. In Chapter 5, XGBoost was also the model with the highest Integrated Compliance Score. Therefore, in the main empirical setting of the thesis, the model with the best SAFE AI evaluation is also the model with the best portfolio-level performance.

The extended analysis of 150 configurations gives a more complete interpretation. At aggregate level, a higher Integrated Compliance Score is associated with a higher Sharpe ratio and lower maximum drawdown. This means *that SAFE AI compliance is not only a model-level measure, because it is also connected with portfolio outcomes*. This is the main empirical finding of the chapter.

However, the relation is not equally strong inside each model family. For the Sharpe ratio, the relationship is positive and statistically significant only for Ridge Regression. For XGBoost and the Neural Network, instead, the relationship is not statistically significant. The results are different when maximum drawdown is considered. In that case, the relationship is stronger for the Neural Network, while for XGBoost it is negative but not significant. For Ridge Regression, the evidence is less clear. Overall, the Integrated Compliance Score gives useful information, but it is not enough on its own to identify the best configuration within each model family.

The most important configuration in the full analysis is **xgboost_47**. This configuration has the highest Integrated Compliance Score and also the highest Sharpe ratio among all 150

configurations. It also has a low maximum drawdown, equal to 16.3%. However, it does not have the lowest maximum drawdown overall, because the lowest value is obtained by **mlp_45**. For this reason, **xgboost_47** can be considered the strongest overall configuration, but not the best configuration for every single performance metric.

The turnover results add an important limitation. At aggregate level, a higher Integrated Compliance Score is associated with higher average turnover. The same relation is found for XGBoost and for the Neural Network. This means that more compliant configurations often require more trading. This is important because, as shown in Chapter 4, higher turnover can reduce performance when transaction costs are considered.

Overall, the second research question can be answered positively for the main case: *XGBoost has the strongest SAFE AI profile and also the best portfolio-level performance*. However, the extended analysis of the empirical SAFE-Performance Frontier shows that *this result is not a general rule for all configurations and all model families*. The final finding is therefore not that higher SAFE AI compliance always produces better portfolios. The finding is that *SAFE AI compliance is useful for portfolio evaluation*, but it must be interpreted together with the Sharpe ratio, maximum drawdown, turnover and transaction costs.

6.7 Chapter summary

This chapter connected the model-level SAFE AI evaluation with the portfolio-level results through the empirical SAFE-Performance Frontier. The objective was to understand whether models and configurations with higher SAFE AI compliance also produce better portfolio outcomes.

The analysis first considered the relation between the Integrated Compliance Score and the Sharpe ratio. At aggregate level, the relation is positive. This means that, when all 150

configurations are considered together, configurations with higher compliance also tend to have higher risk-adjusted performance. This supports the idea that SAFE AI compliance is not only a model-level concept, but can also be connected to portfolio performance.

The second part of the analysis considered maximum drawdown. Also in this case, the aggregate relation goes in the expected direction. Configurations with higher Integrated Compliance Score usually have lower maximum drawdown. This reinforces the result obtained for the Sharpe ratio, because higher compliance is associated not only with better risk-adjusted performance, but also with lower downside risk.

However, the analysis inside each model family gives a more careful interpretation. The relation between compliance and performance is not always statistically significant within Ridge Regression, XGBoost and the Neural Network. This means that the aggregate result should not be interpreted as a rule that holds for every model family and every configuration.

The turnover analysis adds another important element. At aggregate level, and especially for XGBoost and the Neural Network, higher Integrated Compliance Score is associated with higher average turnover. This means that more compliant configurations can also be more costly to implement. For this reason, the SAFE AI profile of a model should not be evaluated separately from its trading intensity and transaction costs.

The answer to the second research question is therefore positive in the main case, but conditional in the extended configuration analysis. XGBoost has both the strongest SAFE AI profile and the best portfolio-level performance in the main empirical setting. However, the SAFE-Performance Frontier shows that this result cannot be generalized to all configurations and all model families.

Overall, the chapter shows that the Integrated Compliance Score is a useful additional criterion for portfolio model evaluation. It helps to connect model quality with portfolio outcomes.

However, it should not be used alone to select the best portfolio model. SAFE AI compliance should be considered together with the Sharpe ratio, maximum drawdown, turnover and transaction costs. Since the frontier is empirical, the results should be interpreted as evidence from the configurations analyzed in this thesis, not as a theoretical rule. Therefore, *the SAFE AI evaluation can support the portfolio selection process, but it cannot replace the direct analysis of portfolio performance and implementation costs.*

7 Conclusions

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7.1 Summary of the thesis

This thesis started from a central problem in portfolio optimization: the estimation of expected returns. In the mean-variance framework introduced by Markowitz (1952), the investor needs two main inputs: *the expected return vector and the covariance matrix*. Previous literature shows that expected returns are difficult to estimate and that small errors in this input can produce large changes in the optimal portfolio weights (Michaud, 1989; Best & Grauer, 1991). For this reason, the thesis compared the Historical Mean Benchmark with three machine learning models: Ridge Regression, the Neural Network and XGBoost. The Historical Mean was used as the traditional benchmark. XGBoost was the main machine learning model, while Ridge Regression and the Neural Network were included as robustness checks.

All four return estimation methods were tested inside the same constrained portfolio optimization framework. The same optimizer, covariance estimator, cardinality constraint, sector constraint, rebalancing rule and transaction cost structure were used for all models. This is important because the comparison focuses mainly on the way in which the expected return vector is estimated. The other parts of the portfolio construction process are kept fixed.

The thesis also included a second level of evaluation. The three machine learning models were evaluated through the SAFE AI framework proposed by Giudici and Raffinetti (2023). The four dimensions considered were robustness, accuracy, fairness and explainability. These dimensions were then combined into the Integrated Compliance Score, denoted as CS4. Finally, the relation between CS4 and portfolio performance was studied through the empirical SAFE-Performance Frontier, using 150 model configurations.

The thesis was therefore organized around two research questions. The *first research question* asked whether XGBoost generates better out-of-sample portfolio performance than the Historical Mean benchmark when both are used in the same constrained portfolio optimization framework. The second research question asked whether the machine learning model with the strongest SAFE AI profile is also the model with the best risk-adjusted portfolio performance. The following sections answer these two research questions.

7.2 Answer to the first research question

The answer to the *first research question* is positive but conditional. In the main setting of the thesis, XGBoost performs better than the Historical Mean benchmark, but the result must be interpreted with some caution.

In the main case, where λ is equal to 1.0, XGBoost is the portfolio with the best overall risk-adjusted performance among the four strategies. It has a higher Sharpe ratio than the Historical Mean benchmark, with a value of 0.9007 against 0.8243. XGBoost also has a higher Sortino ratio and a lower maximum drawdown than the Historical Mean benchmark. This means that the improvement is not only related to return, but also to the risk profile of the portfolio.

However, XGBoost is not the best model in every single metric. Ridge Regression has the lowest maximum drawdown among the four portfolios. This shows that the interpretation must be based on the full set of indicators, not only on one measure. In this sense, XGBoost is the best performing model in the main setting because it gives the best overall balance between return and risk.

The result also remains valid after transaction costs. However, the advantage of XGBoost becomes smaller. This happens because XGBoost has a higher average turnover than the Historical Mean benchmark. Its turnover is 78.15%, while the turnover of the benchmark is 46.46%. This means that XGBoost changes the portfolio more frequently from one rebalancing date to the next one. For this reason, part of its gross advantage is absorbed by trading costs.

With transaction costs equal to 30 basis points, the net Sharpe ratio of XGBoost is still higher than the one of the Historical Mean benchmark. The net Sharpe ratio is 0.7846 for XGBoost and 0.7588 for the benchmark. However, the difference is smaller than before transaction costs. This confirms that XGBoost creates additional value also after costs, but it also shows that this result depends on the level of trading activity.

The lambda sensitivity analysis gives a more careful interpretation of the result. XGBoost performs better than the Historical Mean benchmark mainly when λ is equal to 1.0 and 2.0. For other values of λ , the comparison is less clear or becomes less favorable for XGBoost. Therefore, the *first research question* has a positive answer only in the main setting of the thesis.

XGBoost improves out-of-sample risk-adjusted portfolio performance in the main setting, but the result is not valid for every value of λ .

Ridge Regression and the Neural Network do not show the same improvement. In the main setting, both models have lower Sharpe ratios than the Historical Mean benchmark. Ridge Regression behaves more defensively and records a lower maximum drawdown, but this advantage does not translate into better overall risk-adjusted performance. The Neural Network is the weakest strategy in the portfolio comparison. This result is relevant because it shows that machine learning, by itself, does not guarantee an improvement in portfolio performance. What matters is the specific model used to estimate expected returns, and the predictions must be assessed after they enter the portfolio optimizer.

7.3 Answer to the second research question

The answer to the second research question is positive in the main empirical comparison, but it is also conditional. Also in this case, the result must be interpreted with caution.

XGBoost obtains a higher Integrated Compliance Score, denoted as CS4, than Ridge Regression and the Neural Network, and therefore has the strongest overall SAFE AI profile among the three machine learning models.

This result is obtained under the three aggregation methods used in the thesis: arithmetic mean, geometric mean and root mean squared. This means that the ranking of the models does not depend on one specific aggregation method.

However, XGBoost is not the best model in every single SAFE AI dimension. XGBoost performs better in terms of accuracy and robustness. Ridge Regression performs better in terms of explainability, while the Neural Network performs better in terms of fairness. Therefore, the higher CS4 of XGBoost does not mean that XGBoost dominates the other models in all

dimensions. It means that XGBoost has the best overall balance across the four dimensions considered in the SAFE AI evaluation.

This result is connected to the portfolio results discussed in the previous section. In the main portfolio setting, XGBoost is also the model with the highest Sharpe ratio and the best overall risk-adjusted performance. Therefore, in the comparison between the three main machine learning models, the model with the highest SAFE AI compliance score is also the model with the best portfolio performance.

For this reason, the answer to the second research question is positive in the main comparison. The model preferred by the SAFE AI evaluation is the same model preferred by the portfolio evaluation. This result is important because it shows that, in the main setting of the thesis, the best-performing model is not only the model with the highest Sharpe ratio, but also the model with the most balanced SAFE AI profile.

The empirical SAFE-Performance Frontier presented in Chapter 6 extends this analysis to 150 model configurations, with 50 configurations for each model family. This part of the analysis gives a more detailed interpretation. When all configurations are considered together, higher CS4 values are associated with higher Sharpe ratios and lower maximum drawdowns. This suggests that SAFE AI compliance contains useful information also from a portfolio point of view.

However, this relation is not equally strong inside each model family. The relation between CS4 and Sharpe ratio is statistically significant mainly for Ridge Regression. For XGBoost and the Neural Network, the relation is weaker when the configurations are analyzed within the same model family. In addition, higher compliance can also be associated with higher turnover. This is important because a model can look better in terms of SAFE AI compliance, but still

require more trading activity. Therefore, higher compliance does not automatically mean better performance after transaction costs.

The conclusion for the second research question is therefore positive but conditional. In the main empirical comparison, the model with the highest SAFE AI compliance score is also the model with the best risk-adjusted portfolio performance. However, the empirical SAFE-Performance Frontier shows that this relation is not valid in the same way for all configurations and all model families. For this reason, SAFE AI compliance and portfolio performance should be analyzed together, but one should not be interpreted as a mechanical consequence of the other. The final finding is not that higher SAFE AI compliance always produces better portfolios, but that SAFE AI compliance is useful when it is interpreted together with risk-adjusted performance, drawdown, turnover and transaction costs.

7.4 Main contributions of the thesis

This thesis provides three main contributions.

The first contribution is the controlled comparison between the Historical Mean benchmark and machine-learning-based expected return estimates. All four return estimation methods were tested inside the same constrained portfolio optimization framework. The same covariance estimator, optimizer, portfolio constraints, rebalancing rule and transaction cost structure were used for all models. This is important because the comparison focuses mainly on the method used to estimate expected returns. In this way, the differences in portfolio performance are not due to different optimization rules, but mainly to different estimates of the expected return vector.

The second contribution is the application of the SAFE AI framework to a stock return prediction problem. In this thesis, the SAFE AI dimensions were adapted to a financial setting

where the objects of prediction are stocks and not individuals. Robustness, accuracy, fairness and explainability were therefore interpreted in relation to return prediction models. In particular, fairness was measured across GICS sectors instead of demographic groups. This makes the SAFE AI evaluation suitable for the structure of the dataset used in the thesis.

The third contribution is the connection between model-level SAFE AI compliance and portfolio-level performance. This connection is based on two elements. The first one is the Integrated Compliance Score, denoted as CS4, which summarizes the four SAFE AI dimensions into one synthetic measure. The second one is the empirical SAFE-Performance Frontier, which is used to study the relation between CS4 and portfolio performance across 150 model configurations. This is useful because it connects two parts of the thesis that could otherwise remain separate: the quality of the prediction model and the performance of the portfolio generated from its predictions.

Overall, the thesis contributes by combining expected return estimation, constrained portfolio optimization and SAFE AI evaluation inside the same empirical analysis. The contribution is therefore mainly empirical and methodological. The thesis applies and connects existing SAFE AI concepts to a machine-learning-based portfolio construction problem.

7.5 Limitations of the analysis

The results of this thesis should be interpreted within the specific empirical setting used in the analysis. The study is based on a cleaned universe derived from the S&P 500, a specific out-of-sample period, a fixed portfolio optimization framework and a specific implementation of the SAFE AI evaluation. For this reason, some limitations must be considered.

The first limitation concerns the investment universe. The analysis is based on stocks derived from the S&P 500. This choice gives a consistent and liquid universe for the empirical analysis.

However, the results cannot be automatically extended to other markets. For example, the same results may not hold for small-cap stocks, emerging markets or less liquid securities. In addition, the cleaning process used to obtain the final universe may introduce a data-availability bias, because stocks with insufficient data coverage are excluded from the analysis.

The second limitation is related to the out-of-sample period. The test period covers the years from 2023 to 2025. This period uses recent data, but it remains relatively short. A longer out-of-sample period could lead to different results. The performance of the models could also change in different market conditions, for example during periods of stronger market stress or lower market volatility.

The third limitation concerns the portfolio construction framework. In this thesis, all portfolios are built using the same constrained optimization framework, with a fixed cardinality of ten stocks and specific weight and sector constraints. This makes the comparison between models more controlled. However, different constraints could lead to different portfolio results. For example, a different number of selected stocks, tighter sector limits or explicit turnover constraints could change the performance of the strategies.

The fourth limitation concerns transaction costs. In the empirical analysis, transaction costs are modeled as proportional costs based on turnover. This choice makes it possible to compare the models in a simple and consistent way. However, real trading costs can be more complex. They can depend on bid-ask spreads, commissions, market impact, stock liquidity and trade size. This is relevant because the results show that turnover has an important effect on net performance, especially for the more dynamic models.

The fifth limitation concerns the set of predictors used by the machine learning models. The analysis uses variables related to reversal, momentum, volatility, illiquidity and size. These variables are standard stock-level characteristics and can be computed from the available

market data. However, other variables could also contain useful information. For example, accounting data, macroeconomic indicators, analyst forecasts, ESG scores or alternative data could change the quality of the expected return estimates.

The final limitation concerns the adaptation of SAFE AI and of the empirical SAFE-Performance Frontier to the financial setting of the thesis. Fairness is measured across GICS sectors. This choice is reasonable because sectors represent natural groups inside the investment universe. However, it is not the only possible way to define fairness in a stock prediction problem. Other groups could be based on size, liquidity, volatility, industry, geography or ESG characteristics. For this reason, the SAFE AI results should be interpreted according to the specific definition of fairness used in this thesis. In addition, the empirical SAFE-Performance Frontier is based on the 150 configurations considered in the analysis. Therefore, it should not be interpreted as a theoretical frontier or as evidence that the same relation must hold for all possible model configurations.

7.6 Future research directions

The limitations discussed in the previous section suggest some possible directions for future research.

A first direction is to apply the same empirical design to other investment universes. This thesis focuses on a cleaned universe derived from the S&P 500. Future studies could apply the same methodology to European equities, emerging markets or global equity portfolios. This would make it possible to understand whether the results obtained in this thesis are specific to large and liquid U.S. stocks or whether they also hold in other markets.

A second direction is to extend the out-of-sample period. The test period used in this thesis covers the years from 2023 to 2025. This period is recent, but it remains relatively short. Future

research could use a longer test period or a period that includes different market conditions. This would help to understand whether the results are stable across time and across different market regimes.

Future research could also modify the portfolio construction framework. In this thesis, all models are compared using the same constrained optimization framework, with a fixed cardinality of ten stocks and specific weight and sector constraints. This choice keeps the comparison focused on the expected return estimator. However, other portfolio constraints could be tested. For example, future studies could include explicit turnover constraints, different cardinality levels, tighter sector limits or ESG constraints. This would be useful because the results of this thesis show that turnover has an important effect on net performance.

Another possible extension concerns transaction costs. In this thesis, transaction costs are modeled as proportional costs based on turnover. This is useful for a first comparison, but it does not capture all components of real trading costs. Future research could include bid-ask spreads, commissions, market impact and stock-specific trading costs. This would make the evaluation of net performance more realistic, especially for models such as XGBoost, which generate higher turnover.

The feature set could also be expanded. The machine learning models used in this thesis are based on reversal, momentum, volatility, illiquidity and size. Future research could add other variables, such as accounting data, macroeconomic indicators, analyst forecasts, ESG scores or alternative data. This would make it possible to understand whether a richer information set improves the quality of the expected return estimates.

Another direction concerns the SAFE AI evaluation. In this thesis, fairness is measured across GICS sectors. This is a reasonable choice for a stock prediction problem, but it is not the only possible one. Future research could define fairness using other groups, such as size, liquidity,

volatility, industry, geography or ESG characteristics. This would help to understand whether the SAFE AI ranking of the models changes when a different definition of fairness is used.

Finally, future research could further develop the link between SAFE AI compliance and portfolio performance. The empirical SAFE-Performance Frontier used in Chapter 6 connects model-level compliance with portfolio-level results across different configurations. This approach could be applied to other financial settings, such as factor investing, fixed-income portfolios or credit-risk models. It could also be extended by studying a wider set of configurations that are strong in both dimensions at the same time, with high SAFE AI compliance and good portfolio performance.

7.7 Final concluding remarks

This thesis shows that machine learning can be useful for expected return estimation, but not in an automatic way and not in every portfolio setting. In the main empirical setting, XGBoost improves out-of-sample risk-adjusted portfolio performance compared with the Historical Mean benchmark. However, this result depends on the portfolio configuration and on the value of the risk-aversion parameter λ . It is also associated with higher turnover, which reduces part of the advantage after transaction costs.

For this reason, the value of machine learning should not be evaluated only before the portfolio optimization step. A model can produce useful predictions, but the final result also depends on how these predictions are transformed into portfolio weights. This is one of the main ideas of the thesis. Expected return estimation and portfolio construction are connected, but they are not the same thing.

The thesis also shows that portfolio performance is not the only aspect to consider when machine learning models are used in finance. The SAFE AI evaluation adds a second level of

analysis, focused on the quality of the models that generate the expected return estimates. In this setting, XGBoost is not the best model in every single SAFE AI dimension, but it is the model with the most balanced profile across robustness, accuracy, fairness and explainability. Overall, the thesis suggests that expected return estimation, portfolio optimization and SAFE AI evaluation should be considered together. A model used for investment decisions should not only generate good portfolio results, but should also be evaluated in terms of robustness, accuracy, fairness and explainability. The final conclusion of the thesis is that SAFE AI compliance can support portfolio model evaluation, but it cannot replace the direct analysis of portfolio performance, turnover and transaction costs.

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Appendix

The appendices include additional material used to support the empirical analysis of this thesis.

They are divided into four sections.

Appendix A reports the complete lambda sensitivity tables discussed in Chapter 4, Appendix B shows the empirical pipeline followed in the analysis, from the original Bloomberg dataset to the construction of the SAFE-Performance Frontier, Appendix C presents the SAFE-Performance Frontier scatter plot, Appendix D reports the main software packages and tools used to carry out the empirical work.

Appendix A – Lambda Sensitivity Tables

Tables A.1, A.2 and A.3 report the complete results of the lambda sensitivity analysis described in Section 3.6.4 and discussed in Section 4.5. The analysis considers the grid $\lambda \in \{0.1, 0.2, 0.5, 1.0, 2.0, 5.0, 10.0\}$. For each row, the best-performing model is highlighted in **bold** and marked with an asterisk (*). For the Sharpe Ratio, higher values indicate better performance. For Maximum Drawdown and Average Turnover, lower values are preferred.

λ	Baseline	Ridge	MLP	XGBoost
0.1	1.2767*	0.5152	0.6837	1.1544
0.2	1.2042*	0.4731	0.6372	1.1233
0.5	1.0214*	0.3847	0.3413	0.9854
1.0	0.8243	0.6151	0.3021	0.9007*
2.0	0.8225	0.7482	0.3464	0.8549*
5.0	0.9394*	0.7275	0.3563	0.8368
10.0	0.9512*	0.6641	0.3266	0.8142

Table A.1: Sharpe ratio for each model across $\lambda \in \{0.1, 0.2, 0.5, 1.0, 2.0, 5.0, 10.0\}$. Bold and shaded cells indicate the best (highest) Sharpe ratio per row. Ridge and MLP are robustness checks.

λ	Baseline	Ridge	MLP	XGBoost
0.1	0.1538	0.1457*	0.1652	0.1600
0.2	0.1689*	0.1938	0.2069	0.1954
0.5	0.2126*	0.2206	0.2387	0.2243
1.0	0.2566	0.2258*	0.2557	0.2341
2.0	0.2696	0.2161*	0.2723	0.2250
5.0	0.2685	0.2154*	0.2948	0.2234
10.0	0.2680	0.2149*	0.3119	0.2234

Table A.2: Maximum drawdown for each model across $\lambda \in \{0.1, 0.2, 0.5, 1.0, 2.0, 5.0, 10.0\}$. Bold and shaded cells indicate the best (lowest) drawdown per row. Ridge and MLP are robustness checks.

λ	Baseline	Ridge	MLP	XGBoost
0.1	0.4080*	0.7458	0.9403	0.7533
0.2	0.4269*	0.7800	0.9438	0.7722
0.5	0.4579*	0.7961	0.9352	0.7777
1.0	0.4646*	0.7986	0.9315	0.7815
2.0	0.4475*	0.7975	0.9385	0.7856
5.0	0.4309*	0.7927	0.9466	0.7668
10.0	0.4268*	0.7793	0.9482	0.7604

Table A.3: Average turnover for each model across $\lambda \in \{0.1, 0.2, 0.5, 1.0, 2.0, 5.0, 10.0\}$. Bold and shaded cells indicate the best (lowest) turnover per row. Ridge and MLP are robustness checks.

Appendix B – Empirical Pipeline

Figure B.1 reports the full empirical pipeline used in the thesis. The figure summarizes the main steps of the analysis, starting from the raw historical stock price data and ending with the SAFE-Performance Frontier.



Figure B.1: Empirical pipeline from raw financial data to the SAFE-Performance Frontier.

Appendix C – SAFE-Performance Frontier

Figure C.1 shows the SAFE-Performance Frontier obtained from the 150 parameter combinations considered in the analysis. The configurations refer to Ridge Regression, MLP and XGBoost. Ridge Regression is shown in blue, MLP in orange and XGBoost in green.

Each point in the graph represents one specific model configuration. The x-axis reports the Sharpe Ratio, while the y-axis reports the arithmetic CS4 compliance score. The dashed black line represents the LOWESS smoothed curve, with $\text{frac} = 0.55$. The global Spearman correlation coefficient is equal to $\rho = 0.597$.

The red star identifies the Pareto-dominant configuration, `xgboost_47`. This configuration achieved the highest Sharpe Ratio while also having a compliance score in the top 50% of all configurations.

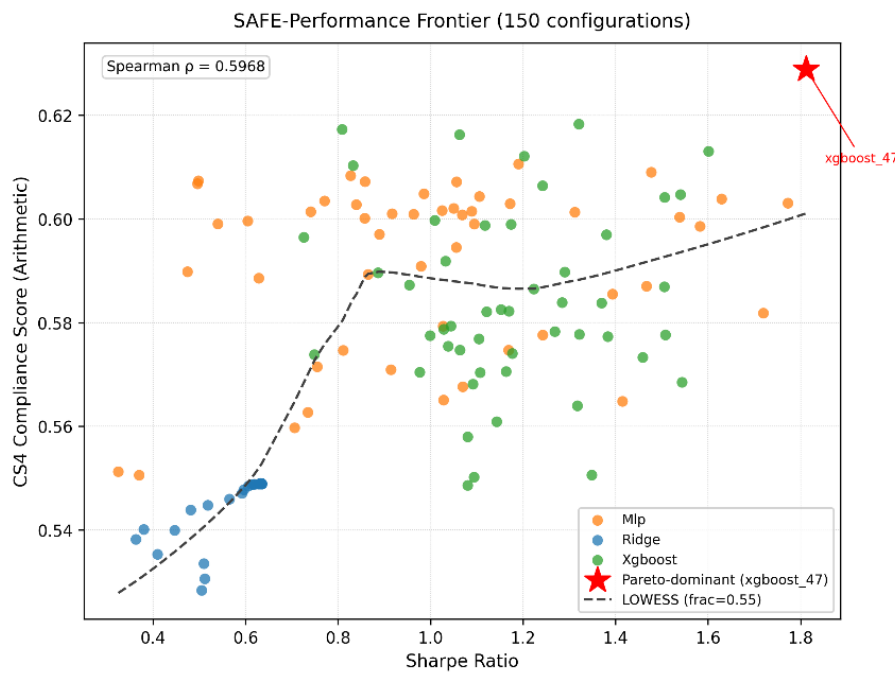


Figure C.1: SAFE-Performance Frontier across 150 model configurations. Each point represents one configuration. The dashed line shows the LOWESS smoothing curve. The red star marks the Pareto-dominant configuration (XGBoost, config 47).

Appendix D – Software and Reproducibility

Table D.1 reports the main software packages and tools used for the empirical analysis. The full workflow was implemented in Python. Bloomberg Terminal was used as the main data source. Gurobi version 13 was used as the MIQP solver for the constrained portfolio optimization problem.

The SAFE AI assessment was carried out using the safeaipackage library developed by Babaci and Giudici (2025). Since the original data come from Bloomberg, the raw dataset cannot be redistributed. However, the main steps of the empirical pipeline, the model configurations and the software environment are reported to make the analysis as transparent and replicable as possible, conditional on access to the same data source.

Tool	Use
Python 3.x	Data processing, modelling, and visualization
pandas / NumPy	Data manipulation and numerical computation
scikit-learn	Ridge Regression and Neural Network
XGBoost	Gradient boosting return prediction
Gurobi 13	Constrained MIQP portfolio optimization
safeaipackage	SAFE AI framework evaluation
Bloomberg Terminal	Primary financial data source (Jan 2010 – Dec 2025)

Table D.1: Software and tools used in the empirical analysis.